

MOMENT ESTIMATES FOR SOME RENORMALIZED PARABOLIC ANDERSON MODELS

XIA CHEN, AURÉLIEN DEYA, CHENG OUYANG, AND SAMY TINDEL

ABSTRACT. The theory of regularity structures enables the definition of the following parabolic Anderson model in a very rough environment: $\partial_t u_t(x) = \frac{1}{2}\Delta u_t(x) + u_t(x) \dot{W}_t(x)$, for $t \in \mathbb{R}_+$ and $x \in \mathbb{R}^d$, where $\dot{W}_t(x)$ is a Gaussian noise whose space time covariance function is singular. In this rough context, we shall give some information about the moments of $u_t(x)$ when the stochastic heat equation is interpreted in the Skorohod as well as the Stratonovich sense. Of special interest is the critical case, for which one observes a blowup of moments for large times.

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2010 *Mathematics Subject Classification.* Primary: 60L30, 60L50, 60F10, 60K37.

Key words and phrases. parabolic Anderson model, regularity structures, Skorohod equation, Stratonovich equation, moment estimate, critical time.

C. Ouyang is supported in part by Simons grant #355480.

S. Tindel is supported by the NSF grant DMS-1952966.

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1. INTRODUCTION

The parabolic Anderson model (sometimes abbreviated as PAM in the sequel) is a linear partial differential equation in a random environment. As for other widely studied objects, many different versions of the model have been analyzed in the literature. In this paper we are concerned with the following continuous version of PAM defined for $(t, x) \in \mathbb{R}_+ \times \mathbb{R}^d$:

$$\partial_t u_t(x) = \frac{1}{2} \Delta u_t(x) + u_t(x) \dot{W}_t(x), \quad (1.1)$$

where Δ stands for the Laplace operator and $\dot{W}$ is a centered Gaussian noise. Equation (1.1) is obviously a stochastic PDE, and the generalized differential element $u_t(x) \dot{W}(x)$ will be interpreted either in the Skorohod sense (for which the product $u_t(x) \diamond \dot{W}_t(x)$ is considered as a Wick product) or in the Stratonovich sense (where the product $u_t(x) \dot{W}_t(x)$ is the usual one). Notice that we consider those two versions of the model for the sake of generality, but also because we will transfer some information from the Skorohod to the Stratonovich equation.

The study of moments for equation (1.1) is at the heart of Anderson's initial motivation in the model. Indeed, the moments of $u_t(x)$ characterize the so-called intermittency phenomenon, as described in e.g. [25]. Moments are also related to the localization of eigenvectors for the Anderson operator $L = \frac{1}{2} \Delta + \dot{W}$, since $u_t(x)$ can be seen as the Laplace transform of the spectral measure of L (see [26, Relation (2.27)]). This is why quantities of the form $\mathbf{E}[|u_t(x)|^p]$, for a given $p > 1$ and for the solution u to (1.1), have been intensively analyzed in the recent past. Let us mention [12, 13, 14] when $\dot{W}$ is a white noise in time and [6, 10, 22] for fractional noises.

The current paper can be seen as an additional step towards moment estimates for the parabolic Anderson model. Namely our study aims at giving some information about the moments of equation (1.1) when $\dot{W}$ is a very rough environment, given as the formal derivative of a multiparametric fractional Brownian motion W . Specifically, consider a centered Gaussian process W indexed by $\mathbb{R}_+ \times \mathbb{R}^d$ and defined on a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$, whose covariance is expressed as:

$$\mathbf{E}[W_t(x) W_s(y)] = R_0(s, t) \prod_{j=1}^d R_j(x_j, y_j),$$

where $\{H_j; 0 \leq j \leq d\}$ is a family of Hurst indices in $(0, 1)$ and the covariance function R_j is defined by

$$R_j(u, v) = \frac{1}{2} (|u|^{2H_j} + |v|^{2H_j} - |u - v|^{2H_j}), \quad u, v \in \mathbb{R}. \quad (1.2)$$

Then the noise $\dot{W}$ driving equation (1.1) has to be thought of as the (ill-defined) derivative $\partial_{t x_1 \dots x_d}^{d+1} W_t(x)$. In this paper we are interested in noises which are rougher than white noise in at least some directions. Otherwise stated we wish some of the H_j 's in (1.2) to be smaller than $\frac{1}{2}$. Recall that the covariance function of $\dot{W}$ is formally written as

$$\mathbf{E} \left[\dot{W}_t(x) \dot{W}_s(y) \right] = \gamma_0(t-s) \gamma(x-y), \quad \text{with} \quad \gamma(x-y) \equiv \prod_{j=1}^d \gamma_j(y_j - x_j), \quad (1.3)$$

where each γ_j is the distributional derivative $\partial_{uv}^2 R_j$. Notice that whenever $H_j < \frac{1}{2}$ the covariance γ_j is a distribution. Therefore we will often express γ_0 and γ in Fourier modes as

$$\gamma_0(t) = \int_{\mathbb{R}} e^{i\lambda t} \mu_0(d\lambda), \quad \text{and} \quad \gamma(x) = \int_{\mathbb{R}^d} e^{i\xi \cdot x} \mu(d\xi), \quad (1.4)$$

where the measures μ_0 and μ on $\mathbb{R}^d$ are respectively defined by

$$\mu_0(d\lambda) = c_0 |\lambda|^{1-2H_0} d\lambda, \quad \text{and} \quad \mu(d\xi) = c_{\mathbf{H}} \prod_{j=1}^d |\xi_j|^{1-2H_j} d\xi, \quad (1.5)$$

where $\mathbf{H}$ denotes the vector $(H_1, \dots, H_d)$ and where $c_0, c_{\mathbf{H}}$ are explicit positive constants. We should already observe at this point that the mere existence of a solution to equation (1.1) in the rough environment given by (1.3) requires a delicate analysis of intersection local times in the Skorohod setting [8], and a cumbersome renormalization procedure for the Stratonovich case (see [15, 16, 20] for some related models).

In order to describe the main results contained in this article let us start with the Skorohod setting for equation (1.1), for which we will assume that $H_0 > \frac{1}{2}$ (for sake of simplicity). Within this framework we define a family of coefficients describing the behavior of our model. Namely set

$$J_* = \left\{ 1 \leq j \leq d; H_j < \frac{1}{2} \right\}, \quad d_* = |J_*|, \quad H_* = \sum_{j \in J_*} H_j, \quad H = \sum_{j=1}^d H_j. \quad (1.6)$$

We also define some similar quantities J^*, d^*, H^* for the indices such that $H_j \geq \frac{1}{2}$. Then the reference [8] exhibits a subcritical regime, for which there is existence and uniqueness of the solution to (1.1) interpreted in the Skorohod sense. This subcritical regime is characterized by the following set of conditions on J_*, d_*, H_* :

$$d - H < 1, \quad \text{and} \quad 4(1 - H_0) + 2(d - H) + (d_* - 2H_*) < 4. \quad (1.7)$$

Denoting by $u^\diamond$ this solution, it is also proved in [8] that $u^\diamond$ admits moments of all orders, namely,

$$\mathbf{E}[|u_t^\diamond(x)|^p] < \infty, \quad \text{for all } t \geq 0, x \in \mathbb{R}^d, \text{ and } p \geq 1.$$

In contrast with this nice situation we will also focus on the so-called critical regime, for which (1.7) is replaced by the following condition on J_*, d_*, H_* :

$$d - H = 1, \quad \text{and} \quad 4(1 - H_0) + (d_* - 2H_*) < 2. \quad (1.8)$$

Under condition (1.8), the moments of $u_t^\diamond(x)$ blow up for large time. This assertion will be quantified precisely in our article, leading to our first contribution (which will be stated more rigorously in Theorem 3.14).

Contribution 1. Assume that condition (1.8) is met, and recall that $u^\diamond$ designates the Skorohod solution of equation (1.1). For all $p > 1$ we define a critical time $t_0(p)$ as

$$t_0(p) = \frac{C_{H_0, \mathbf{H}}}{(p-1)^{1/(2H_0-1)}} \quad (1.9)$$

where $C_{H_0, \mathbf{H}}$ is given by an explicit variational inequality. Then the following holds true:

- (i) For any $p \geq 2$, if $t < t_0(p)$ we have $\mathbb{E}[|u_t^\diamond(x)|^p] < \infty$ for all $x \in \mathbb{R}^d$;
- (ii) For all $p > 1$, if $t > t_0(p)$ the p -th moment of $u_t^\diamond(x)$ blows up.

Remark 1.1. Our Contribution 1 gives a rather complete picture of the moments problem for the Skorohod equation (1.1) in the critical regime. In addition, it is also clear from equation (1.9) that $p \mapsto t_0(p)$ decreases from $+\infty$ to 0 as p varies in $(1, \infty)$. We firmly believe that $t_0(p)$ separates well behaved from ill-behaved p -th moments for $u^\diamond$, and this is what our Contribution 1 asserts for $p \geq 2$. However, item (ii) in Contribution 1 only yields an upper bound for the critical time $t_0(p)$ when $p \in (1, 2)$. Hence a full characterization of $t_0(p)$ for $p < 2$ is still an open problem.

As mentioned above, we also wish to analyze equation (1.1) in the Stratonovich sense. In this case the existence of a process u satisfying (1.1) is an open problem for the rough fractional environment $\dot{W}$ we consider here. Indeed, it is a well-known fact that equation (1.1) needs an extra renormalization procedure in order to be properly defined as soon as $2H_0 + H \leq d + 1$ (recall that H is defined by (1.6) above). In the special case $d = 1$ and for a localized version of the noise, a thorough study of this renormalization procedure within the landmark of regularity structures is carried out in [15, 16]. For $d = 2, 3$ and when $\dot{W}$ is a spatial white noise (namely $H_0 = 1$ and $H_i = \frac{1}{2}$ for $i = 1, \dots, d$ in our setting), the construction of a renormalized solution is achieved in [20]. However, the case of the renormalized parabolic Anderson model on the whole space (for a general dimension and for a general Gaussian noise) is not explicitly available in the literature. A proper definition of equation (1.1) will thus be the aim of our next result (the reader is sent to Theorem 4.11 for a rigorous statement).

Contribution 2. Consider equation (1.1) driven by a Gaussian noise whose covariance function is given by (1.3). We assume that the coefficients $H_0, \mathbf{H} \in (0, 1)^{d+1}$ verify

$$2H_0 + H > d + \frac{1}{2}, \quad \text{or otherwise stated} \quad d - H < 2H_0 - \frac{1}{2}. \quad (1.10)$$

Then a global solution u in a weighted space can be constructed under those conditions.

Remark 1.2. One could argue that our Contribution 2 is a consequence of the very general renormalization procedure put forward in [3], combined with the PAM setting handled in [20]. However, we believe that our explicit, self-contained and relatively simple presentation of the problem is interesting in its own right.

Remark 1.3. In order to reach the region $2H_0 + H > d + \frac{1}{2}$ announced in (1.10), two levels of renormalization are necessary for equation (1.1) in the Stratonovich sense. For sake of conciseness, only the first level of renormalization is explicitly carried out in the current paper. This covers the region $2H_0 + H > d + \frac{2}{3}$, and we refer to Remark 4.19 for an extension to the region $d + \frac{1}{2} < 2H_0 + H \leq d + \frac{2}{3}$.

Once the existence of a solution to equation (1.1) in the Stratonovich sense is established, our next aim will be to get some information about the moments of its solution u . This problem is challenging in a renormalized context, and to the best of our knowledge unaddressed in the literature. We are only aware of the reference [18] about existence of moments in small time for the 2-dimensional spatial white noise case. Our third contribution, which is summarized below, aims at filling this gap.

Contribution 3. Let $u_t(x)$ be the Stratonovich solution of equation (1.1) constructed from the regularity structure formalism. Then the following holds true:

(i) Under the subcritical assumption (1.7), the random variable $u_t(x)$ admits moments of all orders. More specifically, we have

$$\mathbf{E}[|u_t(x)|^p] < \infty, \quad \text{for all } t \geq 0, x \in \mathbb{R}^d, \text{ and } p \geq 1.$$

(ii) In the critical case defined by (1.8) and for a fixed $p > 1$, the random variable $u_t(x)$ admits a moment of order p for t small enough.

Gathering our Contributions 1-2-3, our main goal in this paper is thus to exhibit the complexity of the moments problem for the PAM in very rough environments.

Let us say a few words about the methodology invoked in order to achieve the main contributions summarized above. We have relied on the following tools:

(a) In order to analyze the moments of the Skorohod solution in the critical case (i.e Contribution 1), we hinge on the Feynman-Kac representation of $u^\diamond$. This representation involves intersection local times of a Brownian motion weighted by the covariance function of the noise $\dot{W}$. We then proceed to a thorough analysis of those intersection local times, thanks to a delicate truncation procedure involving a Girsanov type transform on the Brownian paths. Notice that the basic aim of the truncation mechanism is (in the end) to apply some classical asymptotic results for Feynman-Kac functionals. For the case of p -th moments for $p \in (1, 2)$ (that is item (ii) in our Contribution 1), we use an additional ingredient based on $\mathcal{S}$ -transforms from white noise analysis.

(b) As mentioned earlier, the existence of a renormalized solution to the Stratonovich version of (1.1) is obtained within the regularity structures framework. In this context we have to adapt the general theory to our situation in $\mathbb{R}^d$. Then our main task is to effectively construct a K -rough path containing the Gaussian iterated integrals necessary to build proper modeled distributions related to the heat equation. This will be achieved in Appendix A.

(c) The fact that the Stratonovich solution admits moments of any order (Contribution 3 item (i)) stems again from Feynman Kac representations. More precisely, the difference between the Feynman-Kac representations of u and $u^\diamond$ corresponds to some fluctuations of the Brownian intersection local times. Then the bulk of our analysis consists in quantifying the fact that the intersection local time fluctuations are negligible with respect to the term involving $\dot{W}$.

As one can see, Feynman-Kac representations are the key ingredient in order to get our main contributions. Those are combined throughout the paper with regularity structures and Malliavin calculus elements. As a conjecture, let us also recall from Remark 1.1 that $t_0(p)$ defined by (1.9) seems to be the critical time for the moments of $u^\diamond$ when $p < 2$ (as well

as for the moments of the Stratonovich solution u in the critical case). We hope to prove this claim in the future, although it currently seems to resist our method of analysis.

The study is organized as follows. First, in Section 2, we will recall some basics about the Malliavin calculus setting and the Skorohod integration procedure. This preliminary material will in fact set the stage for Section 3, where equation (1.1) will be investigated in the Skorohod sense, leading to the above-described Contribution 1. Then, in Section 4, we will turn to the Stratonovich interpretation of the model and provide a rigorous “regularity structures” analysis of the existence issue in Contribution 2. Our last main Contribution 3 will be the topic of the subsequent Section 5, which will conclude the main part of the paper. Finally, Appendix A will be devoted to the proof of the main technical result of Section 4, namely the construction of a so-called (α, K) -rough path above the fractional noise $\dot{W}$.

Notation 1.4. In the sequel we set $(\Omega, \mathcal{F}, \mathbf{P})$ for the probability space related to W , with $\mathbf{E}$ for the related expected value. We denote by $(\hat{\Omega}, \mathcal{G}, \mathbb{P})$ the probability spaces corresponding to the Brownian motions in Feynman-Kac representations, with a related expected value $\mathbb{E}$. The heat kernel on $\mathbb{R}^d$ is denoted by $p_t(x)$, and recall that

$$p_t(x) = \frac{1}{(2\pi t)^{d/2}} \exp\left(-\frac{|x|^2}{2t}\right). \quad (1.11)$$

Also notice that the inner product of $a, b \in \mathbb{R}^d$ is written as $a \cdot b$ throughout the paper. In the following we will often deal with product measures, for which we adopt the following convention: for $m, n \geq 1$, a measure ν on $\mathbb{R}^m$ and $\xi = (\xi_1, \dots, \xi_n) \in (\mathbb{R}^m)^n$, we set

$$\nu(d\xi) = \nu^{\otimes n}(d\xi) = \prod_{k=1}^n \nu(d\xi_k). \quad (1.12)$$

Finally, recall that we write $\mathbf{H}$ for the vector of Hurst parameters $(H_1, \dots, H_d)$.

2. ELEMENTS OF MALLIAVIN CALCULUS

In this section we recall the basic Malliavin calculus notation which will be invoked in the forthcoming computations. The reader is sent to [22, 28] for more details.

Let us start by introducing some basic notions on Fourier transforms of functions. The space of real valued infinitely differentiable functions with compact support is denoted by $\mathcal{D}(\mathbb{R}^d)$ or $\mathcal{D}$. We write $\mathcal{S}(\mathbb{R}^d)$, or simply $\mathcal{S}$ for the space of Schwartz functions. Its dual, the space of tempered distributions, is $\mathcal{S}'(\mathbb{R}^d)$ or $\mathcal{S}'$. If f is a vector of n tempered distributions on $\mathbb{R}^d$, then we write $f \in \mathcal{S}'(\mathbb{R}^d, \mathbb{R}^n)$. In the sequel we will play with two different kinds of Fourier transforms on $\mathbb{R}^{d+1}$. Namely for a function $f(t, x)$ on $\mathbb{R}^{d+1}$, the Fourier transform on the full space-time domain $\mathbb{R}^{d+1}$ is defined with the normalization

$$\mathcal{F}f(\eta, \xi) = \int_{\mathbb{R}^{d+1}} e^{-i(\eta\tau + \xi \cdot x)} f(t, x) dt dx, \quad (2.1)$$

so that the inverse Fourier transform is given by $\mathcal{F}^{-1}u(\eta, \xi) = \frac{\mathcal{F}f(-t, -\xi)}{(2\pi)^{d+1}}$. Our analysis will also rely on a Fourier transform denoted by $\mathcal{F}^s$, which is defined on the spatial variables

only. It is given by

$$\mathcal{F}^s f(t, \xi) = \int_{\mathbb{R}^d} e^{-i\xi \cdot x} f(t, x) dx. \quad (2.2)$$

2.1. Wiener space related to W . On a complete probability space $(\Omega, \mathcal{F}, \mathbf{P})$ we consider a Gaussian noise W encoded by a centered Gaussian family $\{W(\varphi); \varphi \in \mathcal{D}(\mathbb{R}_+ \times \mathbb{R}^d)\}$. As mentioned in the introduction (see relation (1.4)), the covariance structure of W is given by

$$\mathbf{E}[W(\varphi)W(\psi)] = \int_{\mathbb{R}_+^2} \int_{\mathbb{R}^d} \mathcal{F}^s \varphi(s_1, \xi) \bar{\mathcal{F}}^s \psi(s_2, \xi) \gamma_0(s_1 - s_2) ds_1 ds_2 \mu(d\xi), \quad (2.3)$$

where γ_0 and μ are respectively defined by (1.3) and (1.5), and where $\mathcal{F}^s \varphi$ designates the spatial Fourier transform of φ . One can thus consider W as an isonormal Gaussian family $\{W(\varphi); \varphi \in \mathcal{H}\}$ on a space $\mathcal{H}$ obtained as the completion of Schwartz functions with respect to the inner product given by the right hand side of (2.3).

We refer to [28] for a detailed account of the Malliavin calculus with respect to a Gaussian process, while we focus here on the basic definitions allowing to state our results in the remainder of the article. We will denote by D the derivative operator in the sense of Malliavin calculus. That is, if F is a smooth and cylindrical random variable of the form

$$F = f(W(\phi_1), \dots, W(\phi_n)),$$

with $\phi_i \in \mathcal{H}$, $f \in C_p^\infty(\mathbb{R}_+ \times \mathbb{R}^d)$ (namely f and all its partial derivatives have polynomial growth), then DF is the $\mathcal{H}$ -valued random variable defined by

$$DF = \sum_{j=1}^n \frac{\partial f}{\partial x_j}(W(\phi_1), \dots, W(\phi_n)) \phi_j.$$

The operator D is closable from $L^2(\Omega)$ into $L^2(\Omega; \mathcal{H})$ and we define the Sobolev space $\mathbb{D}^{1,2}$ as the closure of the space of smooth and cylindrical random variables under the norm

$$\|DF\|_{1,2} = (\mathbf{E}[F^2] + \mathbf{E}[\|DF\|_{\mathcal{H}}^2])^{1/2}.$$

For any integer $n \geq 0$ we denote by $\mathcal{H}_n$ the n th Wiener chaos of W . We recall that $\mathcal{H}_0$ is simply $\mathbb{R}$ and for $n \geq 1$, $\mathcal{H}_n$ is the closed linear subspace of $L^2(\Omega)$ generated by the random variables $\{H_n(W(h)); h \in \mathcal{H}, \|h\|_{\mathcal{H}} = 1\}$, where H_n is the n th Hermite polynomial. For any $n \geq 1$, we denote by $\mathcal{H}^{\otimes n}$ (resp. $\mathcal{H}^{\odot n}$) the n th tensor product (resp. the n th symmetric tensor product) of $\mathcal{H}$. Then, the mapping $I_n(h^{\otimes n}) = H_n(W(h))$ can be extended to a linear isometry between $\mathcal{H}^{\odot n}$ (equipped with the modified norm $\sqrt{n!} \|\cdot\|_{\mathcal{H}^{\otimes n}}$) and $\mathcal{H}_n$.

Consider now a random variable $F \in L^2(\Omega)$ which is measurable with respect to the σ -field $\mathcal{F}^W$ generated by W . This random variable can be expressed as

$$F = \mathbf{E}[F] + \sum_{n=1}^{\infty} I_n(f_n), \quad (2.4)$$

where the series converges in $L^2(\Omega)$. Moreover, the functions $f_n \in \mathcal{H}^{\odot n}$ are determined by F . Identity (2.4) is called the Wiener chaos expansion of F .

2.2. Extended Skorohod integrals. The standard definition of Skorohod's integral is obtained by a duality relation in $L^2(\Omega)$. Namely, we denote by $\delta^\diamond$ the L^2 -adjoint of the derivative operator given by the duality formula

$$\mathbf{E}[\delta^\diamond(u) F] = \mathbf{E}[\langle DF, u \rangle_{\mathcal{H}}], \quad (2.5)$$

for any $F \in \mathbb{D}^{1,2}$ and any element $u \in L^2(\Omega; \mathcal{H})$ in the domain of $\delta^\diamond$. More specifically, the domain of $\delta^\diamond$ is defined as the set of processes $u \in L^2(\Omega; \mathcal{H})$ such that for all cylindrical functions F we have

$$|\mathbf{E}[\langle DF, u \rangle_{\mathcal{H}}]| \leq C_u \|F\|_{L^2(\Omega)}.$$

The operator $\delta^\diamond$ is also called the *Skorohod integral* because in the Brownian motion case, it coincides with an extension of the Itô integral introduced by Skorohod.

In the sequel we consider random variables which are only L^p -integrable with $p \in (1, 2)$. We thus need the following extension of the Skorohod integral with respect to the noise W .

Definition 2.1. Fix $p > 1$. Let $u = \{u(t, y), t \geq 0, y \in \mathbb{R}^d\}$ be a random field such that

$$\| \|u(t, y)\|_{\mathcal{H}} \|_{L^p(\Omega)} < \infty.$$

Then we say that u is Skorohod integrable in L^p , denoted by $u \in \text{Dom}_p \delta^\diamond$, if for any smooth and cylindrical random variable $F \in \mathcal{S}$, we have

$$|\mathbf{E}[\langle DF, u \rangle_{\mathcal{H}}]| \leq c_u \|F\|_{L^q(\Omega)},$$

where $\frac{1}{p} + \frac{1}{q} = 1$. We say that u is Skorohod integrable if $u \in \text{Dom} \delta^\diamond = \cup_{p>1} \text{Dom}_p \delta^\diamond$. If $u \in \text{Dom}_p \delta^\diamond$, the Skorohod integral $\delta^\diamond(u) = \int_{\mathbb{R}_+ \times \mathbb{R}^d} u_s(y) d^\diamond W_s(y)$ is the random variable in $L^p(\Omega)$ defined by the duality relationship

$$\mathbf{E}[\langle DF, u \rangle_{\mathcal{H}}] = \mathbf{E}[\delta^\diamond(u) F], \quad \text{for all } F \in \mathcal{S}.$$

3. SKOROHOD CASE

This section focuses on equation (1.1) interpreted in the Skorohod sense. In Section 3.1 below we first recall some existence and uniqueness results for the equation. Then we will give some information about the moments in the subcritical case.

3.1. Existence and uniqueness results. In this section we first give some existence and uniqueness results for equation (1.1) considered in the Skorohod sense. Those results are mainly taken from [8], although we will also deal with extended solutions in the sense of Definition 2.1. Specifically, we consider equation (1.1) considered in the Skorohod sense, that is

$$\partial_t u_t(x) = \frac{1}{2} \Delta u_t(x) + u_t(x) \diamond \dot{W}_t(x), \quad t \in \mathbb{R}_+, x \in \mathbb{R}^d. \quad (3.1)$$

Recall from Notation 1.4 that we denote by $p_t(x)$ the d -dimensional heat kernel $p_t(x) = (2\pi t)^{-d/2} e^{-|x|^2/2t}$, for any $t > 0, x \in \mathbb{R}^d$. We define the solution of equation (3.1) as follows.

Definition 3.1. Let $T > 0$. Consider a random field $u = \{u_t(x); 0 \leq t \leq T, x \in \mathbb{R}^d\}$ such that there exists $p > 1$ satisfying

$$\sup_{0 \leq s \leq T} \sup_{y \in \mathbb{R}^d} \|u_s(y)\|_{L^p(\Omega)} < \infty. \quad (3.2)$$

Then u is said to be a mild solution to equation (3.1) up to time T with initial condition $u_0 \in C_b(\mathbb{R}^d)$, if for any $t \geq 0$ and $x \in \mathbb{R}^d$, the process $\{u_s(y); s \geq 0, y \in \mathbb{R}^d\}$ is Skorohod integrable in the sense of Definition 2.1, and the following equation holds for any $t \geq 0$ and $x \in \mathbb{R}^d$:

$$u_t(x) = p_t u_0(x) + \int_0^t \int_{\mathbb{R}^d} p_{t-s}(x-y) u_s(y) d^\diamond W_s(y). \quad (3.3)$$

If we impose $p = 2$ in Definition 3.1, then our definition coincides with the usual notion of Skorohod solution for equation (1.1). In the subcritical regime L^p moments are available for all $p \geq 2$, and therefore equation (3.1) can be solved in the usual sense. We recall this result (taken from [8]) for the sake of completeness.

Proposition 3.2. *We assume that the coefficients $H_0, \dots, H_d$ satisfy the subcritical relation (1.7) and that $H_0 > \frac{1}{2}$. Then equation (3.1) admits a unique solution, considered in the sense of Definition 3.1 with $p = 2$.*

Remark 3.3. Proposition 3.2 has been extended in [9] to the case $H_0 < 1/2$.

In contrast with the subcritical situation, in the critical case one can only guarantee the existence of a usual Skorohod solution up to a critical time. This is the content of the following proposition, also borrowed from [8].

Proposition 3.4. *Assume that the critical condition (1.8) on H_0 and $\mathbf{H}$ holds true, where we recall that the vector $\mathbf{H}$ is introduced in Notation 1.4. Then one can find a strictly positive time $t^* = t^*(H_0, \mathbf{H}, d)$ such that there exists a unique $L^2(\Omega)$ -mild solution $u^\diamond$ of equation (3.3) for $t \in [0, t^*)$.*

Proposition 3.4 opens the way to a possible definition of the solution to equation (3.3) in the L^p sense for $p \in (1, 2)$. While the existence of such a solution relies heavily on moment bounds which are obtained through Feynman-Kac representations, uniqueness holds true by just invoking the fact that we are dealing with a linear equation. This is summarized in the following proposition.

Proposition 3.5. *As in Proposition 3.4, we assume that the critical condition (1.8) on H_0 and $\mathbf{H}$ holds true. For $p \in (1, 2)$, suppose that one can find a strictly positive time $t_p^* = t^*(p, H_0, \mathbf{H}, d)$ such that there exists a $L^p(\Omega)$ -mild solution $u^\diamond$ of equation (3.3) on the interval $[0, t_p^*)$. Then this solution is unique.*

Proof. Suppose that u and v are two mild solutions in the L^p sense (given by Definition 3.1) with the same initial condition. Set $w = u - v$. Then, w satisfies

$$w_t(x) = \int_0^t \int_{\mathbb{R}^d} p_{t-s}(x-y) w_s(y) d^\diamond W_s(y). \quad (3.4)$$

Iterating this relation we get

$$w_t(x) = \int_0^t \int_{\mathbb{R}^d} \left(\int_0^s \int_{\mathbb{R}^d} p_{t-s}(x-y) p_{s-r}(y-z) w_r(z) d^\diamond W_r(z) \right) d^\diamond W_s(y). \quad (3.5)$$

That is, $w_t(x)$ is an iterated Skorohod integral. By Proposition 2.6 and Proposition 2.7 in [29] this iterated Skorohod integral coincides with a double Skorohod integral (notice that in

the reference [29] these results are proved for Skorohod integrals in $L^2(\Omega)$, but they can be easily extended to the Skorohod integral in $L^p(\Omega)$ introduced in Definition 2.1). Moreover, using (3.4), (3.5) and iterated versions in higher order chaoses, one can prove that $w_t(x)$ is orthogonal to any element in a finite Wiener chaos. This does not imply immediately that $w_t(x) = 0$, because $w_t(x) \in L^p(\Omega)$ for some $p > 1$. However, given any random variable $F \in L^q(\Omega)$ where $\frac{1}{p} + \frac{1}{q} = 1$, we can approximate F in L^q by a sequence F_n of elements on a finite Wiener chaos. Then, $\mathbf{E}[w_t(x)F] = \lim_{n \rightarrow \infty} \mathbf{E}[w_t(x)F_n] = 0$, which implies $w_t(x) = 0$. $\square$

3.2. Feynman-Kac representation. In order to analyze the moments of $u_t^\diamond(x)$ and compare the Skorohod and Stratonovich settings for equation (1.1), we first give a Feynman-Kac representation for an approximating sequence of the solution $u^\diamond$ to equation (3.1). To this end, let p_ϵ be a $d+1$ dimensional heat kernel given by (1.11). To separate its first coordinate from the others, we write $p_\epsilon(\hat{x}) = p_\epsilon(x_0, x)$ with $\hat{x} = (x_0, x) \in \mathbb{R}^{d+1}$. Consider now the smoothed noise

$$W_t^\epsilon(x) = (p_\epsilon * \dot{W})_t(x) = W(p_\epsilon(t - \cdot, x - \cdot)). \quad (3.6)$$

In the same spirit as (1.3), we denote by γ_0^ϵ and γ^ϵ the corresponding covariance functions (in time and space) of W^ϵ , with μ_0^ϵ and μ^ϵ their spectral measures. Similar to (1.5), it is readily checked that

$$\mu_0^\epsilon(d\lambda) = c_0 e^{-\epsilon|\lambda|^2} |\lambda|^{1-2H_0} d\lambda, \quad \text{and} \quad \mu^\epsilon(d\xi) = c_{\mathbf{H}} e^{-\epsilon|\xi|^2} \prod_{j=1}^d |\xi_j|^{1-2H_j} d\xi, \quad (3.7)$$

With the above notions in hand, we set

$$V_t^{\epsilon, B}(x) = \int_0^t W_{t-s}^\epsilon(B_s^x) ds = W(\phi), \quad (3.8)$$

where the second notation $W(\phi)$ corresponds to the Wiener integral introduced in (2.3) and where the function ϕ is given by

$$\phi(s, y) = \int_0^t p_\epsilon(u - s, B_{t-u}^x - y) du. \quad (3.9)$$

Moreover, in relation (3.9) B^x stands for a d -dimensional Brownian motion independent of W with initial condition $x \in \mathbb{R}^d$. The functional (3.8) gives rise to an approximate solution of the Skorohod equation, as stated in the following proposition:

Proposition 3.6. *For $\epsilon > 0$, $t \in \mathbb{R}_+$ and $x \in \mathbb{R}^2$, let $V_t^{\epsilon, B}(x)$ be defined by (3.8) and set*

$$\beta_t^{\epsilon, B} \equiv \int_{[0, t]^2} \int_{\mathbb{R}^{d+1}} e^{-\epsilon(|\xi|^2 + |\lambda|^2)} e^{i(\xi \cdot (B_{s_1} - B_{s_2}) + \lambda(s_2 - s_1))} \mu_0(d\lambda) \mu(d\xi) ds_1 ds_2, \quad (3.10)$$

where we write $B^0 = B$ for notational sake, where μ^ϵ is introduced in (3.7) and where γ_0^ϵ is the inverse Fourier transform of μ_0^ϵ . We define $u_t^{\epsilon, \diamond}$ on $\mathbb{R}_+ \times \mathbb{R}^d$ by:

$$u_t^{\epsilon, \diamond}(x) = \mathbb{E} \left[e^{V_t^{\epsilon, B}(x) - \frac{1}{2} \beta_t^{\epsilon, B}} \right], \quad (3.11)$$

where we recall that the expectation $\mathbb{E}$ has been introduced in Notation 1.4. Then the process $u^{\varepsilon, \diamond}$ is the unique solution to the following Skorohod type equation:

$$\partial_t u_t^{\varepsilon, \diamond}(x) = \frac{1}{2} \Delta u_t^{\varepsilon, \diamond}(x) + u_t^{\varepsilon, \diamond}(x) \diamond \dot{W}^\varepsilon(x), \quad t \in \mathbb{R}_+, x \in \mathbb{R}^d, \quad (3.12)$$

where $\dot{W}^\varepsilon$ is the smoothed noise $\dot{W}^\varepsilon = \dot{W} * p_\varepsilon$ defined in (3.6).

Proof. Similarly to the proof of Proposition 5.2 in [23], we can show that

$$u_t^{\varepsilon, \diamond}(x) = \mathbb{E} \left[\exp \left(V_t^{\varepsilon, B}(x) - \frac{1}{2} \mathbf{E} \left[\left| V_t^{\varepsilon, B}(x) \right|^2 \right] \right) \right].$$

Now a direct application of (2.3) reveals that

$$\begin{aligned} \mathbf{E} \left[\left| V_t^{\varepsilon, B}(x) \right|^2 \right] &= \int_{[0, t]^2} \int_{\mathbb{R}^{d+1}} e^{-\varepsilon(|\xi|^2 + |\lambda|^2)} e^{i(\xi \cdot (B_{s_1} - B_{s_2}) + \lambda(s_2 - s_1))} \mu_0(d\lambda) \mu(d\xi) ds_1 ds_2 \\ &= \beta_t^{\varepsilon, B}, \end{aligned} \quad (3.13)$$

which proves our claim. $\square$

We now prove the convergence of the regularized Feynman-Kac representation to the solution of the Skorohod equation (3.1).

Proposition 3.7. *Let $u^{\varepsilon, \diamond}$ be the process defined by (3.11), and recall that $u^\diamond$ designates the solution to equation (3.1) as given in Proposition 3.2 or Proposition 3.4. We assume that one of the following situations is met:*

- (i) *The subcritical condition (1.7) is satisfied and t is any positive number.*
- (ii) *The critical assumption (1.8) prevails and $t \in [0, t^*)$, where t^* is defined in Proposition 3.4.*

Then for all $x \in \mathbb{R}^d$, the random variable $u_t^{\varepsilon, \diamond}(x)$ converges to $u_t^\diamond(x)$ in $L^2(\Omega)$ as $\varepsilon \rightarrow 0$.

Proof. We will prove the proposition under condition (ii), the other case being handled similarly. In order to show the L^2 -convergence of $u_t^{\varepsilon, \diamond}$, we consider the quantity α_t (also denoted by α_t^{12} in the sequel) defined by

$$\alpha_t \equiv \int_{[0, t]^2} \int_{\mathbb{R}^d} e^{i(\xi \cdot (B_{s_1}^1 - B_{s_2}^2) + \lambda(s_1 - s_2))} \mu_0(d\lambda) \mu(d\xi) ds_1 ds_2 \quad (3.14)$$

In the above, B^1, B^2 are two independent Brownian motions starting from x , which are also independent of the noise W . Then in order to prove that $u_t^{\varepsilon, \diamond}(x)$ converges in $L^2(\Omega)$, it is sufficient to verify that

$$\mathbb{E}[e^{\alpha_t}] < \infty \quad \text{and} \quad \lim_{\varepsilon_1, \varepsilon_2 \rightarrow 0} \mathbf{E}[u_t^{\varepsilon_1, \diamond}(x) u_t^{\varepsilon_2, \diamond}(x)] = \mathbb{E}[e^{\alpha_t}]. \quad (3.15)$$

In order to prove (3.15), we resort to expression (3.11), which yields

$$\mathbf{E}[u_t^{\varepsilon_1, \diamond}(x) u_t^{\varepsilon_2, \diamond}(x)] = \mathbf{E} \left[\mathbb{E} \left[\exp \left(V_t^{\varepsilon_1, B^1}(x) + V_t^{\varepsilon_2, B^2}(x) - \frac{1}{2} \left(\beta_t^{\varepsilon_1, B^1} + \beta_t^{\varepsilon_2, B^2} \right) \right) \right] \right].$$

Applying Fubini's theorem we thus get:

$$\begin{aligned} \mathbf{E} [u_t^{\varepsilon_1, \diamond}(x) u_t^{\varepsilon_2, \diamond}(x)] \\ = \mathbf{E} \left[\exp \left(\frac{1}{2} \mathbf{E} \left[\left(V_t^{\varepsilon_1, B^1}(x) + V_t^{\varepsilon_2, B^2}(x) \right)^2 \right] - \frac{1}{2} \left(\beta_t^{\varepsilon_1, B^1} + \beta_t^{\varepsilon_2, B^2} \right) \right) \right]. \end{aligned} \quad (3.16)$$

Moreover, recall that $\beta_t^{\varepsilon, B} = \mathbf{E}[|V_t^{\varepsilon, B}(x)|^2]$ according to (3.13). Plugging this identity into (3.16), we thus get

$$\mathbf{E} [u_t^{\varepsilon_1, \diamond}(x) u_t^{\varepsilon_2, \diamond}(x)] = \mathbf{E} \left[e^{\alpha_t^{\varepsilon_1, \varepsilon_2}} \right],$$

where the quantity $\alpha_t^{\varepsilon_1, \varepsilon_2}$ is defined by

$$\begin{aligned} \alpha_t^{\varepsilon_1, \varepsilon_2} &= \mathbf{E} \left[V_t^{\varepsilon_1, B^1}(x) V_t^{\varepsilon_2, B^2}(x) \right] \\ &= \int_{[0, t]^2} \int_{\mathbb{R}^{d+1}} e^{-\frac{(\varepsilon_1 + \varepsilon_2)}{2} (|\xi|^2 + |\lambda|^2)} e^{i(\xi \cdot (B_{s_1}^1 - B_{s_2}^2) + \lambda(s_2 - s_1))} \mu_0(d\lambda) \mu(d\xi) ds_1 ds_2. \end{aligned} \quad (3.17)$$

Notice that for fixed $\varepsilon_1, \varepsilon_2 > 0$, the fact that $\alpha_t^{\varepsilon_1, \varepsilon_2}$ is well-defined stems easily from the presence of the exponential term $e^{-(\varepsilon_1 + \varepsilon_2)|\xi|^2}$ in the right hand side of (3.17). In addition, [8, inequality (3.1)] implies that

$$\mathbf{E}[e^{\alpha_t}] < \infty,$$

for $t < t^*$. Since

$$\int_{[0, t]^2} \mathbf{E} \left[e^{i(\xi \cdot (B_{s_1}^1 - B_{s_2}^2) + \lambda(s_1 - s_2))} \right] ds_1 ds_2 = \left| \int_0^t \mathbf{E} \left[e^{i(\xi \cdot B_{s_1}^1 + \lambda s_1)} \right] ds_1 \right|^2 \geq 0,$$

an easy monotone convergence argument together with a Taylor series expansion of the exponential function yields

$$\lim_{\varepsilon_1, \varepsilon_2 \rightarrow 0} \mathbf{E} [u_t^{\varepsilon_1, \diamond}(x) u_t^{\varepsilon_2, \diamond}(x)] = \lim_{\varepsilon_1, \varepsilon_2 \rightarrow 0} \mathbf{E} \left[e^{\alpha_t^{\varepsilon_1, \varepsilon_2}} \right] = \mathbf{E} [e^{\alpha_t}],$$

which is our claim (3.15). We have thus obtained the L^2 convergence of $u^{\varepsilon, \diamond}$.

We now claim that the process $u_t^\diamond(x)$ defined as the limit of $u_t^{\varepsilon, \diamond}(x)$ is a mild solution to equation (3.1) in the sense of Definition 3.1 with $p = 2$. First notice that (3.2) holds for the process $u^\diamond$ because of the relation

$$\sup_{\varepsilon > 0} \sup_{t \leq t^*} \sup_{x \in \mathbb{R}^2} \|u_t^{\varepsilon, \diamond}(x)\|_{L^2(\Omega)} < \infty,$$

which stems from (3.15). Then owing to relation (2.5), $u^\diamond$ satisfies (3.3) in the Skorohod sense if for any smooth and cylindrical random variable F , we have

$$\mathbf{E} [F(u_t^\diamond(x) - p_t u_0(x))] = \mathbf{E} [\langle DF, G \rangle_{\mathcal{H}}], \quad (3.18)$$

where the process G is defined by

$$G_s(y) = p_{t-s}(x - y) u_s^\diamond(y) \mathbf{1}_{[0, t]}(s).$$

Relation (3.18) is obtained by taking limits on a similar equation for $u_t^{\varepsilon, \diamond}(x)$ (see [22, Theorem 3.6] for a similar argument). $\square$

3.3. Moments estimates in the subcritical case. In this section we recall some moment estimates and representations established in [8] for the Skorohod equation. We label those results here since they will be invoked in order to get integrability results for the Stratonovich equation. We start with a bound on the L^p moments.

Proposition 3.8. *Assume that the subcritical condition (1.7) is satisfied, and let $u^\diamond$ be the unique solution of equation (3.1). Then for all $t \geq 0$ and $p \geq 1$ we have*

$$\mathbf{E}[|u_t^\diamond(x)|^p] = c_{p,t} < \infty.$$

Let us now define a generalization of the quantity α_t defined by (3.14), which is used for moment representations. Namely for two coordinates $1 \leq j_1, j_2 \leq p$ we set

$$\alpha_t^{j_1 j_2} = \int_{[0,t]^2} \gamma(B_{s_1}^{j_1} - B_{s_2}^{j_2}) \gamma_0(s_1 - s_2) ds_1 ds_2, \quad (3.19)$$

where γ and γ_0 are the (generalized) covariance function given by (1.3), and $B^j, j = 1, \dots, p$ are independent Brownian motions starting from x . With this notation in hand and thanks to the representation (3.11) for the approximation $u^{\varepsilon, \diamond}$ of $u^\diamond$, we also get the following representation for the moments of $u_t^\diamond(x)$.

Proposition 3.9. *We suppose that the subcritical condition (1.7) is fulfilled, and recall that $u^\diamond$ is the unique solution of equation (3.1). Then for all $t \geq 0$ and any integer $p \geq 2$ we have*

$$\mathbf{E}[|u_t^\diamond(x)|^p] = \mathbb{E} \left[\exp \left(\sum_{1 \leq j_1 < j_2 \leq p} \alpha_t^{j_1 j_2} \right) \right], \quad (3.20)$$

where $\alpha_t^{j_1 j_2}$ is introduced in (3.19). Moreover, for any $\lambda > 0$ one has $\mathbb{E}[\exp(\lambda \alpha_t^{j_1 j_2})] < \infty$.

3.4. Critical moments for $p \geq 2$. We assume throughout this section that the critical condition (1.8) is satisfied. In this case we will see that the moments of $u^\diamond$ blow up after a critical time t^* and we shall identify the time t^* in some cases. Before we proceed to the proof of this fact, let us introduce some functional spaces and inequalities of interest.

Definition 3.10. *Let $W^{1,2}(\mathbb{R}^d)$ be the usual Sobolev space of order $(1, 2)$ in $\mathbb{R}^d$. We define a subset $\mathcal{A}_d$ of functions defined on $[0, 1] \times \mathbb{R}^d$ as follows:*

$$\mathcal{A}_d = \left\{ g : [0, 1] \times \mathbb{R}^d \rightarrow \mathbb{R}; \quad g(s, \cdot) \in W^{1,2}(\mathbb{R}^d) \text{ and } \int_{\mathbb{R}^d} g^2(s, x) dx = 1 \text{ for all } s \in [0, 1] \right\}.$$

It is readily checked that there exists a constant C such that the following inequality holds for all $g \in \mathcal{A}_d$:

$$\int_{[0,1]^2} \int_{(\mathbb{R}^d)^2} \gamma_0(s-t) \gamma(x-y) g^2(s, x) g^2(t, y) dx dy ds dt \leq C^4 \int_{[0,1] \times \mathbb{R}^d} |\nabla_x g(s, x)|^2 ds dx. \quad (3.21)$$

Remark 3.11. When the noise W is time independent, that is when $H_0 = 1$ (hence $\gamma_0 \equiv 1$), equality (3.21) can be established using the same method as in [1]. The general case then follows from [7, Lemma 5.2].

Notation 3.12. We call $\kappa = \kappa(H_0, \mathbf{H})$ the best constant in inequality (3.21).

Remark 3.13. The classical Gagliardo-Nirenberg inequality asserts that for $p(d-2) \leq d$ we have

$$\|f\|_{L^{2p}(\mathbb{R}^d)} \leq \kappa \|\nabla f\|_{L^2(\mathbb{R}^d)}^{\frac{d(p-1)}{2p}} \cdot \|f\|_{L^2(\mathbb{R}^d)}^{1-\frac{d(p-1)}{2p}}. \quad (3.22)$$

The best constant κ in the above inequality is called the Gagliardo-Nirenberg constant. Now in equation (3.21) consider the special case when $d = 2, p = 2$, and $H_0 = 1, H_1 = H_2 = 1/2$. In this situation it is clear that the quantity κ defined in Notation 3.12 coincides with the Gagliardo-Nirenberg constant in relation (3.22).

We can now state the main result of this section, which identifies the exact time of blowup for the integer moments of $u_t(x)$ in the critical case.

Theorem 3.14. *We suppose that condition (1.8) is met for our indices H_0 and H . For any integer $p \geq 2$ we define a critical time $t_0(p)$ by*

$$t_0(p) = \left(\frac{1}{\kappa^4(p-1)} \right)^{1/(2H_0-1)},$$

where we recall that κ is the constant introduced in Notation 3.12. For $\epsilon > 0$, let $u^{\epsilon, \diamond}$ be the unique solution to equation (3.12). Then the following assertions holds true:

- (i) If $t < t_0 \equiv t_0(2) = \kappa^{-4/(2H_0-1)}$, then for every $x \in \mathbb{R}^d$ the sequence $\{u_t^{\epsilon, \diamond}(x), \epsilon > 0\}$ converges in $L^2(\Omega)$ to an element $u_t^\diamond(x)$. The process $\{u_t^\diamond(x) : t \in [0, t_0), x \in \mathbb{R}^d\}$ is said to be the solution of equation (3.1).
- (ii) For any $p \geq 2$, if $t < t_0(p)$ the sequence $\{u_t^{\epsilon, \diamond}(x); \epsilon > 0\}$ also converges in $L^p(\Omega)$ to $u_t^\diamond(x)$.
- (iii) For $p \in (1, \infty)$, if $t > t_0(p)$ and $x \in \mathbb{R}^d$ we have

$$\lim_{\epsilon \rightarrow 0} \|u_t^{\epsilon, \diamond}(x)\|_{L^p(\Omega)} = \infty.$$

Remark 3.15. In the critical case when $p \in [1, 2)$ the notion of solution to equation (3.1) or (3.3) is not as clear as in Section 3.2. However, let us mention the following facts:

- (a) If $p \geq 2$ and $t < t_0(p)$, then we can take limits in equation (3.12) and show that the limit $\{u_t^\diamond(x); t \geq 0, x \in \mathbb{R}^d\}$ still solves the Skorohod equation (3.3). We have not provided details for the sake of conciseness.
- (b) Whenever $p \in (1, \infty)$ and $t > t_0(p)$ one can not take limits in equation (3.3), even if one resorts to the extended Skorohod setting of Definition 2.1. However, item (iii) in Theorem 3.14 asserts that if we could give a meaning to equation (3.3) by a regularization procedure, then its solution $u_t(x)$ would not belong to $L^p(\Omega)$.
- (c) Our current techniques do not allow to assert the L^p -convergence of u^ϵ when $p \in (1, 2)$ and $t < t_0(p)$. Neither are we able to prove the existence of a L^p -solution of (3.3) for $p < 2$, even if we invoke the extended Skorohod setting. On the other hand, the definition of $t_0(p)$ in Theorem 3.14 clearly allows p to be less than 2. It is therefore reasonable to conjecture that when $p \in (1, 2)$ and $t < t_0(p)$, u^ϵ converges in L^p to the unique solution of equation (3.3) in the extended Skorohod setting. Provided this can be achieved, Proposition 3.5 establishes the uniqueness of the aforementioned extended solution.

The proof of Theorem 3.14 is split in the sections below.

3.5. Proof of the L^p -convergence in Theorem 3.14. In this section we work under the critical assumption (1.8). Before we proceed to the proof of the relation $\mathbf{E}[|u_t^\diamond(x)|^p] < \infty$ for t small enough, we first state a general sub-additivity result which is useful for our next computations. It is borrowed from [8, Theorem 6.1].

Lemma 3.16. *Let $\nu(d\xi)$ be a measure on $\mathbb{R}^d$, and consider two $\mathbb{R}^d$ -valued independent Brownian motions B and $\tilde{B}$. For $t \geq 0$ we introduce the random variable $\mathcal{H}_t$ defined by*

$$\mathcal{H}_t = \int_{\mathbb{R}^d} \nu(d\xi) \left[\int_0^t e^{i\xi \cdot B_s} ds \right] \left[\int_0^t e^{-i\xi \cdot \tilde{B}_s} ds \right].$$

Then for any $t_1, t_2 > 0$ and $\theta \in \mathbb{R}_+$, the following inequality holds true:

$$\sum_{n=0}^{\infty} \frac{\theta^n}{n!} \left\{ \frac{1}{(t_1 + t_2)^n} \mathbb{E}[\mathcal{H}_{t_1+t_2}^n] \right\} \leq \left(\sum_{n=0}^{\infty} \frac{\theta^n}{n!} \left\{ \frac{1}{t_1^n} \mathbb{E}[\mathcal{H}_{t_1}^n] \right\} \right) \left(\sum_{n=0}^{\infty} \frac{\theta^n}{n!} \left\{ \frac{1}{t_2^n} \mathbb{E}[\mathcal{H}_{t_2}^n] \right\} \right) \quad (3.23)$$

whenever the right hand side is finite.

Remark 3.17. We could obviously have stated (3.23) in the following exponential form,

$$\mathbb{E} \left[\exp \left(\frac{\theta \mathcal{H}_{t_1+t_2}}{t_1 + t_2} \right) \right] \leq \mathbb{E} \left[\exp \left(\frac{\theta \mathcal{H}_{t_1}}{t_1} \right) \right] \mathbb{E} \left[\exp \left(\frac{\theta \mathcal{H}_{t_2}}{t_2} \right) \right].$$

However, equation (3.23) is the one which will be used for our computations below.

We will now separate the upper bound computations for $u_t^\diamond(x)$ into a L^2 bound and L^p bounds for $p > 2$.

3.5.1. Convergence for $p = 2$. Our aim in this section is to prove item (i) in Theorem 3.14. According to the Feynman-Kac representation (3.20), an upper bound on the L^2 moments of $u_t^\diamond(x)$ amounts to prove the following relation

$$\mathbb{E}[\exp(\alpha_t^{12})] < \infty, \quad \text{for all } t < \frac{1}{\kappa^4/(2H_0-1)} \equiv t_0(2), \quad (3.24)$$

where we recall that the constant κ is defined in Notation 3.12. Also notice that the random variable α_t^{12} introduced in (3.19) will be written as

$$\alpha_t^{12} = \int_{[0,t]^2} \gamma_0(s-r) \gamma(B_s - \tilde{B}_r) ds dr, \quad (3.25)$$

for two independent Brownian motions B and $\tilde{B}$. In addition, notice that only the large positive values of α_t^{12} might be responsible for the blowup of exponential moments. Therefore relation (3.24) can be easily deduced from the following asymptotic tail behavior

$$\limsup_{b \rightarrow \infty} \frac{1}{b} \log \mathbb{P} \{ \alpha_t^{12} \geq b \} \leq -\frac{1}{\kappa^4 t^{2H_0-1}}. \quad (3.26)$$

We now proceed to prove (3.26), and we divide our proof in several steps.

Step 1: Scaling arguments. We recall once again that γ_0 and γ are introduced in (1.3). Formally they are given by

$$\gamma_0(u) = C_{H_0}|u|^{-\alpha_0}, \quad \text{and} \quad \gamma(x) = C_{\mathbf{H}} \prod_{j=1}^d |x_j|^{-\alpha_j}, \quad \text{where} \quad \alpha_j = 2 - 2H_j. \quad (3.27)$$

With this formal expression in mind, we consider $b > 0$ and set $\sigma = \frac{b}{t}s, \rho = \frac{b}{t}r$ in expression (3.25). Invoking the usual Brownian scaling and the fact that $\sum_{j=1}^d (H_j - 1) = -1$ under our critical assumption (1.8), we let the patient reader check that

$$\alpha_t^{12} \stackrel{(d)}{=} b^{-1} t^{2H_0-1} \int_{[0,b]^2} \gamma_0(b^{-1}(\sigma - \rho)) \gamma(B_\sigma - \tilde{B}_\rho) d\sigma d\rho. \quad (3.28)$$

As mentioned in the introduction, expressions like (3.28) are better expressed in Fourier modes, thanks to (1.4) and (1.5). We get

$$\alpha_t^{12} \stackrel{(d)}{=} b^{-1} t^{2H_0-1} \int_{\mathbb{R}^{d+1}} \mu_0(d\lambda) \mu(d\xi) \left[\int_0^b e^{i(\lambda b^{-1}s + \xi \cdot B_s)} ds \right] \left[\int_0^b e^{-i(\lambda b^{-1}s + \xi \cdot \tilde{B}_s)} ds \right].$$

In order to write the above expression in a more compact way, let us define the following random variables,

$$\eta_b(\lambda, \xi) = \int_0^b e^{i(\lambda b^{-1}s + \xi \cdot B_s)} ds, \quad \text{and} \quad \tilde{\eta}_b(\lambda, \xi) = \int_0^b e^{-i(\lambda b^{-1}s + \xi \cdot \tilde{B}_s)} ds, \quad (3.29)$$

and denote by Z_b the weighted integral of η_b and $\tilde{\eta}_b$, namely

$$Z_b = \int_{\mathbb{R}^{d+1}} \eta_b(\lambda, \xi) \tilde{\eta}_b(\lambda, \xi) \mu_0(d\lambda) \mu(d\xi).$$

We end up with the following identity in law

$$\alpha_t^{12} \stackrel{(d)}{=} b^{-1} t^{2H_0-1} Z_b. \quad (3.30)$$

We now go back to our main objective. Plugging (3.30) into (3.26), a few elementary algebraic manipulations yield that (3.24) can be reduced to the following bound,

$$\limsup_{b \rightarrow \infty} \frac{1}{b} \log \mathbb{P} \left\{ |Z_b|^{1/2} \geq \frac{b}{t^{H_0-1/2}} \right\} \leq -\frac{1}{\kappa^4 t^{2H_0-1}}. \quad (3.31)$$

Moreover, since t is an arbitrary positive number above, we simply set $u = t^{-(H_0-1/2)}$. We get that (3.24) is implied by the inequality

$$\limsup_{b \rightarrow \infty} \frac{1}{b} \log \mathbb{P} \left\{ |Z_b|^{1/2} \geq ub \right\} \leq -\frac{u^2}{\kappa^4}, \quad (3.32)$$

which should hold for all $u > 0$.

As a last preliminary step, we set up a cutoff procedure on the random variable Z_b . Namely for $\delta > 0$ we define

$$\mu_0^\delta(d\lambda) = e^{-\delta|\lambda|^2} \mu_0(d\lambda), \quad \text{and} \quad \bar{\mu}_0^\delta(d\lambda) = (1 - e^{-\delta|\lambda|^2}) \mu_0(d\lambda). \quad (3.33)$$

Accordingly, we also set

$$Z_b^\delta = \int_{\mathbb{R}^{d+1}} \eta_b(\lambda, \xi) \tilde{\eta}_b(\lambda, \xi) \mu_0^\delta(d\lambda) \mu(d\xi), \quad \text{and} \quad \bar{Z}_b^\delta = \int_{\mathbb{R}^{d+1}} \eta_b(\lambda, \xi) \tilde{\eta}_b(\lambda, \xi) \bar{\mu}_0^\delta(d\lambda) \mu(d\xi), \quad (3.34)$$

and notice that $Z_b = Z_b^\delta + \bar{Z}_b^\delta$. We will treat Z_b^δ and $\bar{Z}_b^\delta$ separately in (3.32).

Step 2: Identification of a negligible term. In this step we prove that the contribution of $\bar{Z}_b^\delta$ in (3.32) is negligible. Specifically, recalling that $\bar{Z}_b^\delta$ is defined by (3.34), given any $\epsilon > 0$ and $L > 0$ we prove that

$$\limsup_{b \rightarrow \infty} \frac{1}{b} \log \mathbb{P} \{ |\bar{Z}_b^\delta|^{1/2} \geq \epsilon b \} \leq -L, \quad (3.35)$$

when δ is small enough. As a first step in this direction, introduce an additional parameter $\hat{\alpha}_0 \in (\alpha_0, 1)$, where we recall that α_0 is defined by (3.27). According to the definition (3.33) of $\bar{\mu}_0^\delta$ we have

$$\bar{\mu}_0^\delta(d\lambda) = \left(1 - e^{-\delta|\lambda|^2}\right) \mu_0(d\lambda) \leq \left(1 - e^{-\delta|\lambda|^2}\right)^{\frac{\hat{\alpha}_0 - \alpha_0}{2}} \mu_0(d\lambda).$$

Owing to the elementary relation $1 - e^{-x} \leq x$ for $x \geq 0$, recalling from (1.5) that $\mu_0(d\lambda) = c_0 |\lambda|^{-(1-\alpha_0)} d\lambda$, and setting

$$\hat{\mu}_0(d\lambda) = c_0 |\lambda|^{-(1-\hat{\alpha}_0)} d\lambda,$$

we get

$$\bar{\mu}_0^\delta(d\lambda) \leq \delta^{\frac{\hat{\alpha}_0 - \alpha_0}{2}} |\lambda|^{\hat{\alpha}_0 - \alpha_0} \mu_0(d\lambda) = \delta^{\frac{\hat{\alpha}_0 - \alpha_0}{2}} \hat{\mu}_0(d\lambda). \quad (3.36)$$

We now compute the n -th moment of the random variable $\bar{Z}_b^\delta$. Starting from definition (3.34) and invoking our convention (1.12) on product measures, for all $n \geq 1$ we have

$$\mathbb{E} [(\bar{Z}_b^\delta)^n] = \int_{(\mathbb{R}^{d+1})^n} \mathbb{E} \left[\prod_{k=1}^n \eta_b(\lambda_k, \xi_k) \tilde{\eta}_b(\lambda_k, \xi_k) \right] \bar{\mu}_0^\delta(d\lambda) \mu(d\xi). \quad (3.37)$$

In addition the families $\{\eta_b(\lambda_k, \xi_k); k \leq n\}$ and $\{\tilde{\eta}_b(\lambda_k, \xi_k); k \leq n\}$ above are i.i.d, due to definition (3.29) and the fact that $B, \tilde{B}$ are two independent Brownian motion. Hence relation (3.37) can be written as

$$\mathbb{E} [(\bar{Z}_b^\delta)^n] = \int_{(\mathbb{R}^{d+1})^n} \left| \mathbb{E} \left[\prod_{k=1}^n \eta_b(\lambda_k, \xi_k) \right] \right|^2 \bar{\mu}_0^\delta(d\lambda) \mu(d\xi). \quad (3.38)$$

Note that from the right-hand side of the above identity, the n -th moment of $\bar{Z}_b^\delta$ is non-negative. Plugging inequality (3.36) in the above identity, we have

$$\mathbb{E} [(\bar{Z}_b^\delta)^n] \leq \delta^{\frac{\hat{\alpha}_0 - \alpha_0}{2} n} \int_{(\mathbb{R}^{d+1})^n} \left| \mathbb{E} \left[\prod_{k=1}^n \eta_b(\lambda_k, \xi_k) \right] \right|^2 \hat{\mu}_0(d\lambda) \mu(d\xi). \quad (3.39)$$

In order to bound the right-hand side of (3.39), we first go back to the definition (3.29) of η_b and set $\lambda := b^{-1}\lambda$ therein. We let the patient reader check that the scaling can be read

in (3.39) as

$$\mathbb{E}[(\bar{Z}_b^\delta)^n] \leq \delta^{\frac{\hat{\alpha}_0 - \alpha_0}{2}n} b^{n\hat{\alpha}_0} \int_{(\mathbb{R}^{d+1})^n} \left| \mathbb{E} \left[\prod_{k=1}^n \int_0^b e^{i(\lambda_k s_k + \xi_k \cdot \tilde{B}_{s_k})} ds_k \right] \right|^2 \hat{\mu}_0(d\lambda) \mu(d\xi). \quad (3.40)$$

Next we notice that the right-hand side of (3.40) is upper bounded in [8, relation (3.1)]. Indeed, with the correspondence $\hat{H}_0 \equiv 2^{-1}(2 - \hat{\alpha}_0)$ and with relation (1.8) in mind, we choose $\hat{\alpha}_0 > \alpha_0$ such that $4(1 - \hat{H}_0) + (d_* - 2H_*) < 2$. Then a direct application of [8, relation (3.1)] yields

$$\mathbb{E}[(\bar{Z}_b^\delta)^n] \leq C^n \delta^{\frac{(\hat{\alpha}_0 - \alpha_0)n}{2}} (n!) b^{n\hat{\alpha}_0} b^{n(1 - \hat{\alpha}_0)} = C^n (n!) \delta^{\frac{(\hat{\alpha}_0 - \alpha_0)n}{2}} b^n. \quad (3.41)$$

Starting from this inequality we can easily get a similar bound for $\mathbb{E}[|\bar{Z}_b^\delta|^n]$ by changing the constant C in the right-hand side of (3.41). Namely, when n is even, note that $\mathbb{E}[(\bar{Z}_b^\delta)^n] = \mathbb{E}[|\bar{Z}_b^\delta|^n]$ due to the fact that our measures μ and μ_0 are symmetric. For any odd number $n = 2k + 1$, we just invoke the Cauchy-Schwartz inequality, which yields

$$\mathbb{E}[|\bar{Z}_b^\delta|^{2k+1}] \leq \left(\mathbb{E}[(\bar{Z}_b^\delta)^{2k}] \right)^{1/2} \left(\mathbb{E}[(\bar{Z}_b^\delta)^{2k+2}] \right)^{1/2}.$$

We let the reader check that this slight elaboration yields (3.41) with $\mathbb{E}[|\bar{Z}_b^\delta|^n]$ on the left-hand side. Therefore, inserting (3.41) into a Taylor expansion for $x \mapsto e^x$, we get that, for some constant $C > 0$,

$$Q \equiv \sup_{b \geq 1} \mathbb{E} \left[\exp \left(\frac{|\bar{Z}_b^\delta|}{C \delta^{\frac{\hat{\alpha}_0 - \alpha_0}{2}} b} \right) \right] < \infty. \quad (3.42)$$

We can now go back to our claim (3.35). Indeed, plugging (3.42) into a standard application of Chebyshev's inequality, we obtain

$$\mathbb{P}\{|\bar{Z}_b^\delta|^{1/2} \geq \epsilon b\} = \mathbb{P}\{|\bar{Z}_b^\delta| \geq \epsilon^2 b^2\} \leq Q \exp \left(-\frac{\epsilon^2 b}{C \delta^{(\hat{\alpha}_0 - \alpha_0)/2}} \right),$$

from which (3.35) is easily deduced by picking a small enough δ . Taking the decomposition $Z_b = Z_b^\delta + \bar{Z}_b^\delta$ given by (3.34), our objective (3.32) is thus reduced to show the following bound for all $u, \delta > 0$ (see e.g. [17, Lemma 1.2.15] for a general result yielding a proper identification of exponentially negligible terms),

$$\limsup_{b \rightarrow \infty} \frac{1}{b} \log \mathbb{P}\{|Z_b^\delta|^{1/2} \geq ub\} \leq -\frac{u^2}{\kappa^4}. \quad (3.43)$$

Step 3: Cutoff procedure in space. In order to prove (3.43), we further decompose Z_b^δ as follows.

$$Z_b^\delta = Z_b^{\delta, M} + \bar{Z}_b^{\delta, M}, \quad (3.44)$$

where $M \geq 1$ is an additional parameter and $Z_b^{\delta, M}, \bar{Z}_b^{\delta, M}$ are respectively defined by

$$Z_b^{\delta, M} = \int_{\mathbb{R} \times [-M, M]^d} \eta_b(\lambda, \xi) \tilde{\eta}_b(\lambda, \xi) \mu_0^\delta(d\lambda) \mu(d\xi), \quad (3.45)$$

and

$$\bar{Z}_b^{\delta,M} = \int_{\mathbb{R} \times ([-M,M]^d)^c} \eta_b(\lambda, \xi) \tilde{\eta}_b(\lambda, \xi) \mu_0^\delta(d\lambda) \mu(d\xi). \quad (3.46)$$

Similarly to the previous step, we will now prove that there exists $l > 0$ such that

$$\limsup_{b \rightarrow \infty} \frac{1}{b} \log \mathbb{P} \left\{ |\bar{Z}_b^{\delta,M}|^{1/2} \geq \epsilon b \right\} \leq -l. \quad (3.47)$$

To this aim we will first upper bound the moments of $\bar{Z}_b^{\delta,M}$ as in (3.41). Namely, along the same lines as for (3.38) we have

$$\mathbb{E} \left[\left(\bar{Z}_b^{\delta,M} \right)^n \right] = \int_{(\mathbb{R} \times ([-M,M]^d)^c)^n} \left| \mathbb{E} \left[\prod_{k=1}^n \eta_b(\lambda_k, \xi_k) \right] \right|^2 \mu_0^\delta(d\lambda) \mu(d\xi), \quad (3.48)$$

which shows in particular that $\mathbb{E}[(\bar{Z}_b^{\delta,M})^n] \geq 0$ for all $n \geq 1$. Furthermore, recall from (3.29) that

$$\begin{aligned} \mathbb{E} \left[\prod_{k=1}^n \eta_b(\lambda_k, \xi_k) \right] &= \mathbb{E} \left[\int_{[0,b]^n} \prod_{k=1}^n e^{-(i\lambda_k b^{-1} s_k + \xi_k \cdot B_{s_k})} ds \right] \\ &= \int_{[0,b]^n} e^{-i\lambda_k b^{-1} s_k} \mathbb{E} \left[\prod_{k=1}^n e^{-i\xi_k \cdot B_{s_k}} \right] ds. \end{aligned}$$

One can then trivially bound the terms $|e^{-i\lambda_k s_k}|$ by 1 in the right-hand side above and resort to the fact that $\mathbb{E} \left[\prod_{k=1}^n e^{-i\xi_k \cdot B_{s_k}} \right] \geq 0$. This yields

$$\left| \mathbb{E} \left[\prod_{k=1}^n \eta_b(\lambda_k, \xi_k) \right] \right| \leq \int_{[0,b]^n} \mathbb{E} \left[\prod_{k=1}^n e^{-i\xi_k \cdot B_{s_k}} \right] ds.$$

Reporting this inequality into (3.48) we get

$$\mathbb{E}[(\bar{Z}_b^{\delta,M})^n] \leq [\mu_0^\delta(\mathbb{R})]^n \mathbb{E}[(\mathcal{H}_b^c(M))^n], \quad (3.49)$$

where we have set

$$\mathcal{H}_b^c(M) = \int_{([-M,M]^d)^c} \left(\int_0^b e^{i\xi \cdot B_s} ds \right) \left(\int_0^b e^{-i\xi \cdot \tilde{B}_s} ds \right) \mu(d\xi). \quad (3.50)$$

We now wish to apply subadditivity properties of $\mathcal{H}_b^c(M)$, such as (3.23), in order to obtain relation (3.47). A first step in this direction is to apply Chebyshev's inequality, which asserts that for all $\epsilon, b > 0$ and $k \geq 1$ we have

$$(\epsilon^2 b^2 N)^{2k} \mathbb{P} \left\{ |\bar{Z}_b^{\delta,M}|^{1/2} \geq \epsilon b \right\} \leq N^{2k} \mathbb{E} \left[|\bar{Z}_b^{\delta,M}|^{2k} \right] = N^{2k} \mathbb{E} \left[(\bar{Z}_b^{\delta,M})^{2k} \right], \quad (3.51)$$

where we have introduced an additional parameter $N \geq 0$ to be specified later on. We sum inequality (3.51) over k and resort to the elementary inequality

$$\frac{1}{2} \exp(x) \leq \frac{e^x + e^{-x}}{2} = \sum_{k=0}^{\infty} x^{2k} / (2k)!,$$

in order to get

$$\begin{aligned} \frac{1}{2} \exp\{\epsilon^2 b^2 N\} \mathbb{P} \left\{ |\bar{Z}_b^{\delta, M}|^{1/2} \geq \epsilon b \right\} &\leq \sum_{k=0}^{\infty} \frac{(\epsilon^2 b^2 N)^{2k}}{(2k)!} \mathbb{P} \left\{ |\bar{Z}_b^{\delta, M}|^{1/2} \geq \epsilon b \right\} \\ &\leq \sum_{k=0}^{\infty} \frac{N^{2k}}{(2k)!} \mathbb{E} \left[(\bar{Z}_b^{\delta, M})^{2k} \right]. \end{aligned}$$

Therefore invoking the fact that $\mathbb{E}[(\bar{Z}_b^{\delta, M})^k] \geq 0$ for all $k \geq 0$, we have

$$\frac{1}{2} \exp\{\epsilon^2 b^2 N\} \mathbb{P} \left\{ |\bar{Z}_b^{\delta, M}|^{1/2} \geq \epsilon b \right\} \leq \sum_{k=0}^{\infty} \frac{N^k}{k!} \mathbb{E} \left[(\bar{Z}_b^{\delta, M})^k \right],$$

and owing to inequality (3.49), the above becomes

$$\frac{1}{2} \exp\{\epsilon^2 b^2 N\} \mathbb{P} \left\{ |\bar{Z}_b^{\delta, M}|^{1/2} \geq \epsilon b \right\} \leq \sum_{k=0}^{\infty} \frac{(N \mu_0^\delta(\mathbb{R}))^k}{k!} \mathbb{E} \left[(\mathcal{H}_b^c(M))^k \right],$$

where we recall that $\mathcal{H}_b^c(M)$ is defined by (3.50). Summarizing our considerations for this step, we have found that for all $\epsilon, b > 0$ and $N > 0$ we have

$$\begin{aligned} \mathbb{P} \left\{ |\bar{Z}_b^{\delta, M}|^{1/2} \geq \epsilon b \right\} &\leq 2 \exp\{-\epsilon^2 b^2 N\} \sum_{k=0}^{\infty} \frac{(N \mu_0^\delta(\mathbb{R}))^k}{k!} \mathbb{E} \left[(\mathcal{H}_b^c(M))^k \right] \\ &= 2 \exp\{-\epsilon^2 b^2 N\} \sum_{k=0}^{\infty} \frac{(bN \mu_0^\delta(\mathbb{R}))^k}{k!} \left\{ \frac{1}{b^k} \mathbb{E} \left[(\mathcal{H}_b^c(M))^k \right] \right\}. \end{aligned}$$

We can now apply Lemma 3.16 in the following way: we set $\theta = bN \mu_0^\delta(\mathbb{R})$ and we assume that b is an integer (generalizations to an arbitrary positive b are left to the reader). Then iterating (3.23) b times and writing $R = Nb$ we end up with

$$\mathbb{P} \left\{ |\bar{Z}_b^{\delta, M}|^{1/2} \geq \epsilon b \right\} \leq 2 \exp\{-\epsilon^2 b R\} (\mathcal{A}^\delta(M, R))^b, \quad (3.52)$$

where the quantity $\mathcal{A}^\delta(M, R)$ is defined by

$$\mathcal{A}^\delta(M, R) = \sum_{k=0}^{\infty} \frac{(R \mu_0^\delta(\mathbb{R}))^k}{k!} \mathbb{E} \left[(\mathcal{H}_1^c(M))^k \right]. \quad (3.53)$$

Finally, one can use the dominated convergence theorem to show that the right-hand side of relation (3.52) satisfies, for every $R > 0$ small enough,

$$\lim_{M \rightarrow \infty} \mathcal{A}^\delta(M, R) = 1. \quad (3.54)$$

Indeed, first note that the bound (3.1) in [8], already used for our relation (3.41), can be extended to the case when $H_0 = 1$ (i.e., the setting without time dependence). A direct application of this bound (or generalization of (3.41) to $\alpha_0 = \hat{\alpha}_0 = 0$ and $b = 1$) yields that for all $k \geq 1$ we have

$$\mathbb{E}[(\mathcal{H}_1^c(M))^k] \leq \mathbb{E}[(\mathcal{H}_1(\mathbb{R}))^k] \leq C^k k!,$$

where we have set

$$\mathcal{H}_1(\mathbb{R}) = \int_{\mathbb{R}} \left(\int_0^b e^{i\xi \cdot B_s} ds \right) \left(\int_0^b e^{-i\xi \cdot \tilde{B}_s} ds \right) \mu(d\xi).$$

Hence one can choose R small enough such that the following domination of the general term of (3.54) holds true,

$$\mathcal{A}^\delta(M, R) \leq \sum_{k=0}^{\infty} \frac{(R\mu_0^\delta(\mathbb{R}))^k}{k!} \mathbb{E} \left[(\mathcal{H}_1(\mathbb{R}))^k \right] \leq \sum_{k=0}^{\infty} (CR\mu_0^\delta(\mathbb{R}))^k < \infty. \quad (3.55)$$

Moreover, invoking relation (3.50) it is readily checked that the k -th moment of $\mathcal{H}_1^c(M)$ in (3.53) is such that

$$\mathbb{E}[(\mathcal{H}_1^c(M))^k] = \int_{([-M, M]^d)^c} \left| \int_{[0,1]^k} \mathbb{E} \left[\prod_{i=1}^k e^{i\xi_k \cdot B_{s_k}} \right] ds \right|^2 \mu(d\xi).$$

Thus the mapping $M \mapsto \mathbb{E}[(\mathcal{H}_1^c(M))^k]$ is monotone and decreasing. As a direct consequence, relation (3.54) follows by dominated convergence.

Let us now turn to our partial objective (3.47). Namely recast relation (3.52) as

$$\frac{1}{b} \log \mathbb{P} \left\{ |\bar{Z}_b^{\delta, M}|^{1/2} \geq \epsilon b \right\} \leq \frac{\log 2}{b} - \epsilon^2 R + \log \mathcal{A}^\delta(M, R).$$

In the right-hand side above, one can fix $R > 0$ small enough and then take M large enough so that

$$\log \mathcal{A}^\delta(M, R) \leq \epsilon^4.$$

Therefore for ϵ small enough we get

$$\frac{1}{b} \log \mathbb{P} \left\{ |\bar{Z}_b^{\delta, M}|^{1/2} \geq \epsilon b \right\} \leq \frac{\log 2}{b} - \frac{1}{2} \epsilon^2 R,$$

from which (3.47) is easily deduced.

Summarizing our considerations from this step, having (3.43), (3.44) and (3.47) in mind we get that our claim (3.32) is achieved as soon as we can prove

$$\limsup_{b \rightarrow \infty} \frac{1}{b} \log \mathbb{P} \left\{ |Z_b^{\delta, M}|^{1/2} \geq ub \right\} \leq -\frac{u^2}{\kappa^4}, \quad (3.56)$$

where we recall that $Z_b^{\delta, M}$ is defined by (3.45).

Step 4: Expression in terms of the moments of $Z_b^{\delta, M}$. Thanks to a standard use of Chebyshev's inequality, relation (3.56) is achieved as long as we can prove the following Gaussian type bound for $|Z_b^{\delta, M}|^{1/2}$,

$$\limsup_{b \rightarrow \infty} \frac{1}{b} \log \mathbb{E} \left[\exp \left(\theta |Z_b^{\delta, M}|^{1/2} \right) \right] \leq \frac{\kappa^4 \theta^2}{4}, \quad (3.57)$$

which should be valid for all $\theta \geq 0$. In order to separate the variables η_b and $\tilde{\eta}_b$ in the definition (3.45) of $Z_b^{\delta, M}$ we first apply Cauchy-Schwarz inequality and then the elementary

inequality $2\sqrt{ab} \leq a + b$ for all $a, b \geq 0$. We get

$$\begin{aligned} & |Z_b^{\delta, M}|^{1/2} \\ & \leq \frac{1}{2} \left\{ \left(\int_{\mathbb{R} \times [-M, M]^d} |\eta_b(\lambda, \xi)|^2 \mu_0^\delta(d\lambda) \mu(d\xi) \right)^{1/2} + \left(\int_{\mathbb{R} \times [-M, M]^d} |\tilde{\eta}_b(\lambda, \xi)|^2 \mu_0^\delta(d\lambda) \mu(d\xi) \right)^{1/2} \right\}. \end{aligned}$$

Taking into account the fact that η and $\tilde{\eta}$ are independent, this entails

$$\mathbb{E} \left[\exp \left(\theta |Z_b^{\delta, M}|^{1/2} \right) \right] \leq \left\{ \mathbb{E} \left[\exp \left(\frac{\theta}{2} |X_b^{\delta, M}|^{1/2} \right) \right] \right\}^2, \quad (3.58)$$

where the random variable $X_b^{\delta, M}$ is defined by

$$X_b^{\delta, M} = \int_{\mathbb{R} \times [-M, M]^d} |\eta_b(\lambda, \xi)|^2 \mu_0^\delta(d\lambda) \mu(d\xi). \quad (3.59)$$

Putting together (3.57) and (3.58) and setting $\theta = 2\theta$, we are reduced to prove

$$\limsup_{b \rightarrow \infty} \frac{1}{b} \log \mathbb{E} \left[\exp \left(\theta |X_b^{\delta, M}|^{1/2} \right) \right] \leq \frac{\kappa^4 \theta^2}{2}. \quad (3.60)$$

In addition, condition (3.60) can be expressed in terms of the moments of $X_b^{\delta, M}$. Indeed, taking into account the fact that $X_b^{\delta, M}$ is a positive random variable, a direct application of [4, Lemma 1.2.6] asserts that (3.60) is equivalent to the following property:

$$\limsup_{b \rightarrow \infty} \frac{1}{b} \log \sum_{n=0}^{\infty} \frac{\theta^n}{n!} \left(\mathbb{E} \left[(X_b^{\delta, M})^n \right] \right)^{1/2} \leq \kappa^4 \theta^2. \quad (3.61)$$

We will use the formulation (3.61) below in order to replace the Brownian motion B in $X_b^{\delta, M}$ by a Ornstein-Uhlenbeck process.

Step 5: Expression in terms of an Ornstein-Uhlenbeck process. Similarly to a strategy borrowed from [24], we now introduce a family of $\mathbb{R}^d$ -valued Ornstein-Uhlenbeck processes indexed by $\alpha > 0$, denoted by B^α . The process B^α solves the equation

$$dB_t^\alpha = -\alpha B_t^\alpha dt + dB_t, \quad (3.62)$$

where B is our standing d -dimensional Wiener process. Now notice that B can also be seen as an Ornstein-Uhlenbeck type process of the form

$$dB_t = -\alpha B_t dt + dW_t, \quad \text{with} \quad W_t = B_t + \alpha \int_0^t B_s ds.$$

Therefore, if we denote by $\mathbb{P}^\alpha$ the law of B^α , a standard application of Girsanov's theorem yields

$$\frac{d\mathbb{P}^\alpha}{d\mathbb{P}} \Big|_{[0, \tau]} = \exp \left(-\alpha \int_0^\tau B_s \cdot dB_s - \frac{\alpha^2}{2} \int_0^\tau |B_s|^2 ds \right). \quad (3.63)$$

In addition, Itô's formula applied to B entails that for any $\tau > 0$ we have

$$|B_\tau|^2 = 2 \int_0^\tau B_s \cdot dB_s + \tau d.$$

Therefore the exponential term in (3.63) can be recast as

$$\left. \frac{d\mathbb{P}^\alpha}{d\mathbb{P}} \right|_{[0,\tau]} = \exp \left(\frac{1}{2} \left(\alpha\tau d - \alpha|B_\tau|^2 - \alpha^2 \int_0^\tau |B_s|^2 ds \right) \right). \quad (3.64)$$

In particular, it is readily checked that

$$\left. \frac{d\mathbb{P}^\alpha}{d\mathbb{P}} \right|_{[0,\tau]} \leq \exp \left(\frac{1}{2} \alpha\tau d \right). \quad (3.65)$$

Let us also recall a moment comparison inequality which is obtained in [8, relation (6.20)]. Namely for $n \geq 1$ and the random variables $X_b^{\delta,M}$ defined by (3.59) we have

$$\mathbb{E}[(X_b^{\delta,M})^n] \leq \mathbb{E}^\alpha[(X_b^{\delta,M})^n],$$

for all $\alpha > 0$, where $\mathbb{E}^\alpha$ denotes the expectation under $\mathbb{P}^\alpha$. Hence in order to prove (3.61) it will be enough to show a uniform bound in α , namely

$$\limsup_{\alpha \rightarrow 0} \limsup_{b \rightarrow \infty} \frac{1}{b} \log \sum_{n=0}^{\infty} \frac{\theta^n}{n!} \left(\mathbb{E}^\alpha [(X_b^{\delta,M})^n] \right)^{1/2} \leq \kappa^4 \theta^2.$$

Therefore, invoking again the equivalence between (3.60) and (3.61) for positive random variables, we are reduced to prove

$$\limsup_{\alpha \rightarrow 0} \limsup_{b \rightarrow \infty} \frac{1}{b} \log \mathbb{E}^\alpha \left[\exp \left(\theta |X_b^{\delta,M}|^{1/2} \right) \right] \leq \frac{\kappa^4 \theta^2}{2} \quad (3.66)$$

for every $\theta > 0$. We now focus on inequality (3.66) for a generic $\theta > 0$.

Step 6: Space-time cutoff for the Ornstein-Uhlenbeck process. Let us introduce an additional parameter $N > 0$ and write

$$X_b^{\delta,M} = X_b^{\delta,M,N} + \bar{X}_b^{\delta,M,N},$$

with

$$\begin{aligned} X_b^{\delta,M,N} &= \int_{[-N,N] \times [-M,M]^d} |\eta_b(\lambda, \xi)|^2 \mu_0^\delta(d\lambda) \mu(d\xi), \\ \bar{X}_b^{\delta,M,N} &= \int_{[-N,N]^c \times [-M,M]^d} |\eta_b(\lambda, \xi)|^2 \mu_0^\delta(d\lambda) \mu(d\xi). \end{aligned} \quad (3.67)$$

Trivially bounding the oscillating exponential terms by 1 in the definition (3.29) of $\eta_b(\lambda, \xi)$, we obtain

$$|\bar{X}_b^{\delta,M,N}| \leq b^2 \mu_0^\delta([-N, N]^c) \mu([-M, M]^d). \quad (3.68)$$

Therefore thanks to the fact that $\delta > 0$, the right-hand side of (3.68) can be made as small as desired by picking N large enough. Similarly to what has been done in Step 2, one can thus prove an inequality of the same form as (3.35) for $\bar{X}_b^{\delta,M,N}$. We are now reduced to show that (3.66) holds true with $X_b^{\delta,M}$ replaced by $X_b^{\delta,M,N}$.

Next we consider a new parameter $K > 0$ and we fix a value $\alpha > 0$. We further decompose $X_b^{\delta,M,N}$ and write

$$X_b^{\delta,M,N} = X_b^{\delta,M,N} \mathbf{1}_{\Omega_{b,K}} + X_b^{\delta,M,N} \mathbf{1}_{\Omega_{b,K}^c}, \quad \text{where} \quad \Omega_{b,K} = \left\{ \frac{1}{b} \int_0^b |B_s| ds \leq K \right\}. \quad (3.69)$$

We will prove that for K large enough, the quantity $X_b^{\delta,M,N} \mathbf{1}_{\Omega_{b,K}^c}$ is negligible with respect to $X_b^{\delta,M,N} \mathbf{1}_{\Omega_{b,K}}$. Indeed, resorting to (3.64) we have,

$$\mathbb{E}^\alpha \left[\exp \left(\int_0^b |B_s| ds \right) \right] \leq \mathbb{E} \exp \left\{ \int_0^b \left(|B_s| - \frac{\alpha^2}{2} |B_s|^2 \right) ds + \frac{\alpha d}{2} b \right\}.$$

Now we can use the elementary inequality $x - \alpha^2 x^2/2 \leq 1/2\alpha^2$, valid for all $x > 0$, in order to get

$$\mathbb{E}^\alpha \left[\exp \left(\int_0^b |B_s| ds \right) \right] \leq e^{C_\alpha b}, \quad \text{where } C_\alpha = \frac{1}{2\alpha^2} + \frac{\alpha d}{2}.$$

Hence given $l > 0$ one can choose $K > 0$ large enough (say $K = l + 2C_\alpha$) such that uniformly in b we have

$$\mathbb{P}^\alpha(\Omega_{b,K}^c) \leq e^{-lb}, \quad (3.70)$$

where we recall that $\Omega_{b,K}$ is defined by (3.69). In addition, one can trivially bound the quantity $|\eta_b(\lambda, \xi)|^2$ in the definition (3.67) of $X_b^{\delta,M,N}$ by b^2 . This yields

$$\mathbb{E}^\alpha \left[\exp \left(\theta |X_b^{\delta,M,N}|^{1/2} \right) \mathbf{1}_{\Omega_{b,K}^c} \right] \leq \exp \left(C_{\delta,M,N}^{1/2} \theta b - lb \right),$$

where l above can be made arbitrarily large. In conclusion of this step, it is enough to prove that for all $\theta, \delta, M, N > 0$, we have

$$\limsup_{\alpha \rightarrow 0} \limsup_{b \rightarrow \infty} \frac{1}{b} \log \mathbb{E}^\alpha \left[\exp \left(\theta |X_b^{\delta,M,N}|^{1/2} \right) \mathbf{1}_{\Omega_{b,K}} \right] \leq \frac{\kappa^4 \theta^2}{2}. \quad (3.71)$$

Step 7: Linearization procedure. One way to recast the definition (3.67) of $|X_b^{\delta,M,N}|^{1/2}$ is to write

$$|X_b^{\delta,M,N}|^{1/2} = \|\eta_b\|_{\mathcal{G}}, \quad (3.72)$$

where the functional space $\mathcal{G}$ is defined by

$$\mathcal{G} = \{h \in L^2([-N, N] \times [-M, M]^d; \mu_0^\delta \otimes \mu); h(-\lambda, -\xi) = \bar{h}(\lambda, \xi)\}. \quad (3.73)$$

In this step we show how to linearize the $\mathcal{G}$ -norm above when $\omega \in \Omega_{b,K}$. To this aim, going back to (3.29), notice that for $\lambda_1, \lambda_2 \in \mathbb{R}$ and $\xi_1, \xi_2 \in \mathbb{R}^d$ we have

$$\begin{aligned} |\eta_b(\lambda_2, \xi_2) - \eta_b(\lambda_1, \xi_1)| &\leq \int_0^b |e^{i(b^{-1}\lambda_2 + \xi_2 \cdot B_s)} - e^{i(b^{-1}\lambda_1 + \xi_1 \cdot B_s)}| ds \\ &\leq |\lambda_2 - \lambda_1| + \left(\int_0^b |B_s| ds \right) |\xi_2 - \xi_1| \leq |\lambda_2 - \lambda_1| + Kb |\xi_2 - \xi_1|, \end{aligned}$$

where we have invoked the fact that $\omega \in \Omega_{b,K}$, with $\Omega_{b,K}$ given by (3.69), for the last inequality. Therefore, setting

$$\mathcal{C} = \left\{ h(\cdot, \cdot) \in \mathcal{G}; \quad |h(\lambda, \xi)| \leq 1 \text{ and } |h(\lambda_1, \xi_1) - h(\lambda_2, \xi_2)| \leq (1 + K)|(\lambda_1, \xi_1) - (\lambda_2, \xi_2)| \right\},$$

it is easily seen that $b^{-1}\eta_b \in \mathcal{C}$ for all $b \geq 1$, whenever $\omega \in \Omega_{b,K}$. Moreover, some standard uniform continuity arguments show that the closure $\mathcal{K}$ of $\mathcal{C}$ in $\mathcal{G}$ is a compact set. We have thus proved that for all $b \geq 1$ and $\omega \in \Omega_{b,K}$ we have

$$b^{-1}\eta_b \in \mathcal{K}.$$

Our next step will be to construct a finite cover of $\mathcal{K}$. To this aim consider the unit ball in $\mathcal{G}$, denoted by $B_{\mathcal{G}}(0, 1)$. We also consider the following set for any $f \in B_{\mathcal{G}}(0, 1)$:

$$\mathcal{O}_f = \{h \in \mathcal{G}; \langle f, h \rangle_{\mathcal{G}} > \|h\|_{\mathcal{G}} - \epsilon\}. \quad (3.74)$$

Then we claim that the family $\{\mathcal{O}_f; f \in B_{\mathcal{G}}(0, 1)\}$ covers $\mathcal{K}$. Indeed, consider $f \in B_{\mathcal{G}}(0, 1)$ such that $\|f\|_{\mathcal{G}} = 1$ and a constant $a > 0$. We set $h = af$. Then it is readily checked that $\langle f, h \rangle_{\mathcal{G}} = \|h\|_{\mathcal{G}}$. Therefore we get

$$\mathcal{K} \subset \{af; \|f\|_{\mathcal{G}} = 1, a > 0\} \subset \{\mathcal{O}_f; f \in B_{\mathcal{G}}(0, 1)\}.$$

Otherwise stated, $\{\mathcal{O}_f; f \in B_{\mathcal{G}}(0, 1)\}$ covers $\mathcal{K}$. In the sequel we will extract a finite family $\{\mathcal{O}_{f_i}, 1 \leq i \leq m\}$ which still covers $\mathcal{K}$, which is possible since $\mathcal{K}$ is a compact set.

Let us now go back to the random function $b^{-1}\eta_b$, seen as an element of $\mathcal{K}$. From the previous consideration we know that for almost every $\omega \in \Omega_{b,K}$ there exists $j = j(\omega) \in \{1, \dots, m\}$ such that $b^{-1}\eta_b \in \mathcal{O}_{f_j}$. Hence from the very definition (3.74) of $\mathcal{O}_{f_j}$, we get

$$\|b^{-1}\eta_b(\cdot, \cdot)\|_{\mathcal{G}} \leq \epsilon + \max_{1 \leq j \leq m} \langle f_j, b^{-1}\eta_b \rangle_{\mathcal{G}}$$

Plugging this inequality into (3.72) we end up with

$$\begin{aligned} \mathbb{E}^{\alpha} \left[\exp \left(\theta |X_b^{\delta, M, N}|^{1/2} \right) \mathbf{1}_{\Omega_{b,K}} \right] &= \mathbb{E}^{\alpha} \left[\exp \left(\theta \|\eta_b\|_{\mathcal{G}} \right) \mathbf{1}_{\Omega_{b,K}} \right] \\ &\leq \mathbb{E}^{\alpha} \left[\exp \left(\theta \max_{1 \leq j \leq m} \langle f_j, \eta_b \rangle_{\mathcal{G}} + b\theta\epsilon \right) \right] \leq e^{b\theta\epsilon} \sum_{j=1}^m \mathbb{E}^{\alpha} \left[\exp \left(\theta \langle f_j, \eta_b \rangle_{\mathcal{G}} \right) \right]. \end{aligned}$$

Hence some elementary properties of the logarithmic function (see e.g [17, Lemma 1.2.15]) entail

$$\begin{aligned} \limsup_{b \rightarrow \infty} \frac{1}{b} \log \mathbb{E}^{\alpha} \left[\exp \left(\theta |X_b^{\delta, M, N}|^{1/2} \right) \mathbf{1}_{\Omega_{b,K}} \right] \\ \leq \theta\epsilon + \max_{1 \leq j \leq m} \left\{ \limsup_{b \rightarrow \infty} \frac{1}{b} \log \left(\mathbb{E}^{\alpha} \left[\exp \left(\theta \langle f_j, \eta_b \rangle_{\mathcal{G}} \right) \right] \right) \right\}. \end{aligned} \quad (3.75)$$

We will now treat each term in the right-hand side of (3.75) separately.

Step 8: Feynman-Kac type asymptotics. Due to the fact that $\mathcal{G}$ is a subspace of $L^2([-N, N] \times [-M, M]^d; \mu_0^{\delta} \otimes \mu)$, for $j = 1, \dots, m$ we have

$$\langle f_j, \eta_b \rangle_{\mathcal{G}} = \int_{[-N, N] \times [-M, M]^d} f_j(\lambda, \xi) \bar{\eta}_b(\lambda, \xi) \mu_0^{\delta}(d\lambda) \mu(d\xi).$$

Recalling once again the definition (3.29) of η_b , this yields

$$\langle f_j, \eta_b \rangle_{\mathcal{G}} = \int_0^b \tilde{f}_j\left(\frac{s}{b}, B_s\right) ds,$$

where the functions $\tilde{f}_j$ are given by the following relation,

$$\tilde{f}_j(s, x) = \int_{[-N, N] \times [-M, M]^d} f_j(\lambda, \xi) e^{i(\lambda s + \xi \cdot x)} \mu_0^{\delta}(d\lambda) \mu(d\xi). \quad (3.76)$$

Also notice that according to the definition (3.73) of $\mathcal{G}$ we have $f_j(\lambda, \xi) = \overline{\tilde{f}_j}(-\lambda, -\xi)$ for all $j = 1, \dots, m, \lambda \in \mathbb{R}$ and $\xi \in \mathbb{R}^d$. With the expression (3.76) in mind, this yields that $\tilde{f}_j$ is real-valued. Thus inequality (3.65) entails

$$\mathbb{E}^\alpha [\exp(\theta \langle f_j, \eta_b \rangle_{\mathcal{G}})] \leq \exp\left(\frac{\alpha d}{2} b\right) \mathbb{E} \left[\exp\left(\theta \int_0^b \tilde{f}_j\left(\frac{s}{b}, B_s\right) ds\right) \right], \quad (3.77)$$

where the expectation on the right-hand side is now taken with respect to a $\mathbb{R}^d$ -valued Brownian motion B .

With (3.77) in hand, we are now back to a more classical Feynman-Kac computation. Specifically, $\tilde{f}_j$ can be seen as the Fourier transform of a finite and compactly supported measure, as is apparent from (3.76). Therefore it fulfills all the regularity assumptions allowing to apply [11, Proposition 3.1]. Applying this proposition we thus get

$$\begin{aligned} & \lim_{b \rightarrow \infty} \frac{1}{b} \log \mathbb{E} \left[\exp\left(\theta \int_0^b \tilde{f}_j\left(\frac{s}{b}, B_s\right) ds\right) \right] \\ &= \sup_{g \in \mathcal{A}_d} \left\{ \theta \int_0^1 \int_{\mathbb{R}^d} \tilde{f}_j(s, x) g^2(s, x) dx ds - \frac{1}{2} \int_0^1 \int_{\mathbb{R}^d} |\nabla_x g(s, x)|^2 dx ds \right\}, \end{aligned} \quad (3.78)$$

where the space $\mathcal{A}_d$ is introduced in Definition 3.10.

Step 9: Evaluation of the Feynman-Kac asymptotics. Let us analyze the right-hand side of relation (3.78). Owing to the definition of $\tilde{f}_j$ in (3.76), it is readily checked that

$$\int_0^1 \int_{\mathbb{R}^d} \tilde{f}_j(s, x) g^2(s, x) dx ds = \langle f_j, \tilde{\mathcal{F}}(g^2) \rangle_{\mathcal{G}},$$

where we recall that the space $\mathcal{G}$ is given by (3.73), and where we have set

$$\tilde{\mathcal{F}}(g^2)(\lambda, \xi) = \int_0^1 \int_{\mathbb{R}^d} g^2(s, x) e^{-i(\lambda s + \xi \cdot x)} dx ds. \quad (3.79)$$

Furthermore, each f_j sits in the unit ball $B_{\mathcal{G}}(0, 1)$ and thus

$$\left| \int_0^1 \int_{\mathbb{R}^d} \tilde{f}_j(s, x) g^2(s, x) dx ds \right| \leq \|\tilde{\mathcal{F}}(g^2)\|_{\mathcal{G}}.$$

Now it is easily seen that

$$\begin{aligned} & \|\tilde{\mathcal{F}}(g^2)\|_{\mathcal{G}}^2 \\ &= \int_{[-N, N] \times [-M, M]^d} |\tilde{\mathcal{F}}(g^2)(\lambda, \xi)|^2 \mu_0^\delta(d\lambda) \mu(d\xi) \\ &\leq \int_{\mathbb{R}^{d+1}} |\tilde{\mathcal{F}}(g^2)(\lambda, \xi)|^2 \mu_0(d\lambda) \mu(d\xi) = \int_{[0, 1]^2} \int_{\mathbb{R}^d \times \mathbb{R}^d} \gamma_0(s - r) \gamma(x - y) g^2(x) g^2(y) dx dy dr ds. \end{aligned}$$

Plugging this information into (3.78), we have obtained that

$$\begin{aligned} & \lim_{b \rightarrow \infty} \frac{1}{b} \log \mathbb{E} \left[\exp\left(\theta \int_0^b \tilde{f}_j\left(\frac{s}{b}, B_s\right) ds\right) \right] \\ &\leq \sup_{g \in \mathcal{A}_d} \left\{ \theta \left(\int_{[0, 1]^2} \int_{\mathbb{R}^d \times \mathbb{R}^d} \gamma_0(s - r) \gamma(x - y) g^2(x) g^2(y) dx dy dr ds \right)^{1/2} - \frac{1}{2} \|\nabla_x g(s, x)\|_{L^2([0, 1] \times \mathbb{R}^d)}^2 \right\}. \end{aligned} \quad (3.80)$$

Now resorting to (3.21), the right-hand side of (3.80) can be upper bounded by

$$\sup_{g \in \mathcal{A}_d} \left\{ \theta \kappa^2 \|\nabla_x g\|_{L^2([0,1] \times \mathbb{R}^d)} - \frac{1}{2} \|\nabla_x g\|_{L^2([0,1] \times \mathbb{R}^d)}^2 \right\} = \sup_{z > 0} \left\{ \theta \kappa^2 z - \frac{1}{2} z^2 \right\} = \frac{1}{2} \kappa^4 \theta^2,$$

where the last equality stems from a trivial optimization procedure. Summarizing our computations for this step, we have found that for $j = 1, \dots, m$ we have

$$\lim_{b \rightarrow \infty} \frac{1}{b} \log \mathbb{E} \left[\exp \left(\theta \int_0^b \tilde{f}_j \left(\frac{s}{b}, B_s \right) ds \right) \right] \leq \frac{1}{2} \kappa^4 \theta^2. \quad (3.81)$$

Step 10: Conclusion. Recall that our desired upper bound has been successively reduced to (3.71). Moreover, we have seen that the left-hand side of (3.71) is bounded by the right-hand side of (3.75) for any arbitrary small ϵ . Therefore, we are left with the evaluation of

$$A_{\theta, \epsilon, \alpha} \equiv \theta \epsilon + \max_{1 \leq j \leq m} \left\{ \limsup_{b \rightarrow \infty} \frac{1}{b} \log (\mathbb{E}^\alpha [\exp (\theta \langle f_j, \eta_b \rangle_{\mathcal{G}})]) \right\}.$$

Now putting together (3.77) and (3.81) we obtain that

$$A_{\theta, \epsilon, \alpha} \leq \theta \epsilon + \frac{\alpha d}{2} + \frac{1}{2} \kappa^4 \theta^2.$$

Since ϵ and α can be made arbitrarily small, we have shown that relation (3.71) holds true. This finishes the proof of relation (3.24). With (3.24) in hand, Theorem 3.14-(i) is obtained along the same lines as for Proposition 3.7. $\square$

Remark 3.18. According to Remark 3.15-(a), we could prove that the limit of $u^{\varepsilon, \diamond}$ in Theorem 3.14-(i) also solves the mild Skorohod equation (3.1). Moreover, thanks to a slight elaboration of Proposition 3.7 we could also obtain the uniqueness of the solution. We have not included those details for sake of conciseness.

3.5.2. Convergence for $p > 2$. Recall that we have proved item (i) in Theorem 3.14. Namely we have shown the L^p convergence of $u_t^{\varepsilon, \diamond}$ to the solution $u_t^\diamond$ of (3.1) (see Remark 3.18 about a notion of solution) for $p = 2$, with $t < t_0 = t_0(2)$. In the current section we extend this result to a general $p > 2$, that is we prove item (ii) in Theorem 3.14. As in Section 3.5.1, we will first focus on showing that $\mathbf{E}[|u_t^\diamond(x)|^p]$ is finite whenever $t < t_0(p)$.

In order to prove the L^p -boundedness of $u_t^\diamond(x)$, we introduce an additional intensity parameter $\zeta > 0$ and consider the process $u^{(\zeta)}$, solution of the following slight extension of (3.1):

$$\partial_t u_t^{(\zeta)}(x) = \frac{1}{2} \Delta u^{(\zeta)}(x) + \sqrt{\zeta} u^{(\zeta)}(x) \diamond \dot{W}_t(x),$$

with initial condition $u_0(x) = 1$. According to an hypercontractivity inequality shown in [27, Theorem 1], the following inequality holds true for all $p \geq 2$ and $(t, x) \in \mathbb{R}_+ \times \mathbb{R}^d$

$$\|u_t^{(\zeta)}(x)\|_{L^p(\Omega)} \leq \|u_t^{((p-1)\zeta)}(x)\|_{L^2(\Omega)}, \quad (3.82)$$

or otherwise stated for $\zeta = 1$,

$$\mathbf{E}[|u_t^\diamond(x)|^p] \leq \left(\mathbf{E}[|u_t^{(p-1)}(x)|^2] \right)^{p/2}. \quad (3.83)$$

In addition, writing formula (3.20) for $p = 2$ and for a noise $\sqrt{p-1} \dot{W}$, we get

$$\mathbf{E} \left[|u_t^{(p-1)}(x)|^2 \right] = \mathbf{E} \left[\exp \left((p-1) \int_{[0,t]^2} \gamma_0(s-r) \gamma(B_s - \tilde{B}_r) ds dr \right) \right].$$

Now resorting to a simple change of variables and invoking formula (3.20) again, we end up with

$$\begin{aligned} \mathbf{E} \left[|u_t^{(p-1)}(x)|^2 \right] &= \mathbf{E} \left[\exp \left(\int_{[0, (p-1)^{1/(2H_0-1)} t]^2} \gamma_0(s-r) \gamma(B_s - \tilde{B}_r) ds dr \right) \right] \\ &= \mathbf{E} \left[|u_{(p-1)^{1/(2H_0-1)} t}^\diamond(x)|^2 \right]. \end{aligned} \quad (3.84)$$

Gathering (3.83) and (3.84), we have thus obtained

$$\mathbf{E} [|u_t^\diamond(x)|^p] \leq \left(\mathbf{E} \left[|u_{(p-1)^{1/(2H_0-1)} t}^\diamond(x)|^2 \right] \right)^{p/2}. \quad (3.85)$$

Applying Theorem 3.14 - item (i) to the right hand side of (3.85), we obtain that $u_t^\diamond(x) \in L^p(\Omega)$ when $t < t_0(p)$.

We now turn to the proof of the fact that $u_t^{\epsilon, \diamond}(x)$ converges to $u_t^\diamond(x)$ in $L^p(\Omega)$ for $t < t_0(p)$. To this aim, we fix a $p' > p$ such that $t < t_0(p') < t_0(p)$. By the same argument that we used to get (3.85), we obtain the following inequality for $u_t^{\epsilon, \diamond}(x)$,

$$\mathbf{E} \left[|u_t^{\epsilon, \diamond}(x)|^{p'} \right] \leq \left(\mathbf{E} \left[|u_{(p'-1)^{1/(2H_0-1)} t}^{\epsilon, \diamond}(x)|^2 \right] \right)^{p'/2}. \quad (3.86)$$

Furthermore, it is easily seen from the proof of Proposition 3.7 that the $L^2(\Omega)$ -norm of $u_t^{\epsilon, \diamond}(x)$ is dominated by that of $u_t^\diamond(x)$. Therefore relation (3.86) implies

$$\mathbf{E} \left[|u_t^{\epsilon, \diamond}(x)|^{p'} \right] \leq \left(\mathbf{E} \left[|u_{(p'-1)^{1/(2H_0-1)} t}^\diamond(x)|^2 \right] \right)^{p'/2}.$$

Now by our choice of p' (such that $t < t_0(p')$), we conclude

$$\sup_{\epsilon > 0} \mathbf{E} \left[|u_t^{\epsilon, \diamond}(x)|^{p'} \right] < \infty.$$

Therefore, the family $\{|u_t^{\epsilon, \diamond}(x)|^p\}_{\epsilon > 0}$ is uniformly integrable. This together with the fact that $u_t^{\epsilon, \diamond}(x)$ converges to $u_t^\diamond(x)$ in $L^2(\Omega)$ gives us the convergence in $L^p(\Omega)$. The proof of Theorem 3.14 - item (ii) is thus completed.

3.6. Proof of Theorem 3.14 - item (iii). Recall that $u_t^{\epsilon, \diamond}(x)$ is defined in Proposition 3.6. In the previous section, we essentially proved that $u_t^{\epsilon, \diamond}(x)$ converges to $u_t^\diamond(x)$ in L^p when $t < t_0(p)$. In this section, we show that $\|u_t^{\epsilon, \diamond}(x)\|_p$ diverges as $\epsilon \rightarrow 0$ when $t > t_0(p)$.

In order to state a lower bound on the $L^p(\Omega)$ moments of $u^{\epsilon, \diamond}$, we first define a functional space $\mathcal{G}_0$ which generalizes the space $\mathcal{G}$ introduced in (3.73).

Definition 3.19. Let $C_{0,b}(\mathbb{R} \times \mathbb{R}^d)$ be the space of compactly supported and bounded functions on $\mathbb{R} \times \mathbb{R}^d$. We set

$$\mathcal{G}_0 = \{h \in L^2(\mathbb{R} \times \mathbb{R}^d, \mu_0 \otimes \mu) \cap C_{0,b}(\mathbb{R} \times \mathbb{R}^d); h(-\lambda, -\xi) = \bar{h}(\lambda, \xi), \mu_0 \otimes \mu - a.e.\}.$$

We can now state a first lower bound on the moments of $u^{\epsilon, \diamond}$ in terms of a variational quantity.

Proposition 3.20. *For $\epsilon > 0$ consider the solution $u^{\epsilon, \diamond}$ to the regularized Skorohod equation (3.12). We assume that the critical conditions (1.8) are met. Let $p > 1$ and denote its conjugate exponent by q . Then for $t > 0$ and $x \in \mathbb{R}^d$ we have*

$$\begin{aligned} & \liminf_{\epsilon \rightarrow 0} \|u_t^{\epsilon, \diamond}(x)\|_p \\ & \geq \sup_{h \in \mathcal{G}_0} \mathbb{E} \left[\exp \left(\int_0^t \tilde{h}(s, B_s) ds - \frac{q-1}{2} \int_{\mathbb{R}^{d+1}} |h(\lambda, \xi)|^2 \mu_0(d\lambda) \mu(d\xi) \right) \right], \end{aligned} \quad (3.87)$$

where the function $\tilde{h}$ is defined similarly to $\tilde{f}_j$ in (3.76), namely,

$$\tilde{h}(s, x) = \int_{\mathbb{R} \times \mathbb{R}^d} h(\lambda, \xi) e^{i(\lambda s + \xi \cdot x)} \mu_0(d\lambda) \mu(d\xi). \quad (3.88)$$

Remark 3.21. By the local finiteness and the symmetry of μ_0 and μ , the function $\tilde{h}$ is real-valued, bounded and uniformly continuous on $[0, 1] \times \mathbb{R}^d$ whenever $h \in \mathcal{G}_0$.

Proof of Proposition 3.20. Recall that $u_t^{\epsilon, \diamond}(x)$ is also given by expression (3.11), where $V_t^{\epsilon, B}(x)$ is introduced in (3.8). Next we recall the definition of $\mathcal{S}$ -transform on the Wiener space related to our noise W (see [21] for more details about the $\mathcal{S}$ -transform). Having in mind the notation introduced in Section 2.1, the $\mathcal{S}$ -transform of $F \equiv u_t^{\epsilon, \diamond}(x)$ is defined for $\varphi \in \mathcal{H}$ by

$$\mathcal{S}F(\varphi) = \mathbf{E}[F\mathcal{E}_\varphi],$$

where the martingale exponential $\mathcal{E}_\varphi$ is given by

$$\mathcal{E}_\varphi = \exp \left(W(\varphi) - \frac{1}{2} \|\varphi\|_{\mathcal{H}}^2 \right). \quad (3.89)$$

As highlighted in [21, Chapter 2], the $\mathcal{S}$ -transform has to be considered as the equivalent of the Fourier transform on a Wiener space. However in our context we will just use the following basic estimate for $F = u_t^{\epsilon, \diamond}(x)$:

$$\|F\|_p \geq \sup \left\{ \frac{\mathcal{S}F(\varphi)}{\|\mathcal{E}_\varphi\|_q}; \varphi \in \mathcal{H} \right\}, \quad (3.90)$$

where we recall that q is the conjugate of p . We will now analyze the right-hand side of relation (3.90).

In order to evaluate the $\mathcal{S}$ -transform $\mathcal{S}F(\varphi)$ in (3.90), We resort to the Feynman-Kac formula (3.11) for $u_t^{\epsilon, \diamond}(x)$, Fubini's theorem and the isometry (2.3) on our standing Wiener space. Similarly to (3.16)-(3.17), albeit with a time-space Fourier transform, we get

$$\mathcal{S}F(\varphi) = \mathbb{E} \left[\exp \left(\int_{\mathbb{R} \times \mathbb{R}^d} \mathcal{F}\varphi(\lambda, \xi) \bar{\mathcal{F}}\psi_t(\lambda, \xi) \mu_0(d\lambda) \mu(d\xi) \right) \right], \quad (3.91)$$

where the function ψ_t is given by

$$\psi_t(\tau, x) = \int_0^t p_\epsilon(\tau - (t - s), x - B_s) ds.$$

Evaluating the Fourier transform of ψ_t and plugging into (3.91), we thus get

$$\mathcal{S}F(\varphi) = \int_0^t \left[\int_{\mathbb{R} \times \mathbb{R}^d} e^{-\epsilon^2(\lambda^2 + |\xi|^2)/2} e^{i(\lambda(t-s) + \xi \cdot B_t)} \mathcal{F}\varphi(\lambda, \xi) \mu_0(d\lambda) \mu(d\xi) \right] ds. \quad (3.92)$$

Let us now compute the quantity $\|\mathcal{E}_\varphi\|_q$ in (3.90). Owing to the fact that $\|\mathcal{E}_g\|_1 = 1$ for any $g \in \mathcal{H}$, we easily get that

$$\|\mathcal{E}_\varphi\|_q = \exp \left(\frac{q-1}{2} \int_{\mathbb{R} \times \mathbb{R}^d} |\mathcal{F}\varphi(\lambda, \xi)|^2 \mu_0(d\lambda) \mu(d\xi) \right). \quad (3.93)$$

Therefore gathering (3.92) and (3.93) into (3.90), taking limit $\epsilon \rightarrow 0$ in (3.92) and observing that $\mathcal{F}\varphi \in L^2(\mathbb{R} \times \mathbb{R}^d; \mu_0 \otimes \mu)$ whenever $\varphi \in \mathcal{H}$, we end up with our claim (3.87). $\square$

Our next step is to relate the lower bound (3.87) to the space $\mathcal{A}_d$ introduced in Definition 3.10. This is summarized in the following lemma.

Lemma 3.22. *Let us assume that the conditions of Proposition 3.20 are met. We also suppose that the following condition is satisfied,*

$$\sup_{h \in \mathcal{G}_0} \sup_{g \in \mathcal{A}_d} \left\{ \phi_t(h, g) - \frac{1}{2} \int_0^1 \int_{\mathbb{R}^d} |\nabla_x g(s, x)|^2 dx ds \right\} > 0, \quad (3.94)$$

where we recall that $\mathcal{G}_0$ is the space introduced in Definition 3.19 and where the variational quantity $\phi_t(h, g)$ is defined by

$$\begin{aligned} \phi_t(h, g) = & \int_{\mathbb{R} \times \mathbb{R}^d} h(\lambda, \xi) \tilde{\mathcal{F}}g^2(\lambda, \xi) \mu_0(d\lambda) \mu(d\xi) \\ & - \frac{1}{2t^{2H_0-1}(p-1)} \int_{\mathbb{R} \times \mathbb{R}^d} |h(\lambda, \xi)|^2 \mu_0(d\lambda) \mu(d\xi), \end{aligned} \quad (3.95)$$

with $\tilde{\mathcal{F}}g^2$ being the truncated Fourier transform given in (3.79). Then we have that

$$\lim_{\epsilon \rightarrow 0} \|u_t^{\epsilon, \diamond}(x)\|_p = \infty. \quad (3.96)$$

Proof. We start from the right-hand side of (3.87) and consider a generic $h \in \mathcal{G}_0$. Then for $b, t > 0$ we also introduce a family of rescaled functions $\{h_{t,b}; t, b > 0\}$ given by

$$h_{t,b}(\lambda, \xi) = \frac{1}{t^{2H_0-1}} h \left(t\lambda, \left(\frac{t}{b} \right)^{1/2} \xi \right).$$

Each $h_{t,b}$ is an element of $\mathcal{G}_0$ and we will now evaluate the expression (3.87) for those functions. We will take the next two observations into account:

(i) The function $\tilde{h}_{t,b}$ defined by (3.88) can be computed thanks to an elementary change of variable. Owing to an additional Brownian scaling argument, we get

$$\int_0^t \tilde{h}_{t,b}(s, B_s) ds \stackrel{(d)}{=} \int_0^b \tilde{h} \left(\frac{s}{b}, B_s \right) ds, \quad (3.97)$$

(ii) Under our standing assumption $d - H = 1$ we also have

$$\int_{\mathbb{R} \times \mathbb{R}^d} |h_{t,b}(\lambda, \xi)|^2 \mu_0(d\lambda) \mu(d\xi) = \frac{b}{t^{2H_0-1}} \int_{\mathbb{R} \times \mathbb{R}^d} |h(\lambda, \xi)|^2 \mu_0(d\lambda) \mu(d\xi). \quad (3.98)$$

Therefore plugging (3.97) and (3.98) into (3.87), we get that

$$\liminf_{\epsilon \rightarrow 0} \|u_t^{\epsilon, \diamond}(x)\|_p \geq \sup_{b>0} \sup_{h \in \mathcal{G}_0} \mathbb{E} \left[\exp \left(\int_0^b \tilde{h} \left(\frac{s}{b}, B_s \right) ds - \frac{(q-1)b}{2t^{2H_0-1}} \int_{\mathbb{R} \times \mathbb{R}^d} |h(\lambda, \xi)|^2 \mu_0(d\lambda) \mu(ds) \right) \right]. \quad (3.99)$$

We are now in a position to apply [8, Proposition 3.1] and take limits as $b \rightarrow \infty$ in (3.99). Indeed it is readily checked that $\tilde{h}$ is bounded and uniformly continuous whenever $h \in \mathcal{G}_0$. Hence [8, Proposition 3.1] reads

$$\begin{aligned} & \lim_{b \rightarrow \infty} \frac{1}{b} \log \mathbb{E} \left[\exp \left(\int_0^b \tilde{h} \left(\frac{s}{b}, B_s \right) ds \right) \right] \\ &= \sup_{g \in \mathcal{A}_d} \left\{ \int_0^1 \int_{\mathbb{R}^d} \tilde{h}(s, x) g^2(s, x) dx ds - \frac{1}{2} \int_0^1 \int_{\mathbb{R}^d} |\nabla_x g(s, x)|^2 dx ds \right\}. \end{aligned} \quad (3.100)$$

We also trivially have

$$\begin{aligned} & \lim_{b \rightarrow \infty} \frac{1}{b} \log \left(\exp \left(\frac{(q-1)b}{2t^{2H_0-1}} \int_{\mathbb{R} \times \mathbb{R}^d} |h(\lambda, \xi)|^2 \mu_0(d\lambda) \mu(ds) \right) \right) \\ &= \frac{(q-1)}{2t^{2H_0-1}} \int_{\mathbb{R} \times \mathbb{R}^d} |h(\lambda, \xi)|^2 \mu_0(d\lambda) \mu(ds) \end{aligned} \quad (3.101)$$

Gathering (3.100) and (3.101) into (3.99) and considering the limit case $b \rightarrow \infty$, we have obtained that (3.96) holds true whenever the following relation is met for at least a $h \in \mathcal{G}_0$:

$$\begin{aligned} \sup_{g \in \mathcal{A}_d} \left\{ \int_0^1 \int_{\mathbb{R}^d} \tilde{h}(s, x) g^2(s, x) dx ds - \frac{1}{2} \int_0^1 \int_{\mathbb{R}^d} |\nabla_x g(s, x)|^2 dx ds \right. \\ \left. - \frac{q-1}{2t^{2H_0-1}} \int_{\mathbb{R} \times \mathbb{R}^d} |h(\lambda, \xi)|^2 \mu_0(d\lambda) \mu(d\xi) \right\} > 0. \end{aligned} \quad (3.102)$$

In order to go from (3.102) to (3.94), we proceed as follows: recalling the definition of $\tilde{\mathcal{F}}$ in (3.79), we will see in the next Lemma 3.23 that if $g \in \mathcal{A}_d$ then $\tilde{\mathcal{F}}g^2 \in \overline{\mathcal{G}_0}$, where $\overline{\mathcal{G}_0}$ is defined by (3.104). With the expression (3.88) for $\tilde{h}(s, x)$ in mind, a direct application of Parseval's identify yields

$$\int_0^1 \int_{\mathbb{R}^d} \tilde{h}(s, x) g^2(s, x) dx ds = \int_{\mathbb{R} \times \mathbb{R}^d} h(\lambda, \xi) \tilde{\mathcal{F}}g^2(\lambda, \xi) \mu_0(d\lambda) \mu(d\xi). \quad (3.103)$$

Invoking the fact that $q-1 = 1/(p-1)$, we thus deduce that

$$\int_0^1 \int_{\mathbb{R}^d} \tilde{h}(s, x) g^2(s, x) dx ds - \frac{q-1}{2t^{2H_0-1}} \int_{\mathbb{R} \times \mathbb{R}^d} |h(\lambda, \xi)|^2 \mu_0(d\lambda) \mu(d\xi) = \phi_t(h, g),$$

where $\phi_t(h, g)$ is defined by (3.95). Hence it is readily checked that condition (3.102) is equivalent to (3.94), which finishes our proof. $\square$

We now turn to the technical result used in the proof of Lemma 3.22.

Lemma 3.23. *Recall that the space $\mathcal{A}_d$ is given in Definition 3.10. Let $\overline{\mathcal{G}_0}$ be the space defined by*

$$\overline{\mathcal{G}_0} = \{h \in L^2(\mathbb{R} \times \mathbb{R}^d; \mu_0 \otimes \mu); h(-\lambda, -\xi) = \bar{h}(\lambda, \xi) \text{ } \mu_0 \otimes \mu - a.e.\} \quad (3.104)$$

Then for any $g \in \mathcal{A}_d$, we have $\tilde{\mathcal{F}}(g^2) \in \overline{\mathcal{G}_0}$.

Proof. This is a direct consequence of inequality (3.21) and definition of $\tilde{\mathcal{F}}$ in (3.79). $\square$

Now we are ready to prove the main result of this section.

Proof of Theorem 3.14 - item (iii). We have shown that $\|u_t^{\epsilon, \diamond}(x)\|_p$ diverges as $\epsilon \rightarrow 0$ as long as relation (3.94) is satisfied. Note that since $\tilde{\mathcal{F}}(g^2) \in \overline{\mathcal{G}_0}$, the functional $\phi_t(h, g)$ defined in (3.95) is continuous in h with respect to the $L^2(\mathbb{R}^{d+1}, \mu_0 \otimes \mu)$ -norm. Therefore,

$$\sup_{h \in \overline{\mathcal{G}_0}} \phi_t(h, g) = \sup_{h \in \mathcal{G}_0} \phi_t(h, g).$$

Hence, we aim to show that (3.94) is satisfied for a suitable family of functions $h \in \overline{\mathcal{G}_0}$. Namely, we want to find some functions $h \in \overline{\mathcal{G}_0}$ such that

$$\sup_{g \in \mathcal{A}_d} \left\{ \phi_t(h, g) - \frac{1}{2} \int_0^1 \int_{\mathbb{R}^d} |\nabla_x g(s, x)|^2 dx ds \right\} > 0. \quad (3.105)$$

To this end, we will consider a family of functions $\{h^{\theta, g}; \theta \in \mathbb{R}, g \in \mathcal{A}_d\}$. Each $h^{\theta, g}$ is defined by

$$h^{\theta, g}(\lambda, \xi) = \theta [\tilde{\mathcal{F}}(g^2)](-\lambda, -\xi) = \theta [\overline{\tilde{\mathcal{F}}(g^2)}](\lambda, \xi),$$

where we recall that $\tilde{\mathcal{F}}(g^2) \in \overline{\mathcal{G}_0}$ according to Lemma 3.23. For such a function $h^{\theta, g}$ we have

$$\phi_t(h^{\theta, g}, g) = \left(\theta - \frac{\theta^2}{2t^{2H_0-1}(p-1)} \right) \int_{\mathbb{R} \times \mathbb{R}^d} |\tilde{\mathcal{F}}(g^2)(\lambda, \xi)|^2 \mu_0(d\lambda) \mu(d\xi).$$

Thus an elementary computation reveals that

$$\sup_{\theta \in \mathbb{R}} \phi_t(h^{\theta, g}, g) = \frac{(p-1)t^{2H_0-1}}{2} \int_{\mathbb{R} \times \mathbb{R}^d} |\tilde{\mathcal{F}}(g^2)(\lambda, \xi)|^2 \mu(d\lambda) \mu(d\xi).$$

Plugging this information into (3.105) we get

$$\begin{aligned} & \sup_{h \in \overline{\mathcal{G}_0}} \sup_{g \in \mathcal{A}_d} \left\{ \phi_t(h, g) - \frac{1}{2} \int_0^1 \int_{\mathbb{R}^d} |\nabla_x g(s, x)|^2 dx ds \right\} \\ & \geq \sup_{g \in \mathcal{A}_d} \left\{ \frac{(p-1)t^{2H_0-1}}{2} \int_{\mathbb{R} \times \mathbb{R}^d} |\tilde{\mathcal{F}}(g^2)(\lambda, \xi)|^2 \mu(d\lambda) \mu(d\xi) - \frac{1}{2} \int_0^1 \int_{\mathbb{R}^d} |\nabla_x g(s, x)|^2 dx ds \right\}. \end{aligned}$$

According to Notation 3.12 and to the definition of $t_0(p)$ in Theorem 3.14, it is clear that the above quantity is strictly positive as soon as $t > t_0(p)$. This concludes the proof. $\square$

4. THE STRATONOVICH EQUATION

We now would like to study equation (1.1) along a Stratonovich (or pathwise) interpretation. The basic idea behind such an interpretation can be roughly expressed as follows. We consider a sequence $\{\dot{W}^n; n \geq 1\}$ of smooth approximations of $\dot{W}$, which can be thought of as the mollification considered in (3.6) with $\epsilon = 1/n$. With this approximation in hand, let $\{u^n; n \geq 1\}$ be the sequence of classical solutions associated with $\dot{W}^n$, that is u^n is the solution of

$$\partial_t u_t^n(x) = \frac{1}{2} \Delta u_t^n(x) + u_t^n(x) \dot{W}_t^n(x), \quad t \in \mathbb{R}_+, x \in \mathbb{R}^d,$$

understood in the classical Lebesgue sense. Then the Stratonovich solution of (1.1) is morally defined as the limit (assuming it exists) of the sequence u^n .

In some situations where $\dot{W}$ is not too rough, the above heuristic idea can be rigourously formulated within the so-called Young setting (see e.g. [22, Section 5]), which provides us with some intrinsic characterization of the solution (independent of the choice of the approximation). In order to extend these considerations to rougher noises more sophisticated machineries, based on higher-order expansions, must come into the picture. In the sequel, we will thus rely on a framework and on tools derived from the so-called *theory of regularity structures*, as introduced by Hairer in [19]. Note that this strategy was already implemented in [15, 16] so as to handle the one-dimensional non-linear heat model

$$\partial_t u_t(x) = \frac{1}{2} \Delta u_t(x) + \sigma(x, u_t(x)) \dot{W}_t(x), \quad t \in [0, T], x \in \mathbb{R}, \quad (4.1)$$

where $\sigma : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is a smooth bounded function with compact support in its first variable, and T is a small enough time. The latter assumptions clearly do not cover the model under consideration (i.e., equation (1.1)), and accordingly further work is required here. An important novelty to tackle in this situation is the “non-compactness” of the perturbation term $u \dot{W}$, as opposed to $\sigma(\cdot, u) \dot{W}$ in (4.1) or to the assumptions that prevail in [19]. This difference naturally forces us to involve weighted topologies in the analysis (note that this was already the case in [22, Section 5] for the Young setting). As a result, the subsequent developments will more specifically lean on the “weighted regularity structures” recently introduced by Hairer and Labbé in [20].

The rest of the section is organized as follows. We first introduce the setting of the theory, and then formulate a general well-posedness result for the equation, when replacing $\dot{W}$ with a so-called *K-rough path* (Theorem 4.11). We finally combine this general deterministic result with a more specific fractional construction (Theorem 4.15), which yields the expected Stratonovich solution (Definition 4.18).

Note that throughout this section, and following the convention in [19], we will appeal to the parabolic norm defined for every $(s, x) \in \mathbb{R}^{d+1}$ as

$$\|(s, x)\|_{\mathfrak{s}} := \max(\sqrt{|s|}, |x_1|, \dots, |x_d|). \quad (4.2)$$

We will also use the notation $a \lesssim b$ in order to signify that there exists an irrelevant constant c such that $a \leq cb$.

4.1. Weighted Besov topologies and K-rough paths. Let us start with an as-compact-as-possible presentation of the regularity structures framework. As we mentioned above, the

formalism is presented here in its weighted version (following [20]). Of course, we will only focus on its application to the dynamics under consideration, that is to the model

$$\begin{cases} \partial_t u = \frac{1}{2} \Delta u + u \chi, & t \in [0, T], x \in \mathbb{R}^d, \\ u_0(x) = \psi(x), \end{cases} \quad (4.3)$$

with χ a distribution of order $\alpha < 0$ to be specified (at this point, the equation is only formal anyway). This customization of the theory will eventually lead us to the introduction of the fundamental object at the core of the machinery: the K -rough path (see Definition 4.7 below).

The weights considered in the sequel have to satisfy a growth assumption which is summarized in the following definition.

Definition 4.1. *A function $w : \mathbb{R}^d \rightarrow [1, \infty)$ is a weight on $\mathbb{R}^d$ if for every $M > 0$, there exist $c_{1,M}, c_{2,M} > 0$ such that for every $x, y \in \mathbb{R}^d$ with $|x - y| \leq M$, one has*

$$c_{1,M} \leq \frac{w(x)}{w(y)} \leq c_{2,M}.$$

Given a weight $w \in \mathbb{R}^d$, we will henceforth denote by $L_w^\infty(\mathbb{R}^{d+1})$ the space of functions defined by

$$L_w^\infty(\mathbb{R}^{d+1}) = \left\{ f : \mathbb{R}^{d+1} \rightarrow \mathbb{R}; \text{ for all } T > 0, \sup_{(s,x) \in [-T,T] \times \mathbb{R}^d} \frac{|f_s(x)|}{w(x)} < \infty \right\}. \quad (4.4)$$

We also write $\mathcal{C}_w^0(\mathbb{R}^{d+1})$ for the set of continuous functions in $L_w^\infty(\mathbb{R}^{d+1})$.

Let us now turn to the definition of the (weighted) Besov-type spaces of distributions involved in Hairer's theory. Consider first the case of a positive order $\lambda \in (0, 1)$:

Definition 4.2. *Let w be a weight on $\mathbb{R}^d$. For every $\lambda \in (0, 1)$, we will say that a function $\theta : \mathbb{R}^{d+1} \rightarrow \mathbb{R}$ belongs to $\mathcal{C}_w^\lambda(\mathbb{R}^{d+1})$ if for every $T > 0$,*

$$\|\theta\|_{\lambda; T, w} := \sup_{(s,x) \in [-T,T] \times \mathbb{R}^d} \frac{|\theta(s, x)|}{w(x)} + \sup_{((s,x), (t,y)) \in \mathcal{D}_{T,2}} \frac{|\theta(s, x) - \theta(t, y)|}{w(y) \| (s, x) - (t, y) \|_s^\lambda} < \infty,$$

where we recall that the norm $\|\cdot\|_s$ is defined in (4.2) and where the domain $\mathcal{D}_{T,2}$ is defined by

$$\mathcal{D}_{T,2} = \left\{ ((s, x), (t, y)) \in \mathbb{R}^{d+1} \times \mathbb{R}^{d+1}; s, t \in [-T, T], \right. \\ \left. \text{with } (s, x) \neq (t, y) \text{ and } \|(s, x) - (t, y)\|_s \leq 2 \right\}. \quad (4.5)$$

In order to define spaces of negative orders, we first need to recall the following notation for a scaling operator. Namely for all $\delta > 0$, $(s, x), (t, y) \in \mathbb{R}^{d+1}$ and $\psi : \mathbb{R}^{d+1} \rightarrow \mathbb{R}$, denote

$$(\mathcal{S}_{s,x}^\delta \psi)(t, y) := \delta^{-(d+2)} \varphi(\delta^{-2}(t-s), \delta^{-1}(y_1-x_1), \dots, \delta^{-1}(y_d-x_d)). \quad (4.6)$$

Also, for every $\ell \geq 0$, we will need to consider a specific set of compactly supported functions:

$$\mathcal{B}_s^\ell = \{\psi \in \mathcal{C}^\ell(\mathbb{R}^{d+1}); \text{ Supp}(\psi) \subset \mathcal{B}_s(0, 1) \text{ and } \|\psi\|_{\mathcal{C}^\ell} \leq 1\}, \quad (4.7)$$

where $\mathcal{B}_s(0, 1)$ stands for the unit ball in $\mathbb{R}^{d+1}$ for the parabolic norm (4.2). Finally, we denote by $\mathcal{C}_\infty^\ell(\mathbb{R}^{d+1})$ the space of ℓ -times differentiable functions (on $\mathbb{R}^{d+1}$) with bounded derivatives,

and define $\mathcal{D}'_\ell(\mathbb{R}^{d+1})$ as the dual space of $\mathcal{C}_\infty^\ell(\mathbb{R}^{d+1})$. With those additional notions in hand, we now give the definition of distributions with negative Hölder type continuity which is used in the sequel.

Definition 4.3. *Let w be a weight on $\mathbb{R}^d$ as given in Definition 4.1. For every $\alpha < 0$, we will say that a distribution $\chi \in \mathcal{D}'(\mathbb{R}^{d+1})$ belongs to $\mathcal{C}_w^\alpha(\mathbb{R}^{d+1})$ if it belongs to $\mathcal{D}'_{2(d+1)}(\mathbb{R}^{d+1})$ and if for every $T > 0$,*

$$\|\chi\|_{\alpha;T,w} := \sup_{(s,x) \in [-T,T] \times \mathbb{R}^d} \sup_{\varphi \in \mathcal{B}_s^{2(d+1)}} \sup_{\delta \in (0,1]} \frac{|\langle \chi, \mathcal{S}_{s,x}^\delta \varphi \rangle|}{\delta^\alpha w(x)} < \infty. \quad (4.8)$$

Remark 4.4. As can be seen from (4.8) we are considering topologies that are “localized” in time, and global, but “weighted”, in space. Besides, note that the choice of the regularity $2(d+1)$ in the condition $\chi \in \mathcal{D}'_{2(d+1)}$ is somewhat arbitrary. In fact, for the deterministic part of the analysis, we could replace this condition with $\chi \in \mathcal{D}'_r(\mathbb{R}^{d+1})$ for any finite $r \geq 1$, as explained in [19]. The $2(d+1)$ -regularity will only prove useful in the stochastic constructions of Section A (see for instance Lemma A.3).

The following topological spaces, which somehow correspond to “lifted versions” of $\mathcal{C}_w^\alpha(\mathbb{R}^{d+1})$, will later accommodate the central K -rough paths:

Definition 4.5. *Let w be a weight on $\mathbb{R}^d$. For every $\alpha < 0$, we say that a map $\zeta : \mathbb{R}^{d+1} \rightarrow \mathcal{D}'(\mathbb{R}^{d+1})$ belongs to $\mathcal{C}_w^\alpha(\mathbb{R}^{d+1})$ if for every $(s, x) \in \mathbb{R}^{d+1}$, $\zeta_{s,x}$ belongs to $\mathcal{D}'_{2(d+1)}(\mathbb{R}^{d+1})$ and if, for every $T > 0$,*

$$\|\zeta\|_{\alpha;T,w} := \sup_{(s,x) \in [-T,T] \times \mathbb{R}^d} \sup_{\varphi \in \mathcal{B}_s^{2(d+1)}} \sup_{\delta \in (0,1]} \frac{|\langle \zeta_{s,x}, \mathcal{S}_{s,x}^\delta \varphi \rangle|}{\delta^\alpha w(x)} < \infty,$$

where the sets $\mathcal{B}_s^l$ are given by (4.7).

We still need one last technical ingredient in the procedure: the definition of a *localized heat kernel*, which essentially transcribes the singular behavior of the (global) heat kernel around $(0, 0)$.

Definition 4.6. *We call a localized heat kernel any function $K : \mathbb{R}^{d+1} \setminus \{0\} \rightarrow \mathbb{R}$ satisfying the following conditions:*

- (i) *It holds that $p_s(x) = K(s, x) + R(s, x)$, for some “remainder” $R \in \mathcal{C}^\infty(\mathbb{R}^{d+1})$, where we recall that the heat kernel p is defined by (1.11).*
- (ii) *$K(s, x) = 0$ as soon as $s \leq 0$.*
- (iii) *There exists a smooth function $K_0 : \mathbb{R}^{d+1} \rightarrow \mathbb{R}$ with support in $[-1, 1]^{d+1}$ such that for every non-zero $(s, x) \in \mathbb{R}^{d+1}$, one has*

$$K(s, x) = \sum_{\ell \geq 0} 2^{-2\ell} (\mathcal{S}_{0,0}^{2^{-\ell}} K_0)(s, x) \quad \text{and} \quad R(s, x) = \sum_{\ell < 0} 2^{-2\ell} (\mathcal{S}_{0,0}^{2^{-\ell}} K_0)(s, x). \quad (4.9)$$

We are finally in a position to introduce the key object of the machinery, namely a distribution in the second chaos of the noise χ which plays the role of the Lévy area in our context.

Definition 4.7. Let w be a weight on $\mathbb{R}^d$ (see Definition 4.1), let K be a localized heat kernel (see Definition 4.6) and consider $\alpha < 0$. Also, fix $\chi \in \mathcal{C}_w^\alpha(\mathbb{R}^{d+1})$. We call an (α, K) -Lévy area above χ (for the weight w) any map $\mathcal{A} : \mathbb{R}^{d+1} \rightarrow \mathcal{D}'(\mathbb{R}^{d+1})$ satisfying the two following conditions.

(i) K -Chen relation: For all $(s, x), (t, y) \in \mathbb{R}^{d+1}$,

$$\mathcal{A}_{s,x} - \mathcal{A}_{t,y} = [(K * \chi)(t, y) - (K * \chi)(s, x)] \cdot \chi ;$$

(ii) Besov regularity: $\mathcal{A}$ belongs to $\mathcal{C}_w^{2\alpha+2}(\mathbb{R}^{d+1})$, where the space $\mathcal{C}_w^{2\alpha+2}$ is introduced in Definition 4.5.

We call (α, K) -rough path above χ (for the weight w) any pair $\chi = (\chi, \chi^2)$ where $\chi \in \mathcal{C}_w^\alpha(\mathbb{R}^{d+1})$ and χ^2 is an (α, K) -Lévy area above χ (for the weight w). We denote by $\mathcal{E}_{\alpha;w}^K$ the set of such (α, K) -rough paths (for the weight w). If $\chi = (\chi, \chi^2), \zeta = (\zeta, \zeta^2) \in \mathcal{E}_{\alpha;w}^K$, we set

$$\|\chi, \zeta\|_{\alpha;T,w} := \|\chi - \zeta\|_{\alpha;T,w} + \|\chi^2 - \zeta^2\|_{2\alpha+2;T,w} .$$

A global distance on $\mathcal{E}_{\alpha;w}^K$ is then given by

$$d_{\alpha;w}(\chi, \zeta) = \sum_{k \geq 1} 2^{-k} \frac{\|\chi, \zeta\|_{\alpha;k,w}}{1 + \|\chi, \zeta\|_{\alpha;k,w}} . \quad (4.10)$$

By mimicking the arguments of the proof of [15, Proposition 3.1], we immediately deduce the following completeness property:

Lemma 4.8. For every weight w on $\mathbb{R}^d$, every localized heat kernel K and every $\alpha < 0$, $(\mathcal{E}_{\alpha;w}^K, d_{\alpha;w})$ is a complete metric space.

Let us complete Definition 4.7 with two fundamental remarks, that often turn out to be essential in the application of the theory.

Remark 4.9. Recall that the space $L_w^\infty(\mathbb{R}^{d+1})$ is defined by (4.4). In the “regular” situation where $\chi \in L_w^\infty(\mathbb{R}^{d+1})$, there exists a straightforward *canonical K -Lévy area above χ* (for the weight w^2) given by the formula

$$\chi_{s,x}^2(t, y) := [(K * \chi)(t, y) - (K * \chi)(s, x)] \cdot \chi(t, y) . \quad (4.11)$$

The resulting *canonical K -rough path* will be our standard reference in approximation (or continuity) results. The situation can here be compared with Lyons’ rough paths theory, where (classical) rough paths are often obtained as the limit of the canonical rough path given by the set of iterated integrals.

Remark 4.10. Starting from a K -Lévy area χ^2 , any constant c gives rise to another K -Lévy area by setting $\hat{\chi}_{s,x}^2(t, y) := \chi_{s,x}^2(t, y) - c$, which paves the way toward renormalization tricks. In the sequel, we will use the notation

$$\text{Renorm}((\chi, \chi^2), c) := (\chi, \chi^2 - c) \quad (4.12)$$

for such elementary renormalization.

4.2. A general solution map. With the above setting and notation in hand, the following “black box” statement about equation (4.3) can now be derived from a slight adaptation of the considerations and results of [20]:

Theorem 4.11. *[Solution map] Fix an arbitrary time horizon $T > 0$ and a parameter $\alpha \in (-\frac{4}{3}, -1)$. Then there exist a localized heat kernel K , two weights w_1, w_2 on $\mathbb{R}^d$ (that depend on T), and a “solution” map*

$$\Phi = \Phi_{\alpha, w_1, w_2}^{K, T} : \mathcal{E}_{\alpha; w_1}^K \times L^\infty(\mathbb{R}^d) \longrightarrow L^\infty([0, T]; L_{w_2}^\infty(\mathbb{R}^d)), \quad (4.13)$$

where $\mathcal{E}_{\alpha; w_1}^K$ is introduced in Definition 4.7 and L_w^∞ is given by (4.4). The map Φ is such that the following properties are satisfied:

- (i) **Weights.** One has $w_1(x) = (1 + |x|)^{\kappa_1}$ and $w_2(x) = e^{\kappa_2(1+|x|)}$, for some $\kappa_1, \kappa_2 > 0$.
- (ii) **Consistency.** Assume $\chi \in L_{w_1}^\infty(\mathbb{R}^{d+1})$ and $\chi \in \mathcal{E}_{\alpha; w_1}^K$ is the canonical K -rough path above χ with Lévy-area term defined along (4.11). Then for any $\psi \in L^\infty(\mathbb{R}^d)$ one has $\Phi(\chi, \psi) = u$, where u is the classical solution on $[0, T]$ of equation (4.3).
- (iii) **Renormalization.** As in item (ii), consider $\chi \in L_{w_1}^\infty(\mathbb{R}^{d+1})$ and its canonical K -rough path χ . For an initial condition $\psi \in L^\infty(\mathbb{R}^d)$ and $c \in \mathbb{R}$, set $\hat{u} = \Phi(\text{Renorm}(\chi, c), \psi)$, where $\text{Renorm}(\chi, c)$ is defined by (4.12). Then $\hat{u}$ is the classical solution on $[0, T]$ of the equation

$$\begin{cases} \partial_t \hat{u} = \frac{1}{2} \Delta \hat{u} + \hat{u} \chi - c \hat{u}, & t \in [0, T], x \in \mathbb{R}^d, \\ u_0(x) = \psi(x). \end{cases}$$

- (iv) **Continuity.** Let $(\chi, \psi) \in \mathcal{E}_{\alpha, w_1}^K \times L^\infty(\mathbb{R}^d)$ and let $(\chi^n, \psi^n) \in \mathcal{E}_{\alpha; w_1}^K \times L^\infty(\mathbb{R}^d)$ be a sequence such that

$$d_{\alpha, w_1}(\chi^n, \chi) \rightarrow 0 \quad \text{and} \quad \|\psi^n - \psi\|_{L^\infty(\mathbb{R})} \rightarrow 0,$$

where d_{α, w_1} is the distance introduced in (4.10). Then $\Phi(\chi^n, \psi^n)$ converges to $\Phi(\chi, \psi)$ in $L^\infty([0, T]; L_{w_2}^\infty(\mathbb{R}^d))$.

Remark 4.12. We are aware that the corresponding results in [20] are actually expressed in terms of (weighted) *models* and *structure group*, following the general terminology of [19]. However, the transition from our (lighter) notion of an (α, K) -rough path to a *regularity structure* (that is, a model together with a structure group) is a matter of elementary considerations (as detailed in [15, Proposition 2.5]). The only technical point requiring some attention is the control of $K * \chi$, as an element of $\mathcal{C}_{w_1}^{\alpha+2}(\mathbb{R}^{d+1})$, in terms of $\chi \in \mathcal{C}_{w_1}^\alpha(\mathbb{R}^{d+1})$, for $\alpha \in (-\frac{4}{3}, -1)$. In fact, following the lines of the proof of [15, Lemma 2.2], one can easily check that for every weight w on $\mathbb{R}^d$, every $\alpha \in (-2, -1)$, every $\chi \in \mathcal{C}_w^\alpha(\mathbb{R}^{d+1})$ and every time $T > 0$, one has

$$\|K * \chi\|_{\alpha+2; T, w} \lesssim \|\chi\|_{\alpha; T, w}, \quad (4.14)$$

which precisely corresponds to the control we need in order to justify this transition.

4.3. Application to the stochastic fractional model. Let us now go back to the stochastic setting and to the consideration of our fractional noise $\chi := \dot{W}$, where $\dot{W}$ is defined by (1.3). In this context, let us recall that the corresponding W is a space-time fractional Brownian motion of Hurst index $H_0 \in (0, 1)$ in time and $\mathbf{H} = (H_1, \dots, H_d) \in (0, 1)^d$ in space. With the results of Theorem 4.11 in mind, the strategy toward the desired Stratonovich solution for (1.1) is now clear: we need to construct a K -rough path above $\dot{W}$ in the almost sure sense, preferably as the limit of some (renormalized) canonical K -rough path (for the continuity property (iv) in Theorem 4.11 to hold).

To this end, let us start from a smooth approximation $\dot{W}^n$ of $\dot{W}$ obtained through a general mollifying procedure. In other words, we define the approximated noise $\dot{W}^n$ by $\dot{W}^0 := 0$ and for $n \geq 1$,

$$\dot{W}^n := \partial_t \partial_{x_1} \cdots \partial_{x_d} W^n, \quad \text{where } W^n := \rho_n * W \text{ and } \rho_n(s, x) := 2^{n(d+2)} \rho(2^{2n}s, 2^nx), \quad (4.15)$$

for some mollifier $\rho : \mathbb{R}^{d+1} \rightarrow \mathbb{R}_+$ satisfying the following (natural) assumptions:

Assumption (ρ). We consider a smooth, even, and $L^1(\mathbb{R}^{d+1})$ function $\rho : \mathbb{R}^{d+1} \rightarrow \mathbb{R}_+$. In addition we suppose that ρ satisfies

$$(i) \quad \int_{\mathbb{R}^{d+1}} \rho(s, x) ds dx = 1.$$

(ii) The Fourier transform $\mathcal{F}\rho$ is Lipschitz.

(iii) For every $(\tau_0, \tau_1, \dots, \tau_d) \in [0, 1]^{d+1}$, the following upper bound holds true for every $(\lambda, \xi) \in \mathbb{R}^{d+1}$,

$$|\mathcal{F}\rho(\lambda, \xi)| \leq c_\tau |\lambda|^{-\tau_0} |\xi_1|^{-\tau_1} \cdots |\xi_d|^{-\tau_d}. \quad (4.16)$$

Remark 4.13. Assumption (ρ) is trivially satisfied by any smooth, even and compactly-supported function $\rho : \mathbb{R}^{d+1} \rightarrow \mathbb{R}_+$ such that $\int_{\mathbb{R}^{d+1}} \rho(s, x) ds dx = 1$. These conditions also cover the mollifying function considered in [22, Section 3.2] or in [23, Section 5], that is $\rho(s, x) := \varphi(s)p_1(x)$, where $\varphi := \mathbf{1}_{[0,1]}$ and p_1 refers to the Gaussian density (1.11) at time 1. Last but not least, Assumption (ρ) is satisfied by the mollifier that we have considered in the Skorohod analysis of Section 3, that is $\rho(s, x) := p_1(s)p_1(x)$ corresponding to relation (3.6). The latter choice will become our standard reference in the sequel (see Definition 4.18), for clarity in the comparison between Skorohod and Stratonovich solutions.

Once endowed with the approximation $\dot{W}^n$, and as evoked earlier, let us consider the canonical K -rough path $\mathbf{W}^n := (\dot{W}^n, \mathbf{W}^{2,n})$ along Remark 4.9. Namely we set

$$\mathbf{W}_{s,x}^{2,n}(t, y) := \mathcal{I}_{s,x}^n(t, y) \cdot \dot{W}^n(t, y), \quad \text{where } \mathcal{I}_{s,x}^n(t, y) := (K * \dot{W}^n)(t, y) - (K * \dot{W}^n)(s, x). \quad (4.17)$$

Our forthcoming statement regarding the convergence of χ^n will rely, among other ingredients, on the following technical lemma.

Lemma 4.14. *Let ρ be a mollifier satisfying Assumption (ρ), and let $H_0 \in (0, 1)$, $\mathbf{H} = (H_1, \dots, H_d) \in (0, 1)^d$ be such that*

$$2H_0 + H < d + 1, \quad (4.18)$$

where the notation H has been introduced in (1.6). Recall that the heat kernel p is defined by (1.11). Let us set from now on

$$\mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) := \frac{1}{|\lambda|^{2H_0-1}} \prod_{i=1}^d \frac{1}{|\xi_i|^{2H_i-1}}, \quad (4.19)$$

namely $c_0 c_{\mathbf{H}} \mathcal{N}_{H_0, \mathbf{H}}$ is the density of the measure $\mu_0 \otimes \mu$ introduced in (1.5). Then the following integral is finite:

$$\mathcal{J}_{\rho, H_0, \mathbf{H}} := \int_{\mathbb{R}^{d+1}} |\mathcal{F}\rho(\lambda, \xi)|^2 \mathcal{F}p(\lambda, \xi) \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) d\lambda d\xi. \quad (4.20)$$

Proof. According to the definition (1.11) of the heat kernel p and recalling that $\mathcal{F}$ stands for the space-time Fourier transform, it is readily checked that for $(\lambda, \xi) \in \mathbb{R}^{d+1}$ we have

$$\mathcal{F}p(\lambda, \xi) = \left(\frac{|\xi|^2}{2} + i\lambda \right)^{-1}. \quad (4.21)$$

Therefore, the integral under consideration can be bounded as

$$\mathcal{J}_{\rho, H_0, \mathbf{H}} \leq \mathcal{J}_{\infty} + \mathcal{J}_0, \quad (4.22)$$

where we consider a compact region $\mathcal{D}_s$ of $\mathbb{R}^{d+1}$ defined by

$$\mathcal{D}_s := \{(\lambda, \xi) \in \mathbb{R}^{d+1} : \lambda^2 + \xi_1^4 + \dots + \xi_d^4 \leq 1\}, \quad (4.23)$$

and where the quantities $\mathcal{J}_{\infty}, \mathcal{J}_0$ are respectively defined by

$$\begin{aligned} \mathcal{J}_{\infty} &:= \int_{\mathbb{R}^{d+1} \setminus \mathcal{D}_s} \frac{d\lambda d\xi}{(\lambda^2 + \xi_1^4 + \dots + \xi_d^4)^{1/2}} |\mathcal{F}\rho(\lambda, \xi)|^2 \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) \\ \mathcal{J}_0 &:= \int_{\mathcal{D}_s} \frac{d\lambda d\xi}{(\lambda^2 + \xi_1^4 + \dots + \xi_d^4)^{1/2}} |\mathcal{F}\rho(\lambda, \xi)|^2 \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi). \end{aligned} \quad (4.24)$$

We now proceed to the evaluation of those two terms.

In order to estimate $\mathcal{J}_{\infty}$, note that $(\mathbb{R}^{d+1} \setminus \mathcal{D}_s) \subset \cup_{i=0}^d \Lambda_i$, where the regions Λ_i are defined by

$$\Lambda_0 := \left\{ (\lambda, \xi_1, \dots, \xi_d) : \lambda^2 \geq \frac{1}{d+1} \right\} \quad \text{and} \quad \Lambda_i := \left\{ (\lambda, \xi_1, \dots, \xi_d) : \xi_i^4 \geq \frac{1}{d+1} \right\}.$$

According to this decomposition we write

$$\mathcal{J}_{\infty} \leq \sum_{i=0}^d \mathcal{J}_{\infty, i}, \quad (4.25)$$

where the terms $\mathcal{J}_{\infty, i}$ can be written as

$$\mathcal{J}_{\infty, i} := \int_{\Lambda_i} \frac{d\lambda d\xi}{(\lambda^2 + \xi_1^4 + \dots + \xi_d^4)^{1/2}} |\mathcal{F}\rho(\lambda, \xi)|^2 \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi). \quad (4.26)$$

Let us now show how to bound $\mathcal{J}_{\infty, 0}$ above. To this aim we invoke our bound (4.16) in two different ways. Namely we take $\tau_0 = 1$, and $\tau_i = 0$ if $|\xi_i| \leq 1$, while $\tau_i = 1$ if $|\xi_i| \geq 1$.

Together with the trivial inequality $\lambda^2 + \sum_{i=1}^d \xi_i^4 \geq \lambda^2$, the term $\mathcal{J}_{\infty,0}$ given in (4.26) can be bounded as follows

$$\mathcal{J}_{\infty,0} \lesssim \left(\int_{\lambda^2 \geq \frac{1}{d+1}} \frac{d\lambda}{|\lambda|^{2H_0+2}} \right) \prod_{i=1}^d \left\{ \int_{|\xi_i| \leq 1} \frac{d\xi_i}{|\xi_i|^{2H_i-1}} + \int_{|\xi_i| \geq 1} \frac{d\xi_i}{|\xi_i|^{2H_i+1}} \right\} < \infty, \quad (4.27)$$

where the last inequality is immediate. The terms $\mathcal{J}_{\infty,i}$ for $i = 1, \dots, d$ in (4.26) are handled similarly, and we omit the details for the sake of conciseness. Taking into account the upper bound (4.25), we end up with the relation $\mathcal{J}_{\infty} < \infty$.

We now turn to a bound on $\mathcal{J}_0$ defined by (4.24), for which we invoke (4.16) with $\tau_i = 0$, for all $i = 0, \dots, d$. We get

$$\mathcal{J}_0 \lesssim \int_{\mathcal{D}_s \cap \mathbb{R}_+^{d+1}} \frac{d\lambda d\xi}{(\lambda^2 + \xi_1^4 + \dots + \xi_d^4)^{1/2}} \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi). \quad (4.28)$$

To see that the latter integral is indeed finite, let us set $\tilde{\xi}_i := \xi_i^2$, so that $(\lambda, \xi_1, \dots, \xi_d) \in \mathcal{D}_s \cap \mathbb{R}_+^{d+1}$ if and only if $(\lambda, \tilde{\xi}_1, \dots, \tilde{\xi}_d) \in \mathcal{B}(0, 1) \cap \mathbb{R}_+^{d+1}$, where $\mathcal{B}(0, 1)$ stands for the standard Euclidean unit ball. This yields

$$\mathcal{J}_0 \lesssim \int_{\mathcal{B}(0,1) \cap \mathbb{R}_+^{d+1}} \frac{d\lambda d\xi}{(\lambda^2 + \tilde{\xi}_1^2 + \dots + \tilde{\xi}_d^2)^{1/2}} \frac{1}{|\lambda|^{2H_0-1}} \prod_{i=1}^d \frac{1}{|\tilde{\xi}_i|^{H_i}} \lesssim \int_0^1 \frac{dr}{r^{2H_0+H-d}},$$

where we have used spherical coordinates to derive the last inequality. The finiteness of $\mathcal{J}_0$ now follows from the assumption $2H_0 + H < d + 1$.

Summarizing our computations, we have seen that $\mathcal{J}_0 < \infty$ and $\mathcal{J}_{\infty} < \infty$. Recalling relation (4.22), this proves our claim $\mathcal{J}_{\rho, H_0, \mathbf{H}} < \infty$. $\square$

With Lemma 4.14 in hand, we introduce a constant $\mathbf{c}_{\rho, H_0, \mathbf{H}}$ which will feature prominently in the sequel. Namely we set

$$\mathbf{c}_{\rho, H_0, \mathbf{H}} := c_{H_0, \mathbf{H}}^2 \mathcal{J}_{\rho, H_0, \mathbf{H}}, \quad \text{with } c_{H_0, \mathbf{H}} = \left(\prod_{i=0}^d \alpha_{H_i} \right)^{-1/2} \text{ and } \alpha_{H_i} := \int_{\mathbb{R}} d\xi \frac{|e^{i\xi} - 1|^2}{|\xi|^{2H_i+1}}, \quad (4.29)$$

where we recall that $\mathcal{J}_{\rho, H_0, \mathbf{H}}$ is defined by (4.20). We are now ready to state the result giving the construction of a K -rough path above our noise $\dot{W}$.

Theorem 4.15. *Let ρ be a mollifier satisfying Assumption (ρ) . Consider Hurst parameters $H_0 \in (0, 1)$ and $\mathbf{H} = (H_1, \dots, H_d) \in (0, 1)^d$. Define*

$$\tilde{J}_* := \left\{ j \in \{1, \dots, d\} : H_j \leq \frac{1}{2} \right\}. \quad (4.30)$$

We strengthen condition (4.18) in the following way:

$$d + \frac{1}{2} < 2H_0 + H < d + 1, \quad \text{and} \quad |\tilde{J}_*| \leq 1, \quad (4.31)$$

where we recall that H is given by (1.6). In this setting, fix $\alpha \in \mathbb{R}$ such that

$$-\frac{3}{2} < \alpha < -(d+2) + 2H_0 + H. \quad (4.32)$$

For $n \geq 1$, define $\dot{W}^n$ as in (4.15) and set

$$\widehat{W}^n := \text{Renorm}(\mathbf{W}^n, \mathbf{c}_{\rho, H_0, \mathbf{H}}^{(n)}), \quad \text{with } \mathbf{c}_{\rho, H_0, \mathbf{H}}^{(n)} = \mathbf{c}_{\rho, H_0, \mathbf{H}} \cdot 2^{2n(d+1-(2H_0+H))}, \quad (4.33)$$

where the operator *Renorm* is introduced in (4.12) and where the constant $\mathbf{c}_{\rho, H_0, \mathbf{H}}$ is given by (4.29). Then for any weight $w(x) := (1 + |x|)^\kappa$ with $\kappa > 0$ and for the distance $d_{\alpha, w}$ given by (4.10), there exists an (α, K) -rough path $\widehat{W}$ such that almost surely

$$\lim_{n \rightarrow \infty} d_{\alpha, w}(\widehat{W}^n, \widehat{W}) = 0. \quad (4.34)$$

Proof. For the sake of clarity, we have postponed the (technical) proof of this theorem to Appendix A. $\square$

Remark 4.16. Notice that the set $\tilde{J}_*$ in (4.30) is slightly different from the set J_* in (1.6). Indeed, $\tilde{J}_*$ involves the condition $H_i \leq 1/2$, as opposed to $H_j < 1/2$ for J_* . This difference is due to the fact that J_* is involved in Skorohod type computations, as opposed to the current Stratonovich case.

Remark 4.17. Observe that for the sake of completeness we have considered a general $H_0 \in (0, 1)$ in Theorem 4.15, while the Skorohod Section 3 was limited to the case $H_0 > \frac{1}{2}$.

By combining the (deterministic) result of Theorem 4.11 with the (stochastic) construction of Theorem 4.15, we derive the desired Stratonovich interpretation of equation (1.1):

Definition 4.18. Let ρ be the weight given by $\rho(s, x) := p_1(s)p_1(x)$ as considered in Remark 4.13. Let $(H_0, \mathbf{H}) \in (0, 1)^{d+1}$ be a vector of Hurst parameters such that

$$d + \frac{2}{3} < 2H_0 + H < d + 1 \quad \text{and} \quad |\tilde{J}_*| \leq 1, \quad (4.35)$$

where we recall that the set $\tilde{J}_*$ is introduced in (4.30). Besides, fix $\alpha \in \mathbb{R}$ such that

$$-\frac{4}{3} < \alpha < -(d+2) + 2H_0 + H,$$

as well as an arbitrary time horizon $T > 0$ and an initial condition $\psi \in L^\infty(\mathbb{R}^d)$. Then, using the notations of Theorem 4.11 and Theorem 4.15, we call $u := \Phi_{\alpha, w_1, w_2}^{K, T}(\widehat{W}, \psi)$ the renormalized Stratonovich solution of equation (1.1), with initial condition ψ . In particular, u is the (almost sure) limit, in $L^\infty([0, T] \times \mathbb{R}^d)$, of the sequence $(u^n)_{n \geq 1}$ of classical solutions of the equation

$$\begin{cases} \partial_t u^n = \frac{1}{2} \Delta u^n + u^n \dot{W}^n - \mathbf{c}_{\rho, H_0, \mathbf{H}}^{(n)} u^n, & t \in [0, T], x \in \mathbb{R}^d, \\ u_0^n(x) = \psi(x). \end{cases} \quad (4.36)$$

Remark 4.19. Observe that the assumptions on $H_0, \mathbf{H}$ in (4.35) are more restrictive than those in Theorem 4.15. This restriction thus stems from Theorem 4.11, which requires us to pick $\alpha > -\frac{4}{3}$, and not only $\alpha > -\frac{3}{2}$.

As the reader might expect it, the extension of the result of Theorem 4.11 to any $\alpha > -\frac{3}{2}$ is actually possible, at the price of additional “third-order” elements (on top of χ and χ^2) in the definition of a K -rough path (see [16, Definition 2.7] for details when $d = 1$). Therefore, applying this extension to our stochastic model requires to construct additional “third-order”

elements above the fractional noise. This strategy has been implemented in [16] for $d = 1$, and when working with the “compact-in-space” topologies derived from the analysis of (4.1). We firmly believe that the constructions of [16] could be extended to the current setting, that is to any dimension $d \geq 1$ and to the whole space $\mathbb{R}^d$. Part of the analysis might in fact be contained in the general renormalization procedure put forward in [3]. However, given the high sophistication of the elements at stake (see the technical computations in Appendix A), we have here preferred to stick to a second-order analysis valid for any $d \geq 1$ for the sake of conciseness. Higher order expansions are deferred to a future publication.

Remark 4.20. Observe that the first assumption $d + \frac{2}{3} < 2H_0 + H < d + 1$ in (4.35) entails that $|\mathbf{J}_*| \leq 2$, and so the second assumption in (4.35) actually reduces to $|\mathbf{J}_*| \neq 2$. This last condition is still restrictive though, since we may have $d + \frac{2}{3} < 2H_0 + H < d + 1$ and $|\mathbf{J}_*| = 2$ (take $d = 2$, $H_0 = 1 - \varepsilon$, $H_1 = \frac{1}{2} - \varepsilon$, $H_2 = \frac{1}{2} - \varepsilon$, for $\varepsilon > 0$ small enough).

5. MOMENTS ESTIMATES FOR THE STRATONOVICH EQUATION

Similarly to the Skorohod situation (see Section 3), we now would like to show that the moments of the renormalized Stratonovich solution u (as introduced in Definition 4.18) are all finite in the subcritical regime (1.7). With this objective in mind, observe first that the continuity property (iv) of Theorem 4.11 only guarantees *almost sure* convergence of u^n to u , and does not provide any estimate on the moments of u . Therefore we only get an almost convergence as well in Definition 4.18. In order to go further, our strategy will somehow consist in bounding the moments of the Stratonovich solution u in terms of those of the Skorohod solution $u^\diamond$ (at least at the level of their respective approximations), so as to exploit the estimates of Section 3 for the moments of $u^\diamond$.

Although both Proposition 3.8 and Definition 4.18 are valid for $H_0 \leq \frac{1}{2}$ (see [9] for the Skorohod case), for more clarity we will restrict our analysis to the case where $H_0 > \frac{1}{2}$ in the sequel. Besides, for our comparison strategy to be possible, we of course need to focus on situations where both u and $u^\diamond$ are well-defined. It is easy to see, by combining the restrictions in (1.7) and (4.35), that this intersection domain can be described through the three conditions

$$H > d - 1 \quad , \quad d + \frac{2}{3} < 2H_0 + H < d + 1 \quad \text{and} \quad |\tilde{\mathbf{J}}_*| \leq 1, \quad (5.1)$$

where we recall that $\tilde{\mathbf{J}}_* := \{j \in \{1, \dots, d\} : H_j \leq \frac{1}{2}\}$. We assume that the condition (5.1) is met in the remainder of the section.

As mentioned above, assuming condition (5.1) is satisfied, the idea will be to compare $u^\diamond$ and u through the Feynman-Kac representations of their respective approximations. Just as in Section 3 (resp. Section 4) we denote by $u^{n,\diamond}$ (resp. u^n) the approximation of $u^\diamond$ (resp. u). Namely $u^{n,\diamond}$ is the solution of (3.12), while u^n solves (4.36), for every fixed $n \geq 1$. At this point, let us recall that according to [23, Proposition 5.2] (and just as in Proposition 3.6), the Feynman-Kac representation of $u^{n,\diamond}$ is given by

$$u_t^{n,\diamond}(x) = \mathbb{E} \left[\psi(B_t^x) \exp \left(V_t^{n,W}(B^x) - \frac{1}{2} \beta_t^n(B^x) \right) \right], \quad (5.2)$$

where B^x is a standard d -dimensional Wiener process starting at x , independent of W , and where $V_t^{n,W}, \beta_t^n$ are given by

$$V_t^{n,W}(B^x) := \int_0^t du \dot{W}^n(u, B_{t-u}^x), \quad \text{and} \quad \beta_t^n(B^x) = \beta_t^{n,B^x} := \mathbb{E}_W[|V_t^{n,W}(B^x)|^2].$$

On the other hand, it is clear that for every fixed $n \geq 1$ equation (4.36) is a standard linear parabolic equation, for which a classical Feynman-Kac formula can be applied. This yields the expression

$$u_t^n(x) = \mathbb{E} \left[\psi(B_t^x) \exp \left(V_t^{n,W}(B^x) - \mathfrak{c}_{\rho, H_0, \mathbf{H}}^{(n)} \cdot t \right) \right], \quad (5.3)$$

where $\mathfrak{c}_{\rho, H_0, \mathbf{H}}^{(n)}$ stands for the renormalization constant in equation (4.36).

Based on the two representations (5.2) and (5.3), we deduce that the desired comparison between $u^{n,\diamond}$ and u^n morally reduces to a comparison between $\frac{1}{2}\beta_t^n(B^x)$ and $\mathfrak{c}_{\rho, H_0, \mathbf{H}}^{(n)} \cdot t$. The following decomposition will thus be an important step in the procedure:

Proposition 5.1. *Assume that $H_0 > \frac{1}{2}$ and that the conditions in (5.1) are satisfied. Let $\mathfrak{c}_{\rho, H_0, \mathbf{H}}^{(n)}$ be the constant defined by (4.33). Then for every fixed $t \geq 0$, it holds that*

$$\mathfrak{c}_{\rho, H_0, \mathbf{H}}^{(n)} t = \frac{1}{2} \mathbb{E}[\beta_t^n(B^x)] + r_t^n, \quad \text{with} \quad r_t^n \xrightarrow{n \rightarrow \infty} a_{H_0, \mathbf{H}} t^{2H_0 + H - d}, \quad (5.4)$$

for some constant $a_{H_0, \mathbf{H}} > 0$.

Proof. For the sake of clarity, let us first recall that $\mathfrak{c}_{\rho, H_0, \mathbf{H}}^{(n)}$ is defined as

$$\begin{aligned} \mathfrak{c}_{\rho, H_0, \mathbf{H}}^{(n)} &:= \mathfrak{c}_{\rho, H_0, \mathbf{H}} \cdot 2^{2n(d+1-(2H_0+H))} \\ &= c_{H_0, \mathbf{H}}^2 \int_{\mathbb{R}^{d+1}} d\lambda d\xi |\mathcal{F}\rho(2^{2n}\lambda, 2^n\xi)|^2 \mathcal{F}p(\lambda, \xi) \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi), \end{aligned} \quad (5.5)$$

where $\mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi)$ is the quantity defined in (4.19). Keeping this expression in mind, and using the subsequent covariance formula (A.1), we can recast the expression (3.10) for $\beta_t^n(B^x)$ as

$$\begin{aligned} \beta_t^n(B^x) &= \mathbf{E}[|V_t^{n,W}(B^x)|^2] \\ &= c_{H_0, \mathbf{H}}^2 \int_{\mathbb{R}^{d+1}} d\lambda d\xi |\mathcal{F}\rho_n(\lambda, \xi)|^2 \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) \int_{[0,t]^2} dudv e^{i(\lambda(v-u) + \xi \cdot (B_u - B_v))}. \end{aligned}$$

Therefore involving elementary symmetry and integration arguments, we get

$$\begin{aligned} &\frac{1}{2} \mathbb{E}[\beta_t^n(B^x)] \\ &= c_{H_0, \mathbf{H}}^2 \int_{\mathbb{R}^{d+1}} d\lambda d\xi |\mathcal{F}\rho_n(\lambda, \xi)|^2 \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) \int_0^t du \int_0^u dv e^{-i\lambda(u-v)} e^{-\frac{|\xi|^2}{2}(u-v)} \\ &= c_{H_0, \mathbf{H}}^2 \int_{\mathbb{R}^{d+1}} d\lambda d\xi |\mathcal{F}\rho_n(\lambda, \xi)|^2 \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) \int_0^t dv (t-v) e^{-v(\frac{|\xi|^2}{2} + i\lambda)} \\ &= c_{H_0, \mathbf{H}}^2 \int_{\mathbb{R}^{d+1}} d\lambda d\xi |\mathcal{F}\rho_n(\lambda, \xi)|^2 \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) \left[\frac{t}{\frac{|\xi|^2}{2} + i\lambda} - \frac{1}{\frac{|\xi|^2}{2} + i\lambda} \int_0^t dv e^{-v(\frac{|\xi|^2}{2} + i\lambda)} \right]. \end{aligned}$$

Hence recalling (4.21) and (5.5) we get

$$\frac{1}{2}\mathbb{E}[\beta_t^n(B^x)] = \mathfrak{c}_{\rho, H_0, \mathbf{H}}^{(n)} t - r_t^n, \quad (5.6)$$

with a constant r_t^n defined by

$$r_t^n := 2c_{H_0, \mathbf{H}}^2 \int_{\mathbb{R}^{d+1}} d\lambda d\xi |\mathcal{F}\rho_n(\lambda, \xi)|^2 \frac{\mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi)}{|\xi|^2 + 2\iota\lambda} \int_0^t dv e^{-v(\frac{|\xi|^2}{2} + \iota\lambda)}. \quad (5.7)$$

Using the assumption $d + \frac{2}{3} < 2H_0 + H < d + 1$, it is not hard to see that

$$\mathcal{A}_t \equiv \int_{\mathbb{R}^{d+1}} d\lambda d\xi \frac{\mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi)}{||\xi|^2 + 2\iota\lambda|} \left| \int_0^t dv e^{-v(\frac{|\xi|^2}{2} + \iota\lambda)} \right| < \infty. \quad (5.8)$$

Indeed, one can decompose $\mathcal{A}_t$ similarly to the term $\mathcal{J}_{\rho, H_0, \mathbf{H}}$ in (4.22). The arguments in order to bound the integral on a ball (corresponding to $\mathcal{J}_0$ in (4.22)) are very similar to the ones in the proof of Lemma 4.14. Let us here only focus on the integration in the unbounded region $\{|\lambda| \geq 1, |\xi_1| \geq 1, \dots, |\xi_d| \geq 1\}$ in the right hand side of (5.8). In this case, for all $\tau_0, \tau_1, \dots, \tau_d \in [0, 1]$ such that $\sum_{i=0}^d \tau_i = 1$, and recalling again the expression (4.19) for $\mathcal{N}_{H_0, \mathbf{H}}$, we have

$$\begin{aligned} & \int_{|\lambda| \geq 1, |\xi_1| \geq 1, \dots, |\xi_d| \geq 1} d\lambda d\xi \frac{\mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi)}{||\xi|^2 + 2\iota\lambda|} \left| \int_0^t dv e^{-v(\frac{|\xi|^2}{2} + \iota\lambda)} \right| \\ & \lesssim \int_{|\lambda| \geq 1, |\xi_1| \geq 1, \dots, |\xi_d| \geq 1} d\lambda d\xi \frac{\mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi)}{\lambda^2 + \xi_1^4 + \dots + \xi_d^4} \\ & \lesssim \left(\int_{|\lambda| \geq 1} \frac{d\lambda}{|\lambda|^{2H_0 + 2\tau_0 - 1}} \right) \prod_{i=1}^d \int_{|\xi_i| \geq 1} \frac{d\xi_i}{|\xi_i|^{2H_i + 4\tau_i - 1}}. \end{aligned} \quad (5.9)$$

We can now pick the τ_i 's such that $2H_0 + 2\tau_0 > 2$ and $2H_i + 4\tau_i > 2$ for $i = 1, \dots, d$, due to the fact that we have

$$2(2H_0 + 2\tau_0) + \sum_{i=1}^d (2H_i + 4\tau_i) = 2(2H_0 + H) + 4 > 4 + 2d,$$

where the last inequality is ensured by (5.1). Choosing this set of parameters in (5.9), it is thus readily checked that $\mathcal{A}_t < \infty$. This proves relation (5.8).

Let us summarize our considerations up to this point: thanks to relation (5.6), our claim (5.4) amounts to prove that $\lim_n r_t^n = a_{H_0, \mathbf{H}} t^{2H_0 + H - d}$, where r_t^n is defined by (5.7). Moreover, relation (5.8) enables a standard application of the dominated convergence theorem in (5.7). Resorting to an additional elementary scaling argument, we obtain $\lim_n r_t^n = r_t$, where

$$r_t := 2c_{H_0, \mathbf{H}}^2 \tilde{\mathcal{A}}_1 t^{2H_0 + H - d}, \quad \text{with} \quad \tilde{\mathcal{A}}_1 = \int_{\mathbb{R}^{d+1}} d\lambda d\xi \frac{\mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi)}{|\xi|^2 + 2\iota\lambda} \int_0^1 dv e^{-v(\frac{|\xi|^2}{2} + \iota\lambda)}. \quad (5.10)$$

This achieves the proof of (5.4).

It only remains us to check that the above quantity $\tilde{\mathcal{A}}_1$ is strictly positive. To this end, we can first write

$$\tilde{\mathcal{A}}_1 = \int_{\mathbb{R}^d} d\xi \int_0^1 dv \left(\prod_{i=1}^d \frac{e^{-v \frac{|\xi_i|^2}{2}}}{|\xi_i|^{2H_i-1}} \right) \left(\int_{\mathbb{R}} \frac{d\lambda}{|\lambda|^{2H_0-1}} \frac{e^{-v\lambda}}{|\xi|^2 + 2i\lambda} \right).$$

Next, since $H_0 > \frac{1}{2}$, a standard inverse Fourier transform procedure allows to assert that there exists a constant $d_{H_0} > 0$ satisfying, for all fixed $\xi \in \mathbb{R}_*^d$, and $v \in (0, 1)$,

$$\int_{\mathbb{R}} \frac{d\lambda}{|\lambda|^{2H_0-1}} \frac{e^{-v\lambda}}{|\xi|^2 + 2i\lambda} = d_{H_0} \int_0^\infty \frac{ds}{|v-s|^{2-2H_0}} e^{-\frac{|\xi|^2}{2}s} > 0.$$

This is enough to show that $\tilde{\mathcal{A}}_1 > 0$, which finishes our proof. $\square$

Based on relations (5.2), (5.3) and (5.4), the fact that one can transfer subcritical estimates from the Skorohod to the Stratonovich equation can be explained in the following way: under the subcritical condition (5.1), the fluctuations of the random variable Y_t^n given by

$$Y_t^n := \beta_t^n(B^x) - \mathbb{E}[\beta_t^n(B^x)] \quad (5.11)$$

are much smaller than the fluctuation of the random variable $V_t^{n,W}(B)$ featuring in (5.2) and (5.3). This assertion is quantified in the following lemma.

Lemma 5.2. *Assume that $H_0 > \frac{1}{2}$ and that the conditions in (5.1) are satisfied. Let Y_t^n be the random variables defined by (5.11). Then for all $\theta \geq 0$ and $t \geq 0$, we have*

$$\sup_{n \geq 1} \mathbb{E}[e^{\theta Y_t^n}] < \infty. \quad (5.12)$$

Proof. We first fix $n \geq 1$. Our proof is inspired by the computations in [22, Theorem 4.6], where the exponential integrability of a random variable similar to Y_t^n is investigated under more restrictive assumptions on $H_1, \dots, H_d$. As in [22], our analysis will be based on Le Gall's decomposition for β_t^n . More specifically, we start the following construction: for $N \geq 1$ and $k = 1, \dots, 2^{N-1}$ we set

$$J_{N,k} = \left[\frac{(2k-2)t}{2^N}, \frac{(2k-1)t}{2^N} \right), I_{N,k} = \left[\frac{(2k-1)t}{2^N}, \frac{2kt}{2^N} \right), \text{ and } A_{N,k} = J_{N,k} \times I_{N,k}.$$

Then we can decompose $\beta_t^n(B^x)$ as

$$\beta_t^n(B^x) = \sum_{N=1}^{\infty} \sum_{k=1}^{2^{N-1}} a_{N,k}^n, \text{ where } a_{N,k}^n = \int_{A_{N,k}} \gamma_0^n(r-s) \gamma^n(B_r^x - B_s^x) dr ds,$$

where γ_0^n (resp. γ^n) is the inverse Fourier transform of the measure $\mu_0^n := \mu_0^{2^{-4n}}$ (resp. $\mu^n := \mu^{2^{-2n}}$) defined by (3.7). Furthermore, owing to the shape of $A_{N,k}$'s combined with the independence of the Brownian increments, one can write

$$a_{N,k}^n \stackrel{(d)}{=} a_N^n, \quad \text{with} \quad a_N^n = \int_{[0, t/2^N]^2} \gamma_0^n(t+s) \gamma^n(B_r - \tilde{B}_s) ds dr, \quad (5.13)$$

where B and $\tilde{B}$ are two independent Brownian motions starting from the origin. It can be shown, exactly along the same lines as in [22, equation (4.13)], that the exponential

integrability of Y_t^n amounts to the following estimate: for any arbitrarily small $\varepsilon > 0$ there exists a constant $C_\varepsilon > 0$ such that for all $m \geq 1$ we have

$$\mathbb{E}[(a_N^n)^m] \leq C_\varepsilon m! \left(\frac{C\varepsilon}{2^N} \right)^{\alpha m}, \quad (5.14)$$

with $\alpha > 1/2$. In fact, we should point out that the exponential integrability in [22] was obtained thanks to (5.14) with $\alpha = 1$. However, by a closer examination of the proof in [22], it is clear that the exponential integrability (5.12) follows as long as $\alpha > 1/2$ in (5.14). The remainder of our proof is thus devoted to justify (5.14).

Recall from (3.7) that μ_0^n and μ^n are dominated by μ_0 and μ respectively, where μ_0 and μ are introduced in (1.5). Therefore for any $\tau > 0$ we have

$$\begin{aligned} & \mathbb{E} \left[\int_{[0,\tau]^2} \gamma_0^n(t+s) \gamma^n(B_r - \tilde{B}_s) ds dr \right]^m \\ &= \int_{(\mathbb{R}^{d+1})^m} \mu_0^n(d\lambda) \mu^n(d\xi) \left| \int_{[0,\tau]^m} \left(\mathbb{E} \prod_{k=1}^m e^{i(\lambda_k s_k + \xi_k \cdot B_{s_k})} \right) ds \right|^2 \\ &\leq \int_{(\mathbb{R}^{d+1})^m} \mu_0(d\lambda) \mu(d\xi) \left| \int_{[0,\tau]^m} \left(\mathbb{E} \prod_{k=1}^m e^{i(\lambda_k s_k + \xi_k \cdot B_{s_k})} \right) ds \right|^2 \\ &= \mathbb{E} \left[\int_{[0,\tau]^2} \gamma_0(t+s) \gamma(B_r - \tilde{B}_s) ds dr \right]^m. \end{aligned} \quad (5.15)$$

On the other hand, by [8, Inequality (3.1)], we have

$$\mathbb{E} \left[\int_{[0,\tau]^2} \gamma_0(t-s) \gamma(B_r - \tilde{B}_s) ds dr \right]^m \leq C^m (m!)^{d-H} \tau^{(2H_0+H-d)m}, \quad (5.16)$$

where $C > 0$ is a constant independent of $\tau > 0$. Now combining (5.13), (5.15) and (5.16), together with the trivial relation that $\gamma_0(t+s) \leq \gamma_0(t-s)$, we have for all $m \geq 1$,

$$\mathbb{E}[a_N^n]^m \leq C^m (m!)^{d-H} \left(\frac{t}{2^N} \right)^{(2H_0+H-d)m}.$$

Note that the above control of moments of a_N^n is uniform in n . Moreover, under our assumption $d - H < 1$, we can conclude that for any $\varepsilon > 0$ the above becomes

$$\mathbb{E}[a_N^n]^m \leq C^m (m!) \left(\frac{\varepsilon t}{2^N} \right)^{(2H_0+H-d)m},$$

when m is large enough. Finally, observe that we assume $2H_0 + H - d > 1/2$ (cf (5.1)). Condition (5.14) is therefore satisfied, and the integrability of Y_t^n (uniformly in n) follows. This finishes the proof of (5.12). $\square$

We are finally in a position to prove the main result of this section, that is the fact that the Stratonovich solution admits finite moments of any order.

Proposition 5.3. *Assume that $H_0 > \frac{1}{2}$ and that the conditions in (5.1) are satisfied. Let u be the solution of the Stratonovich equation as given by Definition (4.18), with initial condition $\psi \equiv 1$. Then for all $t \geq 0$ and $p \geq 1$ we have*

$$\mathbf{E}[|u_t(x)|^p] = c_{p,t} < \infty.$$

Proof. Recall from (5.2) and (5.3) that

$$u_t^n(x) = \mathbb{E} \left[\exp \left(V_t^{n,W}(B^x) - \mathbf{c}_{\rho, H_0, \mathbf{H}}^{(n)} \cdot t \right) \right],$$

and

$$u_t^{n,\diamond}(x) = \mathbb{E} \left[\exp \left(V_t^{n,W}(B^x) - \frac{1}{2} \beta_t^n(B^x) \right) \right].$$

Without loss of generality, we assume that p is an integer in the rest of the proof. We therefore write

$$u_t^n(x) = \mathbb{E} \left[e^{V_t^{n,W}(B^x) - \frac{1}{2} \beta_t^n(B^x)} \cdot e^{\frac{1}{2}(\beta_t^n(B^x) - \mathbb{E}[\beta_t^n(B^x)])} \right] \cdot e^{\frac{1}{2} \mathbb{E}[\beta_t^n(B^x)] - \mathbf{c}_{\rho, H_0, \mathbf{H}}^{(n)} \cdot t}. \quad (5.17)$$

Taking expectation in (5.17), we end up with

$$\begin{aligned} \mathbf{E}[(u_t^n(x))^p] &= \mathbf{E} \mathbb{E} \left[\prod_{k=1}^p e^{V_t^{n,W}(B^{k,x}) - \frac{1}{2} \beta_t^n(B^{k,x})} \cdot \prod_{k=1}^p e^{\frac{1}{2}(\beta_t^n(B^{k,x}) - \mathbb{E}[\beta_t^n(B^{k,x})])} \right] \\ &\quad \times \prod_{k=1}^p e^{\frac{1}{2} \mathbb{E}[\beta_t^n(B^{k,x})] - \mathbf{c}_{\rho, H_0, \mathbf{H}}^{(n)} \cdot t}, \end{aligned} \quad (5.18)$$

where $\{B^{k,x}; k = 1, \dots, p\}$ is a family of independent Brownian motions which is also independent from the noise W . Moreover, a standard application of Fubini's theorem in (5.18) yields

$$\begin{aligned} \mathbf{E}[(u_t^n(x))^p] &= \mathbb{E} \left[\mathbf{E} \left[\prod_{k=1}^p e^{V_t^{n,W}(B^{k,x}) - \frac{1}{2} \beta_t^n(B^{k,x})} \cdot \prod_{k=1}^p e^{\frac{1}{2}(\beta_t^n(B^{k,x}) - \mathbb{E}[\beta_t^n(B^{k,x})])} \right] \right] \\ &\quad \times \prod_{k=1}^p e^{\frac{1}{2} \mathbb{E}[\beta_t^n(B^{k,x})] - \mathbf{c}_{\rho, H_0, \mathbf{H}}^{(n)} \cdot t}. \end{aligned} \quad (5.19)$$

Now, notice that

$$\mathbf{E} \left[\prod_{k=1}^p e^{V_t^{n,W}(B^{k,x}) - \frac{1}{2} \beta_t^n(B^{k,x})} \right] = \exp \left(\sum_{1 \leq j_1 < j_2 \leq p} \alpha_t^{j_1 j_2; n} \right), \quad (5.20)$$

where $\alpha_t^{j_1 j_2; n}$ is defined the same way as in (3.19) but with γ, γ_0 replaced by γ^n, γ_0^n . Plugging (5.20) into (5.19), we have

$$\mathbf{E}[(u_t^n(x))^p] = \mathbb{E} \left[e^{\sum_{1 \leq j_1 < j_2 \leq p} \alpha_t^{j_1 j_2; n}} \prod_{k=1}^p e^{\frac{1}{2}(\beta_t^n(B^{k,x}) - \mathbb{E}[\beta_t^n(B^{k,x})])} \right] \prod_{k=1}^p e^{\frac{1}{2} \mathbb{E}[\beta_t^n(B^{k,x})] - \mathbf{c}_{\rho, H_0, \mathbf{H}}^{(n)} \cdot t}.$$

On the other hand, invoking similar considerations as for (5.15), we get that for any $\lambda > 0$

$$\mathbb{E}[\exp(\lambda \alpha_t^{j_1 j_2; n})] \leq \mathbb{E}[\exp(\lambda \alpha_t^{j_1 j_2})].$$

The rest of the proof then follows from Proposition 3.9, Proposition 5.1, Lemma 5.2 and a simple application of Fatou's lemma and Hölder's inequality. $\square$

APPENDIX A. PROOF OF THEOREM 4.15

This section is devoted to the proof of Theorem 4.15, that is to the construction of the (α, K) -rough path $\widehat{\mathbf{W}}$ at the basis of the Stratonovich interpretation of the model (along Definition 4.18).

Therefore, from now on and for the rest of the section, we fix a mollifier ρ , some Hurst indexes $H_0, \mathbf{H}$, and a parameter α such that the assumptions in Theorem 4.15 are all met.

We recall that the convenient notation $\mathcal{N}_{H_0, \mathbf{H}}$ has been introduced in (4.19). Let us also label the following covariance formulas (which immediately generalize (3.7)) for further reference:

Lemma A.1. *Let $\dot{W}^n$ be the smoothed noise defined by (4.15) and recall that the kernel K is defined by (4.9). For every fixed $n \geq 1$, the families $\{\dot{W}^n(t, y); (t, y) \in \mathbb{R}^{d+1}\}$ and $\{K * \dot{W}^n(t, y); (t, y) \in \mathbb{R}^{d+1}\}$ are centered Gaussian processes with respective covariance functions given by the formulas*

$$\mathbb{E}[\dot{W}^n(t, y)\dot{W}^n(\tilde{t}, \tilde{y})] = c_{H_0, \mathbf{H}}^2 \int_{\mathbb{R}^{d+1}} d\lambda d\xi |\mathcal{F}\rho_n(\lambda, \xi)|^2 \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) e^{i(\lambda(t-\tilde{t}) + \xi \cdot (y-\tilde{y}))}, \quad (\text{A.1})$$

and

$$\begin{aligned} & \mathbb{E}[(K * \dot{W}^n)(t, y)(K * \dot{W}^n)(\tilde{t}, \tilde{y})] \\ &= c_{H_0, \mathbf{H}}^2 \int_{\mathbb{R}^{d+1}} d\lambda d\xi |\mathcal{F}\rho_n(\lambda, \xi)|^2 |\mathcal{F}K(\lambda, \xi)|^2 \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) e^{i(\lambda(t-\tilde{t}) + \xi \cdot (y-\tilde{y}))}, \end{aligned} \quad (\text{A.2})$$

where the constant $c_{H_0, \mathbf{H}}$ is the one defined in (4.29) and where the function $\mathcal{N}_{H_0, \mathbf{H}}$ is introduced in (4.19).

Just as in [15, Corollary 3.5], the proof of Theorem 4.15 essentially relies on suitable moments estimates (see Proposition A.4 and Proposition A.8 below). The transition from these estimates to the desired convergence property will then go through the following multiparametric and distributional version of the Garsia-Rodemich-Rumsey Lemma. Observe that this kind of property is one of the key technical ingredients in the theory of regularity structures.

Lemma A.2 (Multiparametric G-R-R lemma). *Fix a regularity parameter β sitting in $(-(d+1), 0)$, as well as a weight w on $\mathbb{R}^d$. Then there exists a finite set Ψ of functions in $\mathcal{C}^{2(d+1)}(\mathbb{R}^{d+1})$ with support in $\mathcal{B}_s(0, 1)$ such that the following property holds true: assume that $\zeta : \mathbb{R}^{d+1} \rightarrow \mathcal{D}'_{2(d+1)}(\mathbb{R}^{d+1})$ is a map with increments of the form*

$$\zeta_{s,x} - \zeta_{t,y} = \sum_{i=1}^r [\theta^i(s, x) - \theta^i(t, y)] \cdot \zeta_{t,y}^{\sharp, i},$$

for some $\theta^i \in \mathcal{C}_w^\mu(\mathbb{R}^{d+1})$ with $\mu \in [0, \min(1, -\beta))$, and some distributions $\zeta_w^{\sharp, i} \in \mathcal{C}_w^\beta(\mathbb{R}^{d+1})$, where we recall that the spaces $\mathcal{C}_w^\beta$ are introduced in Definition 4.3. Then for every $T > 0$,

one has

$$\begin{aligned} & \|\zeta\|_{\beta+\mu;T,w^2} \\ & \lesssim \sup_{\psi \in \Psi} \sup_{n \geq 0} \sup_{(s,x) \in \Lambda_s^n \cap ([-(T+2), T+2] \times \mathbb{R}^d)} 2^{n(\beta+\mu)} \frac{|\langle \zeta_{s,x}, \psi_{s,x}^n \rangle|}{w(x)^2} + \sum_{i=1, \dots, r} \|\theta^i\|_{\mu;T+2,w} \|\zeta^{\sharp,i}\|_{\beta;T+2,w}, \end{aligned} \quad (\text{A.3})$$

where the discrete set Λ_s^n is defined by $\Lambda_s^n := \{(2^{-2n}k_0, 2^{-n}k_1, \dots, 2^{-n}k_d); k_0, k_1, \dots, k_d \in \mathbb{Z}\}$, and where norms for θ^i and $\zeta^{\sharp,i}$ are respectively given in Definition 4.2 and 4.3. For the sake of clarity, we have also used the standard notation $\psi_{s,x}^n := \mathcal{S}_{s,x}^{2^{-n}} \psi$ in the right-hand side of (A.3).

Proof. It is a mere “weighted” adaptation of the arguments of the proof of [15, Lemma 3.2] (which was itself an adaptation of the arguments in [19, Section 3]). For the sake of conciseness, we leave the details behind this slight adaptation as an exercise to the reader. $\square$

As a last preliminary step, we also label the following elementary property for further use:

Lemma A.3. Recall that the sets $\mathcal{B}_s^l$ are given by (4.7). Let ψ be a generic element of $\mathcal{B}_s^{2(d+1)}$ and for all $\alpha_0 \in (0, 1)$, $\alpha \in (0, 1)^d$, consider the function $\mathcal{N}_{\alpha_0, \alpha}$ introduced in (4.19). Then it holds that

$$\int_{\mathbb{R}^{d+1}} d\lambda d\xi \mathcal{N}_{\alpha_0, \alpha}(\lambda, \xi) |\mathcal{F}\psi(\lambda, \xi)| < \infty.$$

A.1. Moment estimate for the first component. In this section we will bound the covariance of $\dot{W}^n$ considered as an element of a space of the form $\mathcal{C}^\alpha$, where α satisfies (4.32).

Proposition A.4. For all $\ell \geq 0$, $n \geq m \geq 0$, $\psi \in \mathcal{B}_s^{2(d+1)}$ and $(s, x) \in \mathbb{R}^{d+1}$, it holds that

$$\mathbb{E}[|\dot{W}^n - \dot{W}^m, \psi_{s,x}^\ell|^2] \lesssim 2^{2\ell(d+2-(2H_0+H)+\varepsilon)} 2^{-m\varepsilon}, \quad (\text{A.4})$$

where the proportional constant in $\lesssim$ does not depend on n, m, ℓ, s, x .

Proof. According to the inner product definition in $\mathbb{R}^{d+1}$, we have

$$\mathbb{E}[\langle \dot{W}^n, \psi_{s,x}^\ell \rangle^2] = \int_{\mathbb{R}^{d+1} \times \mathbb{R}^{d+1}} dt dy d\tilde{t} d\tilde{y} \psi_{s,x}^\ell(t, y) \psi_{s,x}^\ell(\tilde{t}, \tilde{y}) \mathbb{E}[\dot{W}^n(t, y) \dot{W}^n(\tilde{t}, \tilde{y})].$$

Therefore using the covariance formula (A.1) together with the definition (2.1) of Fourier transform, we get

$$\begin{aligned} \mathbb{E}[\langle \dot{W}^n, \psi_{s,x}^\ell \rangle^2] &= c_{H_0, \mathbf{H}}^2 \int_{\mathbb{R}^{d+1} \times \mathbb{R}^{d+1}} dt dy d\tilde{t} d\tilde{y} \psi_{s,x}^\ell(t, y) \psi_{s,x}^\ell(\tilde{t}, \tilde{y}) \\ &\quad \times \int_{\mathbb{R}^{d+1}} d\lambda d\xi |\mathcal{F}\rho_n(\lambda, \xi)|^2 \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) e^{i(\lambda(t-\tilde{t})+\xi \cdot (y-\tilde{y}))} \\ &= c_{H_0, \mathbf{H}}^2 \int_{\mathbb{R}^{d+1}} d\lambda d\xi |\mathcal{F}\rho_n(\lambda, \xi)|^2 |\mathcal{F}\psi_{s,x}^\ell(\lambda, \xi)|^2 \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi). \end{aligned} \quad (\text{A.5})$$

We now recall that ρ_n is a rescaled version of the mollifier given by (4.15), and we have also set $\psi_{s,x}^n = \mathcal{S}_{s,x}^{2^{-n}} \psi$ in the right-hand side of (A.3). Hence we obtain

$$\begin{aligned} & \mathbb{E}[\langle \dot{W}^n, \psi_{s,x}^\ell \rangle^2] \\ &= c_{H_0, \mathbf{H}}^2 \int_{\mathbb{R}^{d+1}} d\lambda d\xi |\mathcal{F}\rho(2^{-2n}\lambda, 2^{-n}\xi)|^2 |\mathcal{F}\psi(2^{-2\ell}\lambda, 2^{-\ell}\xi)|^2 \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi). \end{aligned} \quad (\text{A.6})$$

We now perform the elementary change of variables $\lambda := 2^{-2\ell}\lambda$ and $\xi := 2^{-\ell}\xi$, which yields

$$\begin{aligned} & \mathbb{E}[\langle \dot{W}^n, \psi_{s,x}^\ell \rangle^2] \\ &= c_{H_0, \mathbf{H}}^2 2^{2\ell(d+2-(2H_0+H))} \int_{\mathbb{R}^{d+1}} d\lambda d\xi |\mathcal{F}\rho(2^{-2(n-\ell)}\lambda, 2^{-(n-\ell)}\xi)|^2 |\mathcal{F}\psi(\lambda, \xi)|^2 \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi). \end{aligned} \quad (\text{A.7})$$

Thanks to (4.16), applied with $\tau_0 = \dots = \tau_d = 0$, the Fourier transform of ρ is bounded. Hence we end up with

$$\mathbb{E}[\langle \dot{W}^n, \psi_{s,x}^\ell \rangle^2] \lesssim 2^{2\ell(d+2-(2H_0+H))} \int_{\mathbb{R}^{d+1}} d\lambda d\xi |\mathcal{F}\psi(\lambda, \xi)|^2 \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi). \quad (\text{A.8})$$

According to Lemma A.3 the latter integral is finite, which gives our claim (A.4) for $m = 0$. The general case $m \geq 0$ can then be derived along similar estimates, invoking the fact that $\mathcal{F}\rho$ is a Lipschitz function (see Assumption (ρ)). $\square$

A.2. Moment estimate for the second component. Let us start with two useful estimates on the Fourier transforms of the (fixed) components (K, R) in the decomposition of the heat kernel (see relation (4.8)).

Lemma A.5. *Let K be the localized heat kernel of Definition 4.6. For all fixed $a_0, a_1, \dots, a_d \in [0, 1]$ such that $\sum_{i=0}^d a_i < 1$, one has, for every $(\lambda, \xi) \in \mathbb{R}^{d+1}$,*

$$|\mathcal{F}K(\lambda, \xi)| \lesssim |\lambda|^{-a_0} \prod_{i=1}^d |\xi_i|^{-2a_i}.$$

Proof. Using the expansion of K in (4.9) and recalling the definition (4.6) of $\mathcal{S}_{s,x}^\delta$, we can first write

$$\mathcal{F}K(\lambda, \xi) = \sum_{\ell \geq 0} 2^{-2\ell} \mathcal{F}K_0(2^{-2\ell}\lambda, 2^{-\ell}\xi). \quad (\text{A.9})$$

Then, since K_0 is a smooth compactly-supported function, one has $|\mathcal{F}K_0(\lambda, \xi)| \lesssim |\lambda|^{-\tau_0}$ and $|\mathcal{F}K_0(\lambda, \xi)| \lesssim |\xi_i|^{-\tau_i}$ for all $\tau_0, \tau_1, \dots, \tau_d \geq 0$ and $(\lambda, \xi) \in \mathbb{R}^{d+1}$. Plugging this information into (A.9), we get

$$\begin{aligned} |\mathcal{F}K(\lambda, \xi)| &\leq \sum_{\ell \geq 0} 2^{-2\ell} |\mathcal{F}K_0(2^{-2\ell}\lambda, 2^{-\ell}\xi)|^{a_0} \dots |\mathcal{F}K_0(2^{-2\ell}\lambda, 2^{-\ell}\xi)|^{a_d} \\ &\lesssim |\lambda|^{-a_0} \prod_{i=1}^d |\xi_i|^{-2a_i} \sum_{\ell \geq 0} 2^{-2\ell(1-(a_0+a_1+\dots+a_d))} \lesssim |\lambda|^{-a_0} \prod_{i=1}^d |\xi_i|^{-2a_i}, \end{aligned}$$

which finishes our proof. $\square$

We now turn to a bound concerning the function R in the decomposition (4.9) of the heat kernel p .

Lemma A.6. *Let R be the remainder term associated with the localized heat kernel K (along Definition 4.6). Then, for all fixed $a_0, a_1, \dots, a_d \geq 0$ such that $\sum_{i=0}^d a_i > 1$, one has, for every $(\lambda, \xi) \in \mathbb{R}^{d+1}$,*

$$|\mathcal{F}R(\lambda, \xi)| \lesssim |\lambda|^{-a_0} \prod_{i=1}^d |\xi|^{-2a_i}. \quad (\text{A.10})$$

As a consequence, if $H_0 \in (0, 1)$, $\mathbf{H} = (H_1, \dots, H_d) \in (0, 1)^d$ are such that $2H_0 + H < d + 1$ (which is ensured by our assumption (4.31)), the following relation holds true for the function $\mathcal{N}_{H_0, \mathbf{H}}$ defined by (4.19):

$$\int_{\mathbb{R}^{d+1}} d\lambda d\xi \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) |\mathcal{F}R(\lambda, \xi)| < \infty. \quad (\text{A.11})$$

Proof. Using the expansion of R in (4.9) and relation (4.6) for $\mathcal{S}_{s,x}^\delta$, we can first write

$$\mathcal{F}R(\lambda, \xi) = \sum_{\ell > 0} 2^{2\ell} \mathcal{F}K_0(2^{2\ell} \lambda, 2^\ell \xi).$$

Then, similarly to what we did in the proof of Lemma A.6, we invoke the bound $|\mathcal{F}K_0(\lambda, \xi)| \lesssim |\lambda|^{-\tau_0}$ and $|\mathcal{F}K_0(\lambda, \xi)| \lesssim |\xi_i|^{-\tau_i}$ for all $\tau_0, \tau_1, \dots, \tau_d \geq 0$ and $(\lambda, \xi) \in \mathbb{R}^{d+1}$. We deduce that for any $a_0, \dots, a_d \geq 0$ such that $\sum_{i=0}^d a_i > 1$ we have

$$\begin{aligned} |\mathcal{F}R(\lambda, \xi)| &\leq \sum_{\ell > 0} 2^{2\ell} |\mathcal{F}K_0(2^{2\ell} \lambda, 2^\ell \xi)|^{1/(d+1)} \dots |\mathcal{F}K_0(2^{2\ell} \lambda, 2^\ell \xi)|^{1/(d+1)} \\ &\lesssim |\lambda|^{-a_0} \prod_{i=1}^d |\xi|^{-2a_i} \sum_{\ell > 0} 2^{2\ell(1-(a_0+a_1+\dots+a_d))} \lesssim |\lambda|^{-a_0} \prod_{i=1}^d |\xi|^{-2a_i}. \end{aligned}$$

This proves the assertion (A.10).

We now turn to a bound on the integral introduced in (A.11). To this aim, we split the integral according to the region $\mathcal{D}_s$ defined by (4.23) and we recall that $R = p - K$, which yields

$$\begin{aligned} \int_{\mathbb{R}^{d+1}} d\lambda d\xi \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) |\mathcal{F}R(\lambda, \xi)| &\lesssim \left[\int_{\mathcal{D}_s} d\lambda d\xi \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) |\mathcal{F}p(\lambda, \xi)| \right. \\ &\quad \left. + \int_{\mathcal{D}_s} d\lambda d\xi \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) |\mathcal{F}K(\lambda, \xi)| \right] + \int_{\mathbb{R}^{d+1} \setminus \mathcal{D}_s} d\lambda d\xi \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) |\mathcal{F}R(\lambda, \xi)|. \quad (\text{A.12}) \end{aligned}$$

Next, taking into account the expression (4.21) for the Fourier transform of p , the integral $\int_{\mathcal{D}_s} d\lambda d\xi \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) |\mathcal{F}p(\lambda, \xi)|$ in (A.12) is (essentially) the same as in the right-hand side of (4.28). We have already shown that this integral is finite in the proof of Lemma 4.14. In addition, one can bound $|\mathcal{F}K(\lambda, \xi)|$ by a constant thanks to Lemma A.5, in order to get

$$\int_{\mathcal{D}_s} \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) |\mathcal{F}K(\lambda, \xi)| d\lambda d\xi \lesssim \int_{\mathcal{D}_s} d\lambda d\xi \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) < \infty.$$

Eventually, the finiteness of $\int_{\mathbb{R}^{d+1} \setminus \mathcal{D}_s} d\lambda d\xi \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) |\mathcal{F}R(\lambda, \xi)|$ can be easily derived from relation (A.10). Plugging the information above into (A.12), this completes the proof of (A.11). $\square$

As we will see in the sequel, the renormalization procedure for $\mathbf{W}^{2,n}$ is based on the following decomposition.

Lemma A.7. *Recall that the renormalization constant $\mathbf{c}_{\rho, H_0, \mathbf{H}}^{(n)}$ is defined in (4.33) as*

$$\mathbf{c}_{\rho, H_0, \mathbf{H}}^{(n)} := \mathbf{c}_{\rho, H_0, \mathbf{H}} \cdot 2^{2n(d+1-(2H_0+H))}. \quad (\text{A.13})$$

It can also be expressed, thanks to (4.29) and (4.20), in the following way:

$$\mathbf{c}_{\rho, H_0, \mathbf{H}}^{(n)} = c_{H_0, \mathbf{H}}^2 \int_{\mathbb{R}^{d+1}} |\mathcal{F}\rho_n(\lambda, \xi)|^2 \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) \mathcal{F}p(\lambda, \xi) d\lambda d\xi. \quad (\text{A.14})$$

Then for all $(s, x), (t, y) \in \mathbb{R}^{d+1}$ and $n \geq 1$, one has the decomposition

$$\mathbb{E}[\mathbf{W}_{s,x}^{2,n}(t, y)] = \mathbf{c}_{\rho, H_0, \mathbf{H}}^{(n)} + \mathcal{E}_{s,x}^n(t, y), \quad (\text{A.15})$$

for some function $\mathcal{E}_{s,x}^n$ such that, for every $\varepsilon \in (0, 1)$, $\ell \geq 0$ and $\psi \in \mathcal{B}_s^\ell$ we have

$$|\langle \mathcal{E}_{s,x}^n, \psi_{s,x}^\ell \rangle| \lesssim 2^{2\ell(1+d-(2H_0+H)+\varepsilon)}. \quad (\text{A.16})$$

Moreover, in relation (A.16) the proportional constant does not depend on n, ℓ, s, x .

Proof. With the definition (4.17) of $\mathbf{W}^{2,n}$ in mind, we can obviously write

$$\mathbb{E}[\mathbf{W}_{s,x}^{2,n}(t, y)] = \mathbf{c}_{\rho, H_0, \mathbf{H}}^{(n)} + \mathcal{E}_{s,x}^n(t, y),$$

as stated in (A.15), where we have simply set

$$\mathcal{E}_{s,x}^n(t, y) := \left\{ \mathbb{E}[(K * \dot{W}^n)(t, y) \dot{W}^n(t, y)] - \mathbf{c}_{\rho, H_0, \mathbf{H}}^{(n)} \right\} - \mathbb{E}[(K * \dot{W}^n)(s, x) \dot{W}^n(t, y)]. \quad (\text{A.17})$$

We now analyze the terms

$$Q^n(s, x; t, y) = \mathbb{E}[(K * \dot{W}^n)(s, x) \dot{W}^n(t, y)] \quad (\text{A.18})$$

in the right-hand side of (A.17). To this aim, we resort to (A.1) and (A.2), which enable to write that for every $(t, y) \in \mathbb{R}^{d+1}$

$$Q^n(t, y; t, y) = c_{H_0, \mathbf{H}}^2 \int_{\mathbb{R}^{d+1}} |\mathcal{F}\rho_n(\lambda, \xi)|^2 \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) \mathcal{F}K(\lambda, \xi) d\lambda d\xi.$$

Thus using expression (A.14), we get

$$Q^n(t, y; t, y) - \mathbf{c}_{\rho, H_0, \mathbf{H}}^{(n)} = -c_{H_0, \mathbf{H}}^2 \int_{\mathbb{R}^{d+1}} |\mathcal{F}\rho_n(\lambda, \xi)|^2 \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) \mathcal{F}R(\lambda, \xi) d\lambda d\xi,$$

where R stands for the remainder term in the decomposition of Definition 4.6, item (i). Performing the same elementary manipulations as for (A.6) and (A.7), invoking the inequality $|\mathcal{F}\rho_n(\lambda, \xi)| \lesssim 1$ and Lemma A.6, we thus get

$$\left| Q^n(t, y; t, y) - \mathbf{c}_{\rho, H_0, \mathbf{H}}^{(n)} \right| \lesssim 1 \leq 2^{2\ell(1+d-(2H_0+H)+\varepsilon)},$$

where the last inequality stems from the fact that we assume $2H_0 + H < d + 1$ in (4.31). In addition, along the same lines as for (A.5), one gets the following identity for the term $Q^n(s, x; t, y)$ defined by (A.18)

$$\begin{aligned} & \int_{\mathbb{R}^{d+1}} dt dy Q^n(s, x; t, y) \psi_{s,x}^\ell(t, y) \\ &= c_{H_0, \mathbf{H}}^2 \int_{\mathbb{R}^{d+1}} d\lambda d\xi |\mathcal{F}\rho_n(\lambda, \xi)|^2 \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) \mathcal{F}K(\lambda, \xi) \mathcal{F}\psi_{s,x}^\ell(\lambda, \xi). \end{aligned}$$

Hence owing to the fact that $\psi_{s,x}^\ell = \mathcal{S}_{s,x}^{2^{-\ell}} \psi$ and performing the change of variable $\lambda := 2^{-2l} \lambda, \xi = 2^{-l} \xi$, we get

$$\begin{aligned} & \left| \int_{\mathbb{R}^{d+1}} dt dy Q^n(s, x; t, y) \psi_{s,x}^\ell(t, y) \right| \\ &= c_{H_0, \mathbf{H}}^2 2^{2\ell(d+2-(2H_0+H))} \left| \int_{\mathbb{R}^{d+1}} d\lambda d\xi |\mathcal{F}\rho_n(2^{2\ell}\lambda, 2^\ell\xi)|^2 \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) \mathcal{F}K(2^{2\ell}\lambda, 2^\ell\xi) \mathcal{F}\psi(\lambda, \xi) \right|. \end{aligned}$$

At this point, observe that due to the assumption $2H_0 + H < d + 1$, we can pick $a_0, a_1, \dots, a_d$ in $[0, 1]$ such that $\sum_{i=0}^d a_i = 1 - \varepsilon$, $2H_0 + a_0 - 1 < 1$ and $2H_i + 2a_i - 1 < 1$ for $i = 1, \dots, d$. We can now apply Lemma A.5 with this set of parameters to deduce that

$$\begin{aligned} & \left| \int_{\mathbb{R}^{d+1}} dt dy Q^n(s, x; t, y) \psi_{s,x}^\ell(t, y) \right| \\ & \lesssim 2^{2\ell(d+1-(2H_0+H)+\varepsilon)} \int_{\mathbb{R}^{d+1}} d\lambda d\xi \frac{1}{|\lambda|^{2H_0+a_0-1}} \prod_{i=1}^d \frac{1}{|\xi_i|^{2H_i+2a_i-1}} |\mathcal{F}\psi(\lambda, \xi)|. \end{aligned}$$

Since $2H_0 + a_0 < 2$ and $2H_i + 2a_i < 2$ for $i = 1, \dots, d$, we can finally appeal to Lemma A.3 to assert that the latter integral is finite, which gives the conclusion. $\square$

We can now turn to a bound on the variance of the renormalized K -rough path $\widehat{\mathbf{W}}^n$.

Proposition A.8. *Let $\widehat{\mathbf{W}}^n$ be the renormalized K -rough path defined by (4.33). Then for all $\ell \geq 0$, $n \geq m \geq 0$, $\psi \in \mathcal{B}_s^{2(d+1)}$, $(s, x) \in \mathbb{R}^{d+1}$ and $\varepsilon \in (0, 1)$, it holds that*

$$\mathbb{E}[|\langle \widehat{\mathbf{W}}_{s,x}^{2,n} - \widehat{\mathbf{W}}_{s,x}^{2,m}, \psi_{s,x}^\ell \rangle|^2] \lesssim 2^{4\ell(1+d-(2H_0+H)+\varepsilon)} 2^{-m\varepsilon}, \quad (\text{A.19})$$

where the proportional constant in (A.19) does not depend on n, m, ℓ, s, x .

Proof. For the sake of conciseness, we will only focus on the case $m = 0$, i.e. we will show the uniform estimate

$$\mathbb{E}[|\langle \widehat{\mathbf{W}}_{s,x}^{2,n}, \psi_{s,x}^\ell \rangle|^2] \lesssim 2^{4\ell(1+d-(2H_0+H)+\varepsilon)}.$$

The proof in the general case $m \geq 0$ could in fact be obtained through elementary adaptations of the subsequent estimates, using the fact that $\mathcal{F}\rho$ is Lipschitz (see e.g. the arguments of [15] for more details on the transition from $m = 0$ to $m \geq 0$).

Observe first that due to Wick's formula for products of Gaussian random variables (and using the notation of (4.17)), we can write

$$\begin{aligned} & \mathbb{E}[|\langle \mathbf{W}_{s,x}^{2,n}, \psi_{s,x}^\ell \rangle|^2] \\ &= \iint_{\mathbb{R}^{d+1} \times \mathbb{R}^{d+1}} dt dy d\tilde{t} d\tilde{y} \psi_{s,x}^\ell(t, y) \psi_{s,x}^\ell(\tilde{t}, \tilde{y}) \mathbb{E}[\mathcal{I}_{s,x}^n(t, y) \dot{W}^n(t, y) \mathcal{I}_{s,x}^n(\tilde{t}, \tilde{y}) \dot{W}^n(\tilde{t}, \tilde{y})] \\ &= (\langle \mathbb{E}[\mathbf{W}_{s,x}^{2,n}], \psi_{s,x}^\ell \rangle)^2 + \mathcal{U}_{s,x}^{\ell,n} + \mathcal{V}_{s,x}^{\ell,n}, \end{aligned}$$

where we have set

$$\mathcal{U}_{s,x}^{\ell,n} := \iint_{\mathbb{R}^{d+1} \times \mathbb{R}^{d+1}} dt dy d\tilde{t} d\tilde{y} \psi_{s,x}^\ell(t, y) \psi_{s,x}^\ell(\tilde{t}, \tilde{y}) \mathbb{E}[\mathcal{I}_{s,x}^n(t, y) \mathcal{I}_{s,x}^n(\tilde{t}, \tilde{y})] \mathbb{E}[\dot{W}^n(t, y) \dot{W}^n(\tilde{t}, \tilde{y})]$$

and

$$\mathcal{V}_{s,x}^{\ell,n} := \iint_{\mathbb{R}^{d+1} \times \mathbb{R}^{d+1}} dt dy d\tilde{t} d\tilde{y} \psi_{s,x}^\ell(t, y) \psi_{s,x}^\ell(\tilde{t}, \tilde{y}) \mathbb{E}[\mathcal{I}_{s,x}^n(t, y) \dot{W}^n(\tilde{t}, \tilde{y})] \mathbb{E}[\dot{W}^n(t, y) \mathcal{I}_{s,x}^n(\tilde{t}, \tilde{y})].$$

Based on this decomposition, we get that

$$\begin{aligned} & \mathbb{E}[|\langle \widehat{\mathbf{W}}_{s,x}^{2,n}, \psi_{s,x}^\ell \rangle|^2] \\ &= \mathbb{E}[|\langle \mathbf{W}_{s,x}^{2,n} - \mathbf{c}_{\rho, H_0, \mathbf{H}}^{(n)}, \psi_{s,x}^\ell \rangle|^2] \\ &= (\langle \mathbb{E}[\mathbf{W}_{s,x}^{2,n}], \psi_{s,x}^\ell \rangle)^2 + \mathcal{U}_{s,x}^{\ell,n} + \mathcal{V}_{s,x}^{\ell,n} - 2\langle \mathbb{E}[\mathbf{W}_{s,x}^{2,n}], \psi_{s,x}^\ell \rangle \langle \mathbf{c}_{\rho, H_0, \mathbf{H}}^{(n)}, \psi_{s,x}^\ell \rangle + \langle \mathbf{c}_{\rho, H_0, \mathbf{H}}^{(n)}, \psi_{s,x}^\ell \rangle^2 \\ &= (\langle \mathbb{E}[\mathbf{W}_{s,x}^{2,n}] - \mathbf{c}_{\rho, H_0, \mathbf{H}}^{(n)}, \psi_{s,x}^\ell \rangle)^2 + \mathcal{U}_{s,x}^{\ell,n} + \mathcal{V}_{s,x}^{\ell,n} \\ &= (\langle \mathcal{E}_{s,x}^n, \psi_{s,x}^\ell \rangle)^2 + \mathcal{U}_{s,x}^{\ell,n} + \mathcal{V}_{s,x}^{\ell,n}, \end{aligned}$$

where we have used Lemma A.7 (and the notation therein) to derive the last identity. Owing to (A.16), our claim (A.19) is thus reduced to check that

$$|\mathcal{U}_{s,x}^{\ell,n}| \lesssim 2^{4\ell(1+d-(2H_0+H)+\varepsilon)} \quad \text{and} \quad |\mathcal{V}_{s,x}^{\ell,n}| \lesssim 2^{4\ell(1+d-(2H_0+H)+\varepsilon)}. \quad (\text{A.20})$$

The remainder of the proof is devoted to prove (A.20).

To this end, recall that $\mathcal{I}_{s,x}^n$ is defined by (4.17), which, together with relation (A.2), yields

$$\begin{aligned} \mathbb{E}[\mathcal{I}_{s,x}^n(t, y) \mathcal{I}_{s,x}^n(\tilde{t}, \tilde{y})] &= c_{H_0, \mathbf{H}}^2 \int_{\mathbb{R}^{d+1}} d\lambda d\xi |\mathcal{F}\rho_n(\lambda, \xi)|^2 |\mathcal{F}K(\lambda, \xi)|^2 \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) \\ &\quad [e^{i(\lambda(t-\tilde{t})+\xi \cdot (y-\tilde{y}))} - e^{i(\lambda(t-s)+\xi \cdot (y-x))} - e^{i(\lambda(s-\tilde{t})+\xi \cdot (x-\tilde{y}))} + 1] \end{aligned}$$

Combining this expression with formula (A.1) for $\mathbb{E}[\dot{W}^n(t, y)\dot{W}^n(\tilde{t}, \tilde{y})]$, we easily deduce that

$$\begin{aligned}
& \mathcal{U}_{s,x}^{\ell,n} \\
&= c_{H_0,\mathbf{H}}^2 \iint_{\mathbb{R}^{d+1} \times \mathbb{R}^{d+1}} d\lambda d\xi d\tilde{\lambda} d\tilde{\xi} |\mathcal{F}\rho_n(\lambda, \xi)|^2 |\mathcal{F}\rho_n(\tilde{\lambda}, \tilde{\xi})|^2 |\mathcal{F}K(\lambda, \xi)|^2 \mathcal{N}_{H_0,\mathbf{H}}(\lambda, \xi) \mathcal{N}_{H_0,\mathbf{H}}(\tilde{\lambda}, \tilde{\xi}) \\
&\quad \left[|\mathcal{F}\psi_{s,x}^\ell(\lambda + \tilde{\lambda}, \xi + \tilde{\xi})|^2 - \overline{\mathcal{F}\psi_{s,x}^\ell(\lambda + \tilde{\lambda}, \xi + \tilde{\xi})} \mathcal{F}\psi_{s,x}^\ell(\tilde{\lambda}, \tilde{\xi}) e^{-i(\lambda s + \xi \cdot x)} \right. \\
&\quad \left. - \mathcal{F}\psi_{s,x}^\ell(\lambda + \tilde{\lambda}, \xi + \tilde{\xi}) \overline{\mathcal{F}\psi_{s,x}^\ell(\tilde{\lambda}, \tilde{\xi})} e^{i(\lambda s + \xi \cdot x)} + |\mathcal{F}\psi_{s,x}^\ell(\lambda, \xi)|^2 \right] \\
&= c_{H_0,\mathbf{H}}^2 \iint_{\mathbb{R}^{d+1} \times \mathbb{R}^{d+1}} d\lambda d\xi d\tilde{\lambda} d\tilde{\xi} |\mathcal{F}\rho_n(\lambda, \xi)|^2 |\mathcal{F}\rho_n(\tilde{\lambda}, \tilde{\xi})|^2 |\mathcal{F}K(\lambda, \xi)|^2 \mathcal{N}_{H_0,\mathbf{H}}(\lambda, \xi) \mathcal{N}_{H_0,\mathbf{H}}(\tilde{\lambda}, \tilde{\xi}) \\
&\quad \left[|\mathcal{F}\psi_{0,0}^\ell(\lambda + \tilde{\lambda}, \xi + \tilde{\xi})|^2 - \overline{\mathcal{F}\psi_{0,0}^\ell(\lambda + \tilde{\lambda}, \xi + \tilde{\xi})} \mathcal{F}\psi_{0,0}^\ell(\tilde{\lambda}, \tilde{\xi}) \right. \\
&\quad \left. - \mathcal{F}\psi_{0,0}^\ell(\lambda + \tilde{\lambda}, \xi + \tilde{\xi}) \overline{\mathcal{F}\psi_{0,0}^\ell(\tilde{\lambda}, \tilde{\xi})} + |\mathcal{F}\psi_{0,0}^\ell(\lambda, \xi)|^2 \right] \\
&= c_{H_0,\mathbf{H}}^2 \iint_{\mathbb{R}^{d+1} \times \mathbb{R}^{d+1}} d\lambda d\xi d\tilde{\lambda} d\tilde{\xi} |\mathcal{F}\rho_n(\lambda, \xi)|^2 |\mathcal{F}\rho_n(\tilde{\lambda}, \tilde{\xi})|^2 |\mathcal{F}K(\lambda, \xi)|^2 \mathcal{N}_{H_0,\mathbf{H}}(\lambda, \xi) \mathcal{N}_{H_0,\mathbf{H}}(\tilde{\lambda}, \tilde{\xi}) \\
&\quad \left| \mathcal{F}\psi_{0,0}^\ell(\lambda + \tilde{\lambda}, \xi + \tilde{\xi}) - \mathcal{F}\psi_{0,0}^\ell(\tilde{\lambda}, \tilde{\xi}) \right|^2. \quad (\text{A.21})
\end{aligned}$$

Along similar arguments, we obtain first

$$\begin{aligned}
& \mathcal{V}_{s,x}^{\ell,n} = c_{H_0,\mathbf{H}}^2 \\
& \iint_{\mathbb{R}^{d+1} \times \mathbb{R}^{d+1}} d\lambda d\xi d\tilde{\lambda} d\tilde{\xi} |\mathcal{F}\rho_n(\lambda, \xi)|^2 |\mathcal{F}\rho_n(\tilde{\lambda}, \tilde{\xi})|^2 \mathcal{F}K(\lambda, \xi) \overline{\mathcal{F}K(\tilde{\lambda}, \tilde{\xi})} \mathcal{N}_{H_0,\mathbf{H}}(\lambda, \xi) \mathcal{N}_{H_0,\mathbf{H}}(\tilde{\lambda}, \tilde{\xi}) \\
& \quad \left[\overline{\mathcal{F}\psi_{0,0}^\ell(\lambda + \tilde{\lambda}, \xi + \tilde{\xi}) - \mathcal{F}\psi_{0,0}^\ell(\lambda, \xi)} \right] \left[\mathcal{F}\psi_{0,0}^\ell(\lambda + \tilde{\lambda}, \xi + \tilde{\xi}) - \mathcal{F}\psi_{0,0}^\ell(\tilde{\lambda}, \tilde{\xi}) \right],
\end{aligned}$$

and we can now apply Cauchy-Schwarz inequality to derive the estimate

$$\begin{aligned}
& |\mathcal{V}_{s,x}^{\ell,n}| \\
& \leq c_{H_0,\mathbf{H}}^2 \iint_{\mathbb{R}^{d+1} \times \mathbb{R}^{d+1}} d\lambda d\xi d\tilde{\lambda} d\tilde{\xi} |\mathcal{F}\rho_n(\lambda, \xi)|^2 |\mathcal{F}\rho_n(\tilde{\lambda}, \tilde{\xi})|^2 |\mathcal{F}K(\lambda, \xi)|^2 \mathcal{N}_{H_0,\mathbf{H}}(\lambda, \xi) \mathcal{N}_{H_0,\mathbf{H}}(\tilde{\lambda}, \tilde{\xi}) \\
& \quad \left| \mathcal{F}\psi_{0,0}^\ell(\lambda + \tilde{\lambda}, \xi + \tilde{\xi}) - \mathcal{F}\psi_{0,0}^\ell(\tilde{\lambda}, \tilde{\xi}) \right|^2. \quad (\text{A.22})
\end{aligned}$$

Combining (A.21)-(A.22) with the uniform bound $|\mathcal{F}\rho_n(\lambda, \xi)| \lesssim 1$, we have thus shown that uniformly in $(s, x) \in \mathbb{R}^{d+1}$ and $n \geq 1$ the following holds true:

$$|\mathcal{U}_{s,x}^{\ell,n}| + |\mathcal{V}_{s,x}^{\ell,n}| \lesssim \mathcal{S}^\ell, \quad (\text{A.23})$$

where the quantity $\mathcal{S}^\ell$ is given by

$$\begin{aligned}
\mathcal{S}^\ell &:= c_{H_0,\mathbf{H}}^4 \iint_{\mathbb{R}^{d+1} \times \mathbb{R}^{d+1}} d\lambda d\xi d\tilde{\lambda} d\tilde{\xi} |\mathcal{F}K(\lambda, \xi)|^2 \mathcal{N}_{H_0,\mathbf{H}}(\lambda, \xi) \mathcal{N}_{H_0,\mathbf{H}}(\tilde{\lambda}, \tilde{\xi}) \\
&\quad \times \left| \mathcal{F}\psi_{0,0}^\ell(\lambda + \tilde{\lambda}, \xi + \tilde{\xi}) - \mathcal{F}\psi_{0,0}^\ell(\tilde{\lambda}, \tilde{\xi}) \right|^2.
\end{aligned}$$

Moreover, an easy scaling argument performed on $\psi_{0,0}^\ell = \mathcal{S}_{0,0}^{2\ell} \psi$ shows that

$$\mathcal{S}^\ell = c_{H_0,\mathbf{H}}^4 2^{4\ell(d+2-(2H_0+H))} \tilde{\mathcal{S}}^\ell,$$

where

$$\begin{aligned} \tilde{\mathcal{S}}^\ell = \iint_{\mathbb{R}^{d+1} \times \mathbb{R}^{d+1}} d\lambda d\xi d\tilde{\lambda} d\tilde{\xi} |\mathcal{F}K(2^{2\ell}\lambda, 2^\ell\xi)|^2 \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) \mathcal{N}_{H_0, \mathbf{H}}(\tilde{\lambda}, \tilde{\xi}) \\ \times |\mathcal{F}\psi(\lambda + \tilde{\lambda}, \xi + \tilde{\xi}) - \mathcal{F}\psi(\tilde{\lambda}, \tilde{\xi})|^2. \end{aligned} \quad (\text{A.24})$$

Plugging this information into (A.23) and then (A.20) we are now reduced to show that for any $\epsilon \in (0, 1)$ we have

$$\tilde{\mathcal{S}}^\ell \lesssim 2^{-4\ell(1-\epsilon)}. \quad (\text{A.25})$$

We shall prove assertion (A.25) in the next two subsections, separating the case $H_0 \leq 1/2$ and $H_0 > 1/2$ (note that due to our restriction $2H_0 + H > d + \frac{1}{2}$, one has in any case $H_0 > \frac{1}{4}$). $\square$

A.2.1. Case $\frac{1}{4} < H_0 \leq \frac{1}{2}$. Our aim in this section is to prove (A.25) when $H_0 \in (\frac{1}{4}, \frac{1}{2}]$. In this case, notice that the condition $2H_0 + H > d + \frac{1}{2}$ entails that the set $\tilde{J}_*$ defined by (4.30) satisfies $|\tilde{J}_*| = 0$. More specifically, we have $H_i > \frac{1}{2}$ for $i = 1, \dots, d$.

With (A.24) in mind, our bound on $\tilde{\mathcal{S}}^\ell$ relies on a proper control of the difference

$$|\mathcal{F}\psi(\lambda + \tilde{\lambda}, \xi + \tilde{\xi}) - \mathcal{F}\psi(\tilde{\lambda}, \tilde{\xi})|.$$

To this aim, let us introduce some additional notation. Namely for $s \geq 0$ and $\lambda, \tilde{\lambda} \in \mathbb{R}^d$ we set

$$\mathcal{T}_s(\lambda) = \left(\int_{\mathbb{R}^{d+1}} dt dy \partial_{tx_1 \dots x_d} \psi(t, y) \left| \int_0^s e^{-i\lambda u} du \right|^{d+1} \right)^{1/(d+1)},$$

and

$$\mathcal{Q}_s(\lambda, \tilde{\lambda}) = \left(\int_{\mathbb{R}^{d+1}} dt dy \partial_{tx_1 \dots x_d} \psi(t, y) \left| \int_0^s du \int_0^u dv e^{-i(\tilde{\lambda}u + \lambda v)} \right|^{d+1} \right)^{1/(d+1)}.$$

Then for $i = 1, \dots, d$ we define

$$\mathcal{T}^{(0)}(\lambda) = \mathcal{T}_t(\lambda), \quad \mathcal{Q}^{(0)}(\lambda, \tilde{\lambda}) = \mathcal{Q}_t(\lambda, \tilde{\lambda}), \quad (\text{A.26})$$

$$\mathcal{T}^{(i)}(\lambda) = \mathcal{T}_{y_i}(\lambda), \quad \mathcal{Q}^{(i)}(\lambda, \tilde{\lambda}) = \mathcal{Q}_{y_i}(\lambda, \tilde{\lambda}). \quad (\text{A.27})$$

With this notation in hand, some elementary algebraic manipulations reveal that for all $\lambda, \tilde{\lambda}, \xi, \tilde{\xi}$ in $\mathbb{R}^d$ we have

$$|\mathcal{F}\psi(\lambda + \tilde{\lambda}, \xi + \tilde{\xi}) - \mathcal{F}\psi(\tilde{\lambda}, \xi + \tilde{\xi})| \lesssim |\lambda| \mathcal{Q}^{(0)}(\lambda, \tilde{\lambda}) \prod_{i=1}^d \mathcal{T}^{(i)}(\xi_i + \tilde{\xi}_i). \quad (\text{A.28})$$

Along the same lines, for $i = 1, \dots, d$ we also get

$$\begin{aligned} & |\mathcal{F}\psi(\tilde{\lambda}, \tilde{\xi}_1, \dots, \tilde{\xi}_{i-1}, \xi_i + \tilde{\xi}_i, \xi_{i+1} + \tilde{\xi}_{i+1}, \dots, \xi_d + \tilde{\xi}_d) \\ & \quad - \mathcal{F}\psi(\tilde{\lambda}, \tilde{\xi}_1, \dots, \tilde{\xi}_{i-1}, \tilde{\xi}_i, \xi_{i+1} + \tilde{\xi}_{i+1}, \dots, \xi_d + \tilde{\xi}_d)| \\ & \lesssim \mathcal{T}^{(0)}(\tilde{\lambda}) \prod_{j=1}^{i-1} \mathcal{T}^{(j)}(\tilde{\xi}_j) \left(|\xi_i| \cdot \mathcal{Q}^{(i)}(\xi_i, \tilde{\xi}_i) \right) \prod_{j=i+1}^d \mathcal{T}^{(j)}(\xi_j + \tilde{\xi}_j). \end{aligned} \quad (\text{A.29})$$

We now label a lemma on the functions $\mathcal{T}^{(i)}$ and $\mathcal{Q}^{(i)}$ which is useful in the sequel. It is borrowed from [15, Lemma 3.10 and 3.11].

Lemma A.9. *Fix $\psi \in \mathcal{C}^{d+1}(\mathbb{R}^{d+1}; \mathbb{R})$ with compact support, $i \in \{0, 1, \dots, d\}$, and let $\mathcal{T}^{(i)}, \mathcal{Q}^{(i)}$ be the functions defined by (A.26)-(A.27). Then, for all $\lambda_1, \lambda_2, \beta_1, \beta_2 \in (0, 2)$ such that $\lambda_1 + \lambda_2 > 3$ and $\beta_1 + \beta_2 > 1$, the following integrals are finite:*

$$\int_{\mathbb{R}^2} dx_1 dx_2 \frac{|\mathcal{T}^{(i)}(x_1 + x_2)|^2}{|x_1|^{\lambda_1-1} |x_2|^{\lambda_2-1}}, \quad \text{and} \quad \int_{\mathbb{R}^2} dx_1 dx_2 \frac{|\mathcal{Q}^{(i)}(x_1, x_2)|^2}{|x_1|^{\beta_1-1} |x_2|^{\beta_2-1}}.$$

With those notations and preliminary results in hand, let us go back to (A.24). Invoking (A.28) and (A.29), our claim (A.25) amounts to show that

$$\begin{aligned} & \iint_{\mathbb{R}^{d+1} \times \mathbb{R}^{d+1}} d\lambda d\xi d\tilde{\lambda} d\tilde{\xi} |\mathcal{F}K(2^{2\ell}\lambda, 2^\ell\xi)|^2 \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) \mathcal{N}_{H_0, \mathbf{H}}(\tilde{\lambda}, \tilde{\xi}) \\ & \times \left(|\lambda|^2 \mathcal{Q}^{(0)}(\lambda, \tilde{\lambda})^2 \right) \prod_{i=1}^d \left(\mathcal{T}^{(i)}(\xi_i + \tilde{\xi}_i) \right)^2 \lesssim 2^{-4\ell(1-\varepsilon)}, \quad (\text{A.30}) \end{aligned}$$

and that for every fixed $i = 1, \dots, d$, we have

$$\begin{aligned} \mathcal{J}^{i, \ell} &:= \iint_{\mathbb{R}^{d+1} \times \mathbb{R}^{d+1}} d\lambda d\xi d\tilde{\lambda} d\tilde{\xi} |\mathcal{F}K(2^{2\ell}\lambda, 2^\ell\xi)|^2 \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) \mathcal{N}_{H_0, \mathbf{H}}(\tilde{\lambda}, \tilde{\xi}) \left(\mathcal{T}^{(0)}(\tilde{\lambda}) \right)^2 \\ & \times \prod_{j=1}^{i-1} \left(\mathcal{T}^{(j)}(\tilde{\xi}_j) \right)^2 \left(|\xi_i|^2 \mathcal{Q}^{(i)}(\xi_i, \tilde{\xi}_i)^2 \right) \prod_{j=i+1}^d \left(\mathcal{T}^{(j)}(\xi_j + \tilde{\xi}_j) \right)^2 \lesssim 2^{-4\ell(1-\varepsilon)}. \quad (\text{A.31}) \end{aligned}$$

We will now prove (A.30) and (A.31) separately.

Proof of (A.30). We can apply Lemma A.5 and recall the definition (4.19) of $\mathcal{N}_{H_0, \mathbf{H}}$ in order to assert that for all $a_0, a_1, \dots, a_d \in [0, 1]$ such that $a_0 + a_1 + \dots + a_d < 1$, the integral in (A.30) is bounded (up to a constant) by

$$\begin{aligned} & 2^{-4\ell \sum_{i=0}^d a_i} \left(\int_{\mathbb{R} \times \mathbb{R}} d\lambda d\tilde{\lambda} \frac{\mathcal{Q}^{(0)}(\lambda, \tilde{\lambda})^2}{|\lambda|^{(2a_0+2H_0-2)-1} |\tilde{\lambda}|^{2H_0-1}} \right) \\ & \times \prod_{i=1}^d \left(\int_{\mathbb{R} \times \mathbb{R}} d\xi_i d\tilde{\xi}_i \frac{\mathcal{T}^{(i)}(\xi_i + \tilde{\xi}_i)^2}{|\xi_i|^{4a_i+2H_i-1} |\tilde{\xi}_i|^{2H_i-1}} \right). \quad (\text{A.32}) \end{aligned}$$

The whole point now is that we can find parameters $a_0, a_1, \dots, a_d \in [0, 1]$ such that $a_0 + a_1 + \dots + a_d = 1 - \varepsilon$ and such that the integrals involved in the above expression are all finite. In order to justify this claim, we can refer to Lemma A.9. According to this property, the first integral in (A.32) is finite whenever $2a_0 + 2H_0 > 3 - 2H_0$ and $2a_0 + 2H_0 < 4$. Moreover, since $a_0 < 1$, we have $2a_0 + 2H_0 < 2 + 2H_0 < 4$. Summarizing those elementary considerations and similar ones for the second integral in (A.32), we get that (A.32) is a finite expression as long as

$$\begin{cases} 3 - 2H_0 < 2a_0 + 2H_0 < 2 + 2H_0 \\ \max(2H_i, 3 - 2H_i) < 4a_i + 2H_i < 2 \quad \text{for } i = 1, \dots, d. \end{cases} \quad (\text{A.33})$$

Provided (A.33) is met, we thus have that the expression (A.32) is bounded, up to a constant, by $2^{4\ell(1-\varepsilon)}$. This proves (A.30).

We now show that the condition in (A.33) can indeed be fulfilled under our standing assumptions. Indeed,

- (i) Since $H_0 > \frac{1}{4}$ and $H_i > \frac{1}{2}$ for $i = 1, \dots, d$, the conditions in (A.33) are easily shown to be satisfied for some values of $a_i \in (0, 1)$.
- (ii) The conditions (A.33) can also be made consistent with the desired assumption $\sum_{i=0}^d a_i = 1 - \epsilon$ for $\epsilon > 0$. In order to verify this assertion, sum the constraints in (A.33). This yields

$$\begin{aligned} 2(3 - 2H_0) + \sum_{i=1}^d \max(2H_i, 3 - 2H_i) \\ < 2(2a_0 + 2H_0) + \sum_{i=1}^d (4a_i + 2H_i) < 2(2 + 2H_0) + 2d. \end{aligned} \quad (\text{A.34})$$

We now decompose the sums over H_i along the set $\mathcal{A} := \{i \in \{1, \dots, d\} : H_i > \frac{3}{4}\}$ and resort to the assumption $\sum_{i=0}^d a_i = 1 - \epsilon$. Recalling our notation $H = \sum_{i=1}^d H_i$, we end up with

$$6 - 4H_0 + 2 \sum_{i \in \mathcal{A}} H_i + \sum_{i \notin \mathcal{A}} (3 - 2H_i) < 4(1 - \epsilon) + 2(2H_0 + H) < 4 + 2d + 4H_0. \quad (\text{A.35})$$

Then some elementary manipulations allow to write this condition as

$$2 - 2(2H_0 + H) + 3d + |\mathcal{A}| - 4 \sum_{i \in \mathcal{A}} (1 - H_i) + 4\epsilon < 2(2H_0 + H) < 2d + 4H_0 + 4\epsilon, \quad (\text{A.36})$$

which is indeed true (at least for $\epsilon > 0$ small enough) under our assumption $2H_0 + H > d + \frac{1}{2}$ (note that the upper bound $2(2H_0 + H) < 2d + 4H_0 + 4\epsilon \Leftrightarrow 2 \sum_{i=1}^d H_i < 2d + 4\epsilon$ is always trivially satisfied). This finishes the proof of (A.30).

Proof of (A.31). Let us fix $i \in \{1, \dots, d\}$. Here we need to split the integral $\mathcal{J}^{i,\ell}$ in (A.31) along the following domains in the λ variable:

$$D_- := \{\lambda \in \mathbb{R} : |\lambda| \leq 1\} \quad , \quad D_+ := \{\lambda \in \mathbb{R} : |\lambda| \geq 1\}.$$

We will also split the ξ variable in the following way: for every $\mathbf{s} \in \{-, +\}^{i-1}$, we set

$$D_{\mathbf{s}}^{(i)} := \{\xi \in \mathbb{R}^d : \text{for } k = 1, \dots, i-1, |\xi_k| \leq 1 \text{ when } \mathbf{s}_k = -, |\xi_k| \geq 1 \text{ when } \mathbf{s}_k = +\}. \quad (\text{A.37})$$

In fact, for symmetry reasons, the ξ -integral in $\mathcal{J}^{i,\ell}$ can be reduced to integrals over sets of the form $D_{\mathbf{s}}^{(i)}$ for $\mathbf{s} = \mathbf{s}^{(i,j)} = \{-\}^j \times \{+\}^{i-1-j}$. Therefore we will achieve (A.31) if we can

properly bound the integrals $\mathcal{J}_{(j),\pm}^{i,\ell}$ defined by

$$\begin{aligned} \mathcal{J}_{(j),\pm}^{i,\ell} := & \iint_{(D_{\pm} \times D_{\mathbf{s}(i,j)}^{(i)}) \times \mathbb{R}^{d+1}} d\lambda d\xi d\tilde{\lambda} d\tilde{\xi} |\mathcal{FK}(2^{2\ell}\lambda, 2^\ell\xi)|^2 \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) \mathcal{N}_{H_0, \mathbf{H}}(\tilde{\lambda}, \tilde{\xi}) \\ & \times \left(\mathcal{T}^{(0)}(\tilde{\lambda}) \right)^2 \prod_{k=1}^{i-1} \left(\mathcal{T}^{(1)}(\tilde{\xi}_k) \right)^2 \left(|\xi_i|^2 \mathcal{Q}^{(i)}(\xi_i, \tilde{\xi}_i)^2 \right) \prod_{k=i+1}^d \left(\mathcal{T}^{(k)}(\xi_k + \tilde{\xi}_k) \right)^2. \end{aligned} \quad (\text{A.38})$$

We now proceed to bound the terms $\mathcal{J}_{(j),+}^{i,\ell}$ and $\mathcal{J}_{(j),-}^{i,\ell}$.

Bound for $\mathcal{J}_{(j),+}^{i,\ell}$. In order to bound $\mathcal{J}_{(j),+}^{i,\ell}$ in (A.38), we proceed similarly to (A.32). Namely we apply Lemma A.5 to assert that for all $a_0, a_1, \dots, a_d \in [0, 1]$ such that $a_0 + a_1 + \dots + a_d < 1$,

$$\begin{aligned} \mathcal{J}_{(j),+}^{i,\ell} \lesssim & 2^{-4\ell(a_0+a_1+\dots+a_d)} \left(\int_{|\lambda| \geq 1} \frac{d\lambda}{|\lambda|^{2a_0+2H_0-1}} \right) \left(\int_{\mathbb{R}} d\tilde{\lambda} \frac{\mathcal{T}^{(0)}(\tilde{\lambda})^2}{|\tilde{\lambda}|^{2H_0-1}} \right) \prod_{r=1}^{i-1} \left(\int_{\mathbb{R}} d\tilde{\xi}_r \frac{\mathcal{T}^{(r)}(\tilde{\xi}_r)^2}{|\tilde{\xi}_r|^{2H_r-1}} \right) \\ & \times \prod_{k=1}^j \left(\int_{|\xi_k| \leq 1} \frac{d\xi_k}{|\xi_k|^{4a_k+2H_k-1}} \right) \prod_{m=j+1}^{i-1} \left(\int_{|\xi_m| \geq 1} \frac{d\xi_m}{|\xi_m|^{4a_m+2H_m-1}} \right) \\ & \times \left(\int_{\mathbb{R} \times \mathbb{R}} d\xi_i d\tilde{\xi}_i \frac{\mathcal{Q}^{(i)}(\xi_i, \tilde{\xi}_i)^2}{|\xi_i|^{(4a_i+2H_i-2)-1} |\tilde{\xi}_i|^{2H_i-1}} \right) \prod_{p=i+1}^d \left(\int_{\mathbb{R} \times \mathbb{R}} d\xi_p d\tilde{\xi}_p \frac{\mathcal{T}^{(p)}(\xi_p + \tilde{\xi}_p)^2}{|\xi_p|^{4a_p+2H_p-1} |\tilde{\xi}_p|^{2H_p-1}} \right). \end{aligned} \quad (\text{A.39})$$

Based on the criteria of Lemma A.9, we get the following conditions on the parameters $a_0, a_1, \dots, a_d$ (so as to ensure that the integrals in (A.39) are all finite, and also that each a_i belongs to $(0, 1)$):

$$\begin{cases} 2 < 2a_0 + 2H_0 < 2 + 2H_0 \\ 2H_k < 4a_k + 2H_k < 2 & \text{for } k = 1, \dots, j \\ 2 < 4a_m + 2H_m < 4 + 2H_m & \text{for } m = j+1, \dots, i-1 \\ \max(2H_i, 3 - 2H_i) < 4a_i + 2H_i < 4 \\ \max(2H_p, 3 - 2H_p) < 4a_p + 2H_p < 2 & \text{for } p = i+1, \dots, d. \end{cases} \quad (\text{A.40})$$

As in the proof of (A.30)-item (ii), we still have to verify that parameters $a_0, \dots, a_d$ can be chosen so that $\sum_{k=0}^d a_k = 1 - \varepsilon$. To this aim, we use the same strategy as for (A.34)-(A.36). Namely we sum all the constraints in (A.38) and we decompose the sum along the set $\mathcal{A}_i := \{p \in \{i+1, \dots, d\} : H_p > 3/4\}$. We let the patient reader check that we get the following set of constraints for $1 \leq j < i \leq d$:

$$a_{ij}^+ < 4(1 - \varepsilon) + 2(2H_0 + H) < b_{ij}^+, \quad (\text{A.41})$$

with two parameters a_{ij}^+, b_{ij}^+ defined by

$$\begin{aligned} a_{ij}^+ &= 4 + 2 \sum_{k=1}^j H_k + 2(i-1-j) \\ &\quad + \left\{ 2H_i \mathbf{1}_{\{H_i > \frac{3}{4}\}} + (3-2H_i) \mathbf{1}_{\{H_i \leq \frac{3}{4}\}} \right\} + 2 \sum_{p \in \mathcal{A}_i} H_p + \sum_{\substack{p=i+1, \dots, d \\ p \notin \mathcal{A}_i}} (3-2H_p), \\ b_{ij}^+ &= 4 + 2(2H_0 + H) \\ &\quad + \left[2 \left(j - \sum_{k=1}^j H_k \right) + 4(i-1-j) + (4-2H_i) + 2 \left((d-i) - \sum_{p=i+1}^d H_p \right) \right]. \end{aligned}$$

In order to see that (A.41) can indeed be satisfied, notice first that the inequality

$$4(1-\varepsilon) + 2(2H_0 + H) < b_{ij}^+ \quad (\text{A.42})$$

is easily checked, thanks to the fact that $H_j < 1$ for $j = 1, \dots, d$. Indeed, this simple observation yields $b_{ij}^+ > 4 + 2(2H_0 + H)$, which entails (A.42). In order to prove

$$4(1-\varepsilon) + 2(2H_0 + H) > a_{ij}^+, \quad (\text{A.43})$$

we resort to the fact that each H_l is greater than $1/2$ for $l = 1, \dots, d$. Therefore $3-2H_l < 2$, which gives

$$a_{ij}^+ < 4 + 2j + 2(i-1-j) + 2 + 2|\mathcal{A}_i| + 2(d-i-|\mathcal{A}_i|) = 4 + 2d < 4 + 2(2H_0 + H).$$

This is enough to ensure (A.43). Since (A.42) and (A.43) implies that (A.41) is satisfied, the proof of (A.38) is now achieved for $\mathcal{J}_{(j),+}^{i,\ell}$.

Bound for $\mathcal{J}_{(j),-}^{i,\ell}$. The procedure is similar to what has been done for $\mathcal{J}_{(j),+}^{i,\ell}$. We will thus only highlight the differences for the sake of completeness. First the equivalent of relation (A.39) is

$$\begin{aligned} \mathcal{J}_{(j),-}^{i,\ell} &\lesssim 2^{-4\ell(a_0+a_1+\dots+a_d)} \left(\int_{|\lambda| \leq 1} \frac{d\lambda}{|\lambda|^{2a_0+2H_0-1}} \right) \left(\int_{\mathbb{R}} d\tilde{\lambda} \frac{\mathcal{T}^{(0)}(\tilde{\lambda})^2}{|\tilde{\lambda}|^{2H_0-1}} \right) \prod_{r=1}^{i-1} \left(\int_{\mathbb{R}} d\tilde{\xi}_r \frac{\mathcal{T}^{(0)}(\tilde{\xi}_r)^2}{|\tilde{\xi}_r|^{2H_r-1}} \right) \\ &\quad \prod_{k=1}^j \left(\int_{|\xi_k| \leq 1} \frac{d\xi_k}{|\xi_k|^{4a_k+2H_k-1}} \right) \prod_{m=j+1}^{i-1} \left(\int_{|\xi_m| \geq 1} \frac{d\xi_m}{|\xi_m|^{4a_m+2H_m-1}} \right) \\ &\quad \left(\int_{\mathbb{R} \times \mathbb{R}} d\xi_i d\tilde{\xi}_i \frac{\mathcal{Q}^{(i)}(\xi_i, \tilde{\xi}_i)^2}{|\xi_i|^{(4a_i+2H_i-2)-1} |\tilde{\xi}_i|^{2H_i-1}} \right) \prod_{p=i+1}^d \left(\int_{\mathbb{R} \times \mathbb{R}} d\xi_p d\tilde{\xi}_p \frac{\mathcal{T}^{(p)}(\xi_p + \tilde{\xi}_p)^2}{|\xi_p|^{4a_p+2H_p-1} |\tilde{\xi}_p|^{2H_p-1}} \right), \end{aligned}$$

with the following associated (and consistent) constraints on the parameters $a_0, a_1, \dots, a_d$:

$$\left\{ \begin{array}{ll} 2H_0 < 2a_0 + 2H_0 < 2 \\ 2H_k < 4a_k + 2H_k < 2 & \text{for } k = 1, \dots, j \\ 2 < 4a_m + 2H_m < 4 + 2H_m & \text{for } m = j+1, \dots, i-1 \\ \max(2H_i, 3-2H_i) < 4a_i + 2H_i < 4 \\ \max(2H_p, 3-2H_p) < 4a_p + 2H_p < 2 & \text{for } p = i+1, \dots, d. \end{array} \right. \quad (\text{A.44})$$

In order to check that $a_0, \dots, a_d$ can be chosen such that $\sum_{i=0}^d a_i = 1 - \varepsilon$, the equivalent of relation (A.41) is

$$a_{ij}^- < 4(1 - \varepsilon) + 2(2H_0 + H) < b_{ij}^-, \quad (\text{A.45})$$

with

$$\begin{aligned} a_{ij}^- &= 4H_0 + 2 \sum_{k=1}^j H_k + 2(i - 1 - j) \\ &\quad + \{2H_i \mathbf{1}_{\{H_i > \frac{3}{4}\}} + (3 - 2H_i) \mathbf{1}_{\{H_i \leq \frac{3}{4}\}}\} + 2 \sum_{p \in \mathcal{A}_i} H_p + \sum_{\substack{p=i+1, \dots, d \\ p \notin \mathcal{A}_i}} (3 - 2H_p), \\ b_{ij}^- &= 4 + 2(2H_0 + H) \\ &\quad + \left[2 \left(j - \sum_{k=1}^j H_k \right) + 4(i - 1 - j) + \underbrace{(4 - 4H_0 - 2H_i)}_{>0} + 2 \left((d - i) - \sum_{p=i+1}^d H_p \right) \right]. \end{aligned}$$

We let the reader check that (A.45) holds true thanks to the assumption $2H_0 + H > d + 1/2$. This proves (A.38) for $\mathcal{J}_{(j),-}^{i,\ell}$.

Summarizing our last considerations, we have proved (A.38) for both $\mathcal{J}_{(j),+}^{i,\ell}$ and $\mathcal{J}_{(j),-}^{i,\ell}$. This yields (A.31), which was the last step in order to achieve (A.25) in the case $1/4 < H_0 \leq 1/2$.

A.2.2. Case $H_0 > \frac{1}{2}$. In this section we show how to modify the considerations of Section A.2.1 in order to prove (A.25) when $H_0 > 1/2$. Let us first recall that our standing assumptions are those of Theorem 4.15. In particular, the set $\tilde{J}_*$ defined by (4.30) satisfies $|\tilde{J}_*| \leq 1$. More specifically, at most one of the H_i 's ($i = 1, \dots, d$) can be smaller than $\frac{1}{2}$. For symmetry reasons, we can assume for instance that $H_2 > \frac{1}{2}, \dots, H_d > \frac{1}{2}$, that is only H_1 may be smaller than $\frac{1}{2}$ in the sequel.

In this situation, we will rely on the upper bound (A.29), while (A.28) is replaced by

$$|\mathcal{F}\psi(\lambda + \tilde{\lambda}, \tilde{\xi}) - \mathcal{F}\psi(\tilde{\lambda}, \tilde{\xi})| \lesssim |\lambda| \cdot \mathcal{Q}^{(0)}(\lambda, \tilde{\lambda}) \cdot \prod_{i=1}^d \mathcal{T}^{(i)}(\xi_i + \tilde{\xi}_i), \quad (\text{A.46})$$

where the functions $\mathcal{T}^{(i)}, \mathcal{Q}^{(i)}$ are still those defined through (A.26)-(A.27). Based on (A.29) and (A.46), our claim (A.25) can be reduced to prove the following analogs of (A.30)-(A.31):

$$\begin{aligned} \hat{\mathcal{J}}^\ell &:= \iint_{\mathbb{R}^{d+1} \times \mathbb{R}^{d+1}} d\lambda d\xi d\tilde{\lambda} d\tilde{\xi} |\mathcal{F}K(2^{2\ell}\lambda, 2^\ell\xi)|^2 \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) \mathcal{N}_{H_0, \mathbf{H}}(\tilde{\lambda}, \tilde{\xi}) \\ &\quad \times \left(|\lambda|^2 \mathcal{Q}^{(0)}(\lambda, \tilde{\lambda})^2 \right) \prod_{i=1}^d \left(\mathcal{T}^{(i)}(\xi_i) \right)^2 \lesssim 2^{-4\ell(1-\varepsilon)}, \quad (\text{A.47}) \end{aligned}$$

and for every fixed $i = 1, \dots, d$,

$$\begin{aligned} \hat{\mathcal{J}}^{i,\ell} := & \iint_{\mathbb{R}^{d+1} \times \mathbb{R}^{d+1}} d\lambda d\xi d\tilde{\lambda} d\tilde{\xi} |\mathcal{FK}(2^{2\ell}\lambda, 2^\ell\xi)|^2 \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) \mathcal{N}_{H_0, \mathbf{H}}(\tilde{\lambda}, \tilde{\xi}) \left(\mathcal{T}^{(0)}(\lambda + \tilde{\lambda}) \right)^2 \\ & \times \prod_{j=1}^{i-1} \left(\mathcal{T}^{(j)}(\tilde{\xi}_j) \right)^2 \left(|\xi_i|^2 \mathcal{Q}^{(i)}(\xi_i, \tilde{\xi}_i)^2 \right) \prod_{j=i+1}^d \left(\mathcal{T}^{(j)}(\xi_j + \tilde{\xi}_j) \right)^2 \lesssim 2^{-4\ell(1-\varepsilon)}. \end{aligned} \quad (\text{A.48})$$

As in Section A.2.1, we will prove (A.47) and (A.48) separately.

Proof of (A.47). In order to handle the term $\hat{\mathcal{J}}^\ell$ in (A.47), we introduce a decomposition of the ξ -domain which is similar to (A.37). Namely we write $\mathbb{R}^d := \cup_{\mathbf{s} \in \{-, +\}^d} D_{\mathbf{s}}$, where for $\mathbf{s} \in \{-, +\}^d$ we have

$$D_{\mathbf{s}} := \{ \xi \in \mathbb{R}^d : |\xi_k| \leq 1 \text{ when } \mathbf{s}_k = -, \quad |\xi_k| \geq 1 \text{ when } \mathbf{s}_k = + \}.$$

Furthermore, we can invoke again symmetry arguments and focus on domains of the form $D_{\mathbf{s}^{(j)}}$ where $\mathbf{s}^{(j)} = \{-\}^j \times \{+\}^{d-j}$. We get that (A.47) is fulfilled once we can bound the integral $\hat{\mathcal{J}}_{(j)}^\ell$ defined by

$$\begin{aligned} \hat{\mathcal{J}}_{(j)}^\ell := & \iint_{(\mathbb{R} \times D_{\mathbf{s}^{(j)}}) \times \mathbb{R}^{d+1}} d\lambda d\xi d\tilde{\lambda} d\tilde{\xi} |\mathcal{FK}(2^{2\ell}\lambda, 2^\ell\xi)|^2 \\ & \times \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) \mathcal{N}_{H_0, \mathbf{H}}(\tilde{\lambda}, \tilde{\xi}) \left(|\lambda|^2 \mathcal{Q}^{(0)}(\lambda, \tilde{\lambda}) \right)^2 \prod_{i=1}^d \mathcal{T}^{(i)}(\tilde{\xi}_i)^2. \end{aligned}$$

Next we can apply Lemma A.5 to assert that for all $a_0, a_1, \dots, a_d \in [0, 1]$ such that $a_0 + a_1 + \dots + a_d < 1$, we have

$$\begin{aligned} \hat{\mathcal{J}}_{(j)}^\ell \lesssim & 2^{-4\ell(a_0 + a_1 + \dots + a_d)} \left(\int_{\mathbb{R} \times \mathbb{R}} d\lambda d\tilde{\lambda} \frac{\mathcal{Q}^{(0)}(\lambda, \tilde{\lambda})^2}{|\lambda|^{(2a_0 + 2H_0 - 2) - 1} |\tilde{\lambda}|^{2H_0 - 1}} \right) \prod_{k=1}^j \left(\int_{|\xi_k| \leq 1} \frac{d\xi_k}{|\xi_k|^{4a_k + 2H_k - 1}} \right) \\ & \prod_{m=j+1}^d \left(\int_{|\xi_m| \geq 1} \frac{d\xi_m}{|\xi_m|^{4a_m + 2H_m - 1}} \right) \prod_{p=1}^d \left(\int_{\mathbb{R}} d\tilde{\xi}_p \frac{\mathcal{T}^{(p)}(\tilde{\xi}_p)^2}{|\tilde{\xi}_p|^{2H_p - 1}} \right). \end{aligned} \quad (\text{A.49})$$

Referring again to the results of Lemma A.9, we derive the following constraints on the parameters $a_0, a_1, \dots, a_d$:

$$\begin{cases} \max(2H_0, 3 - 2H_0) < 2a_0 + 2H_0 < 2 + 2H_0 \\ 2H_k < 4a_k + 2H_k < 2 & \text{for } k = 1, \dots, j \\ 2 < 4a_m + 2H_m < 4 + 2H_m & \text{for } m = j + 1, \dots, d. \end{cases} \quad (\text{A.50})$$

Like in the case $H_0 \leq 1/2$, we sum those constraints over k, m in order to check that $\sum_{j=0}^d a_j = 1 - \varepsilon$. We get the following equivalent of relation (A.41):

$$\hat{a}_j < 4(1 - \varepsilon) + 2(2H_0 + H) < \hat{b}_j, \quad (\text{A.51})$$

where

$$\begin{aligned}\hat{a}_j &= \mathbf{1}_{\{H_0 > \frac{3}{4}\}} \left\{ 4H_0 + 2 \sum_{k=1}^j H_k + 2(d-j) \right\} + \mathbf{1}_{\{H_0 < \frac{3}{4}\}} \left\{ 6 - 4H_0 + 2 \sum_{k=1}^j H_k + 2(d-j) \right\}, \\ \hat{b}_j &= 4 + 2(2H_0 + H) + \left[4(d-j) + 2 \left(j - \sum_{k=1}^j H_k \right) \right].\end{aligned}$$

We let the reader check that under our assumptions $H_0 > 1/2$ and $2H_0 + H > d + 1/2$, condition (A.51) is met. Observe for instance that due to $H_0 > 1/2$, we have

$$6 - 4H_0 + 2 \sum_{k=1}^j H_k + 2(d-j) < 6 - 4H_0 + 2d < 4 + 2d < (4 + 2(2H_0 + H)) - 1.$$

This finishes the proof of (A.47).

Proof of (A.48). Like in the proof of (A.31) above, the term $\hat{\mathcal{J}}^{i,\ell}$ introduced in (A.48) is decomposed along the set $D_{\mathbf{s}}^{(i)}$ defined by (A.37). We are reduced to the evaluation of some integrals $\hat{\mathcal{J}}_{(j)}^{i,\ell}$ given by

$$\begin{aligned}\hat{\mathcal{J}}_{(j)}^{i,\ell} &:= \iint_{(\mathbb{R} \times D_{\mathbf{s}}^{(i,j)}) \times \mathbb{R}^{d+1}} d\lambda d\xi d\tilde{\lambda} d\tilde{\xi} |\mathcal{F}K(2^{2\ell}\lambda, 2^\ell\xi)|^2 \mathcal{N}_{H_0, \mathbf{H}}(\lambda, \xi) \mathcal{N}_{H_0, \mathbf{H}}(\tilde{\lambda}, \tilde{\xi}) \\ &\quad \times \mathcal{T}^{(0)}(\lambda + \tilde{\lambda})^2 \prod_{k=1}^{i-1} \left(\mathcal{T}^{(i)}(\tilde{\xi}_i) \right)^2 \left(|\xi_i|^2 \mathcal{Q}^{(i)}(\xi_i, \tilde{\xi}_i)^2 \right) \prod_{k=i+1}^d \left(\mathcal{T}^{(k)}(\xi_k + \tilde{\xi}_k) \right)^2,\end{aligned}$$

where $\mathbf{s}^{(i,j)} := \{-\}^j \times \{+\}^{i-1-j}$. We now follow the same procedure as before and bound this integral by

$$\begin{aligned}\hat{\mathcal{J}}_{(j)}^{i,\ell} &\lesssim 2^{-4\ell(a_0+a_1+\dots+a_d)} \left(\int_{\mathbb{R} \times \mathbb{R}} d\lambda d\tilde{\lambda} \frac{\mathcal{T}^{(0)}(\lambda + \tilde{\lambda})^2}{|\lambda|^{2a_0+2H_0-1} |\tilde{\lambda}|^{2H_0-1}} \right) \\ &\quad \prod_{k=1}^j \left(\int_{|\xi_k| \leq 1} \frac{d\xi_k}{|\xi_k|^{4a_k+2H_k-1}} \right) \prod_{m=j+1}^{i-1} \left(\int_{|\xi_m| \geq 1} \frac{d\xi_m}{|\xi_m|^{4a_m+2H_m-1}} \right) \\ &\quad \left(\int_{\mathbb{R} \times \mathbb{R}} d\xi_i d\tilde{\xi}_i \frac{\mathcal{Q}^{(i)}(\xi_i, \tilde{\xi}_i)^2}{|\xi_i|^{(4a_i+2H_i-2)-1} |\tilde{\xi}_i|^{2H_i-1}} \right) \prod_{p=i+1}^d \left(\int_{\mathbb{R} \times \mathbb{R}} d\xi_p d\tilde{\xi}_p \frac{\mathcal{T}^{(p)}(\xi_p + \tilde{\xi}_p)^2}{|\xi_p|^{4a_p+2H_p-1} |\tilde{\xi}_p|^{2H_p-1}} \right),\end{aligned}$$

and the right-hand side above is finite as long as the following constraints are satisfied:

$$\begin{cases} \max(2H_0, 3 - 2H_0) < 2a_0 + 2H_0 < 2 \\ 2H_k < 4a_k + 2H_k < 2 & \text{for } k = 1, \dots, j \\ 2 < 4a_m + 2H_m < 4 + 2H_m & \text{for } m = j+1, \dots, i-1 \\ \max(2H_i, 3 - 2H_i) < 4a_i + 2H_i < 4 \\ \max(2H_p, 3 - 2H_p) < 4a_p + 2H_p < 2 & \text{for } p = i+1, \dots, d. \end{cases} \quad (\text{A.52})$$

Note that since $H_0 > \frac{1}{2}$, one has $2 > 3 - 2H_0$ and so the first condition can indeed be met. Similarly, since $H_2 > \frac{1}{2}, \dots, H_d > \frac{1}{2}$, one has $2 > 3 - 2H_p$ for $p = i+1, \dots, d$, and so the last

condition makes sense as well. If we sum again the constraints in (A.52) in order to check that $\sum_{i=1}^d a_i = 1 - \varepsilon$, we get

$$\hat{a}_{ij} < 4(1 - \varepsilon) + 2(2H_0 + H) < \hat{b}_{ij}, \quad (\text{A.53})$$

where $\hat{a}_{ij}, \hat{b}_{ij}$ are respectively defined by

$$\begin{aligned} \hat{a}_{ij} = & 2 \max(2H_0, 3 - 2H_0) + 2 \sum_{k=1}^j H_k + 2(i - 1 - j) \\ & + \max(2H_i, 3 - 2H_i) + \sum_{p=i+1}^d \max(2H_p, 3 - 2H_p) \end{aligned}$$

and

$$\hat{b}_{ij} = 4 + 2j + \sum_{m=j+1}^{i-1} (4 + 2H_m) + 4 + 2(d - i).$$

We now check that (A.53) is fulfilled under our assumption $d + 1/2 < 2H_0 + H \leq d + 1$. Indeed, we can first use the fact that $H_0 > 1/2$ and $H_p > 1/2$ for $p = i + 1, \dots, d$, plus the obvious bound $\max(2H_i, 3 - 2H_i) < 3$, in order to assert that

$$\hat{a}_{ij} < 4 + 2j + 2(i - 1 - j) + 3 + 2(d - i) = 4 + (2d + 1) < 4 + 2(2H_0 + H),$$

where the last inequality stems from the assumption $2H_0 + H > d + \frac{1}{2}$. As for the second inequality in (A.53), we can write

$$\begin{aligned} & 4 + 2j + \sum_{m=j+1}^{i-1} (4 + 2H_m) + 4 + 2(d - i) \\ &= 4 + (2d + 2) + 4(i - 1 - j) + 2 \sum_{m=j+1}^{i-1} H_m - 2(i - 1 - j) \\ &> 4 + (2d + 2) + 2(i - 1 - j) \\ &> 4(1 - \varepsilon) + 2(2H_0 + H), \end{aligned}$$

where we have used the fact that $2H_0 + H \leq d + 1$ to derive the last inequality.

This finishes the proof of (A.48), which was our last step in order to achieve Proposition A.8. $\square$

A.3. Conclusion: proof of Theorem 4.15. Let us now see how we can use the moments estimates of Propositions A.4 and A.8 in order to prove the desired convergence (4.34).

First, by applying Lemma A.2 to a constant distribution $\zeta_{s,x} := \dot{W}^n - \dot{W}^m$ (which means that $\theta^i = \zeta^{\sharp,i} = 0$ in Lemma A.2), we get that for every $k, p \geq 1$,

$$\begin{aligned} \mathbb{E} \left[\left\| \dot{W}^n - \dot{W}^m \right\|_{\alpha;k,w}^{2p} \right] &\lesssim \mathbb{E} \left[\sup_{\psi \in \Psi} \sup_{\ell \geq 0} \sup_{(s,x) \in \Lambda_s^\ell \cap ([-(k+2), k+2] \times \mathbb{R}^d)} 2^{2\ell p \alpha} \frac{|\langle \dot{W}^n - \dot{W}^m, \mathcal{S}_{s,x}^{2^{-\ell}} \psi \rangle|^{2p}}{w(x)^2} \right] \\ &\lesssim \sum_{\psi \in \Psi} \sum_{\ell \geq 0} \sum_{(s,x) \in \Lambda_s^\ell \cap ([-(k+2), k+2] \times \mathbb{R}^d)} 2^{2\ell p \alpha} \frac{\mathbb{E} [|\langle \dot{W}^n - \dot{W}^m, \mathcal{S}_{s,x}^{2^{-\ell}} \psi \rangle|^{2p}]}{w(x)^{2p}}, \end{aligned}$$

Furthermore, $\dot{W}^n - \dot{W}^m$ is a Gaussian process. Therefore we have

$$\begin{aligned} \mathbb{E} \left[\left\| \dot{W}^n - \dot{W}^m \right\|_{\alpha;k,w}^{2p} \right] &\lesssim \sum_{\psi \in \Psi} \sum_{\ell \geq 0} \sum_{(s,x) \in \Lambda_s^\ell \cap ([-(k+2), k+2] \times \mathbb{R}^d)} 2^{2\ell p \alpha} \frac{\mathbb{E} [|\langle \dot{W}^n - \dot{W}^m, \mathcal{S}_{s,x}^{2^{-\ell}} \psi \rangle|^{2p}]}{w(x)^{2p}} \\ &\lesssim 2^{-m\varepsilon p} \sum_{\ell \geq 0} 2^{2\ell p(\alpha+d+2-(2H_0+H)+\varepsilon)} \sum_{(s,x) \in \Lambda_s^\ell \cap ([-(k+2), k+2] \times \mathbb{R}^d)} w(x)^{-2p}, \quad (\text{A.54}) \end{aligned}$$

where the last inequality follows from Proposition A.4 and the fact that Ψ is a finite set.

At this point, observe that

$$\begin{aligned} \sum_{(s,x) \in \Lambda_s^\ell \cap ([-(k+2), k+2] \times \mathbb{R}^d)} w(x)^{-2p} &= \left(\sum_{q_0 \in \mathbb{Z}} \mathbf{1}_{\{-(k+2) \leq q_0 2^{-2\ell} \leq k+2\}} \right) \left(\sum_{q \in \mathbb{Z}^d} (1 + 2^{-\ell} |q|)^{-2\kappa p} \right) \\ &\lesssim 2^{2\ell} k \left\{ 1 + 2^{\kappa \ell p} \sum_{q \in \mathbb{Z}^d \setminus \{0\}} |q|^{-2\kappa p} \right\}. \end{aligned}$$

Going back to (A.54) and picking $\varepsilon > 0$ small enough such that $\beta := -\alpha - (d+2) + (2H_0 + H) - \varepsilon > 0$, we have obtained that for every $k, p \geq 1$,

$$\mathbb{E} \left[\left\| \dot{W}^n - \dot{W}^m \right\|_{\alpha;k,w}^{2p} \right] \lesssim k 2^{-m\varepsilon p} \sum_{\ell \geq 0} \left\{ 2^{-2\ell(\beta p - 1)} + 2^{-2\ell((\beta - \kappa)p - 1)} \sum_{q \in \mathbb{Z}^d \setminus \{0\}} |q|^{-2\kappa p} \right\}. \quad (\text{A.55})$$

Without loss of generality, we can here assume that $0 < \kappa < \beta$. Then we can pick $p \geq 1$ large enough so that $(\beta - \kappa)p - 1 > 0$ and $2\kappa p > d$, which ensures that the sum in (A.55) is finite, and so, for every $k \geq 1$ and any such large $p \geq 1$,

$$\mathbb{E} \left[\left\| \dot{W}^n - \dot{W}^m \right\|_{\alpha;k,w}^{2p} \right] \lesssim k 2^{-m\varepsilon p}. \quad (\text{A.56})$$

Using similar arguments (starting from Lemma A.2, and also leaning on (4.14)), we can then turn the estimate of Proposition A.8 into the bound

$$\mathbb{E} \left[\left\| \widehat{\mathbf{W}}^{2,n} - \widehat{\mathbf{W}}^{2,m} \right\|_{2\alpha+2;k,w}^{2p} \right] \lesssim k 2^{-m\varepsilon p}, \quad (\text{A.57})$$

for every $k \geq 1$, every $\varepsilon > 0$ small enough and every $p \geq 1$ large enough.

Combining (A.56) and (A.57), we get that for all $\varepsilon > 0$ small enough and $p \geq 1$ large enough

$$\mathbb{E} [d_{\alpha,w}(\widehat{\mathbf{W}}^n, \widehat{\mathbf{W}}^m)^{2p}] \lesssim 2^{-m\varepsilon p},$$

for all $n \geq m \geq 1$, and accordingly $(\widehat{\mathbf{W}}^n)_{n \geq 1}$ is a Cauchy sequence in $L^p(\Omega; (\mathcal{E}_{\alpha;w}^K, d_{\alpha;w}))$. By Lemma 4.8, we can assert that there exists an element $\widehat{\mathbf{W}} \in \mathcal{E}_{\alpha;w}^K$ satisfying

$$\mathbb{E}[d_{\alpha,w}(\widehat{\mathbf{W}}, \widehat{\mathbf{W}}^m)^{2p}] \lesssim 2^{-m\epsilon p},$$

for every $p \geq 1$ large enough. The desired conclusion, that is the almost sure convergence of $\widehat{\mathbf{W}}^n$ to $\widehat{\mathbf{W}}$ in $(\mathcal{E}_{\alpha;w}^K, d_{\alpha;w})$, immediately follows from Borel-Cantelli lemma.

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XIA CHEN: DEPARTMENT OF MATHEMATICS, UNIVERSITY OF TENNESSEE KNOXVILLE, TN 37996-1300, UNITED STATES.

E-mail address: `xchen@math.utk.edu`

AURÉLIEN DEYA: INSTITUT ELIE CARTAN, UNIVERSITY OF LORRAINE B.P. 239, 54506 VANDOEUVRE-LÈS-NANCY, CEDEX FRANCE.

E-mail address: `Aurelien.Deya@univ-lorraine.fr`

CHENG OUYANG: DEPARTMENT OF MATHEMATICS, STATISTICS AND COMPUTER SCIENCE, UNIVERSITY OF ILLINOIS AT CHICAGO, CHICAGO, UNITED STATES.

E-mail address: `couyang@math.uic.edu`

SAMY TINDEL: DEPARTMENT OF MATHEMATICS, PURDUE UNIVERSITY, 150 N. UNIVERSITY STREET, WEST LAFAYETTE, IN 47907, UNITED STATES.

E-mail address: `stindel@purdue.edu`